Antihypertensive Therapy

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CHAPTER OUTLINE

PHARMACOLOGY OF THE NONDIURETIC ANTIHYPERTENSIVE DRUGS

ANGIOTENSIN-CONVERTING ENZYME INHIBITORS

CLASS MECHANISMS OF ACTION

Angiotensin-converting enzyme (ACE) inhibitors inhibit the activity of ACE, which converts the inactive decapeptide angiotensin I (Ang I) into the potent hormone angiotensin II (Ang II) (Fig. 48.1, Table 48.1; see Chapter 49 for a review of diuretic drugs). Because Ang II plays a crucial role in maintaining and regulating BP levels by promoting vasoconstriction and kidney sodium and water retention, ACE inhibitors are powerful tools for targeting multiple pathways that contribute to hypertension. ACE inhibitors directly reduce circulating and tissue levels of Ang II, thus blocking the potent vasoconstriction induced by the hormone (Table 48.2).[1] The resulting decrease in peripheral vascular resistance is not accompanied by changes in cardiac output or glomerular filtration rate (GFR); the heart rate is unchanged or may be reduced in patients with baseline heart rates higher than 85 beats per minute.[2] A reduction in systemic and local levels of Ang II leads to effects beyond vasodilation that contribute to the antihypertensive efficacy of ACE inhibitors (see Table 48.2).[3-16]

CLASS MEMBERS

Currently, more than 15 ACE inhibitors are in clinical use. Each drug has a unique structure that determines its potency, tissue receptor-binding affinity, metabolism, and prodrug compound, but these drugs have remarkably similar clinical effects (Tables 48.3 and 48.4).[3] The drugs are classified into sulfhydryl, carboxyl, or phosphinyl categories on the basis of the ligand that binds to the ACE–zinc moiety.

Sulfhydryl angiotensin-converting enzyme inhibitors

Captopril is a short-acting sulfhydryl-containing ACE inhibitor available in tablets of 12.5, 25, and 50 mg and 100 mg (see Tables 48.3 and 48.4).[12,17,18] The usual starting dosage for hypertension treatment is 25 mg two or three times daily (see Table 48.3), and the dosage can be titrated at 1- to 2-week intervals.[12] Captopril has 75% bioavailability, with peak onset within 1 hour. The half-life is 2 hours; with long-term administration, the hemodynamic effects are maintained for 3 to 8 hours.[12] Food may decrease captopril absorption by up to 54%, but this decrease is clinically insignificant.[19] Captopril is partially metabolized in the liver into an inactive compound; 95% of the parent compound and metabolites are eliminated in the urine within 24 hours. The elimination half-life increases markedly in patients with creatinine clearances of <20 mL/min/1.73 m^2. In such patients, the initial dosages should be reduced and smaller increments should be used for titration. Hemodialysis removes approximately 35% of the dose.[12,20]

Carboxyl angiotensin-converting enzyme inhibitors

Benazepril hydrochloride is a long-acting, nonsulfhydryl-containing, carboxyl ACE inhibitor that is available as 10- or 20-mg tablets alone or in combination with amlodipine.[18] The usual initial dosage is 10 mg daily, with maintenance dosages of 20 to 40 mg daily. Some patients respond better to twice-daily dosing (see Tables 48.3 and 48.4).[21] The onset of action occurs in 2 to 6 hours; maximal antihypertensive responsiveness occurs in 2 weeks. Benazepril is a prodrug that is rapidly bioactivated in the liver into the active benazeprilat compound, which is 200 times more potent than benazepril. The elimination half-life of benazeprilat is 22 hours. Benazeprilat is excreted primarily in the urine. Dialysis does not remove benazepril, but the initial dose should be no more than 10 mg in patients with a creatinine clearance <60 mL/min/1.73 m^2, and the dose should be reduced to 5 mg in patients with creatinine clearance of <30 mL/min/1.73 m^2.[22]

Enalapril maleate is a nonsulfhydryl prodrug of the long-acting ACE inhibitor enalaprilat.[18,23] Oral preparations are available in tablets of 2.5, 5, 10, and 20 mg. The initial dosage of enalapril is 5 mg daily (see Table 48.3). The usual daily dose is 10 to 40 mg singly or in divided doses. Initial responses occur in 1 hour, and peak serum levels of enalaprilat are achieved in 3 to 4 hours. Enalapril undergoes biotransformation in the liver into the active compound enalaprilat (see Table 48.4). Enalapril is excreted primarily in the urine. Dosages should be reduced by 25% to 50% in patients with end-stage kidney disease (ESKD).[22]

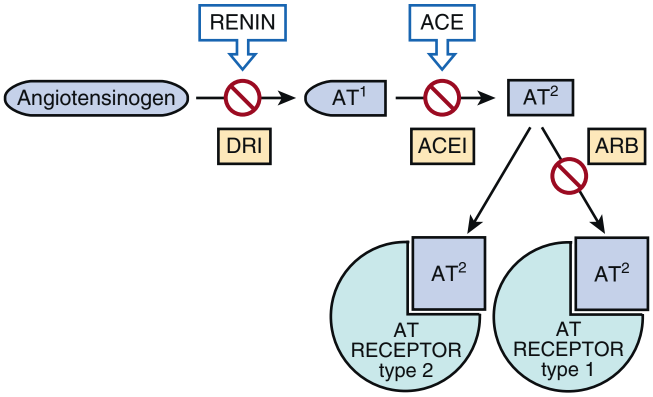

Fig. 48.1 Renin–angiotensin–aldosterone system (RAAS)–effect site for each type of RAAS-blocking drugs. The two major classes of drugs that target the RAAS are the angiotensin-converting enzyme *(ACE)* inhibitors and the selective AT1 receptor blockers *(ARBs)*. Both these drug classes target angiotensin II, but the differences in their mechanisms of action have implications for their effects on other pathways and receptors. Both ACE inhibitors and ARBs are effective antihypertensive agents that have been shown to reduce the risk of cardiovascular and kidney events. Direct inhibition of renin—the most proximal aspect of the RAAS—became clinically feasible in 2007 with the introduction of aliskiren. *AT,* Angiotensin; *DRI,* direct renin inhibitor. (From Robles NR, Cerezo I, Hernandez-Gallego R. Renin-angiotensin system blocking drugs. *J Cardiovasc Pharmacol Ther.* 2014;19:14–33.)

Table 48.1 Pharmacologic Classification of Nondiuretic Antihypertensive Drugs

Angiotensin-converting enzyme inhibitors
Sulfhydryl
Carboxyl
Phosphinyl
Angiotensin II type 1 receptor antagonists
Biphenyl tetrazoles
Nonbiphenyl tetrazoles
Nonheterocyclics
β-Adrenergic antagonists and α_1- and β-adrenergic antagonists
Nonselective β-adrenergic antagonists
Nonselective β-adrenergic antagonists with partial agonist activity
β_1-Selective adrenergic antagonists
β_1-Selective adrenergic antagonists with partial agonist activity
Nonselective β-adrenergic and α_1-adrenergic antagonists
Calcium channel blockers
Benzothiazepines
Dihydropyridines
Diphenylalkylamines
Central α_2-adrenergic agonists
Central and peripheral adrenergic-neuronal blocking agents
Direct-acting vasodilators
Moderately selective peripheral α_1-adrenergic antagonists
Peripheral α_1-adrenergic antagonists
Peripheral adrenergic-neuronal blocking agents
Renin inhibitors
Aldosterone receptor antagonists
Tyrosine hydroxylase inhibitors
Dual endothelin A and B receptor antagonists

Table 48.2 Antihypertensive Mechanisms of Action of Angiotensin-Converting Enzyme Inhibitors

Lower peripheral vascular resistance.
Inhibit the breakdown of vasodilatory bradykinins.
Enhance vasodilatory prostaglandin synthesis.
Improve nitric oxide–mediated endothelial function.
Reverse vascular hypertrophy.
Decrease aldosterone secretion.
Induce natriuresis.
Augment renal blood flow.
Blunt sympathetic nervous system activity and pressor responses.
Inhibit norepinephrine and arginine vasopressin release.
Inhibit baroreceptor reflexes.
Reduce endothelin-1 levels.
Inhibit thirst.
Inhibit oxidation of cholesterol.
Inhibit collagen deposition in target organs.

Lisinopril is a nonsulfhydryl analog of enalaprilat.[18,24] The initial dosage is 10 mg daily, and the usual daily dose is 20 to 40 mg (see Table 48.3). The initial antihypertensive response occurs in 1 hour, peaks at 6 hours, and lasts for 24 hours (see Table 48.4). The maximal effect may not be observed for 24 hours. The elimination half-life is 12 hours. Lisinopril is not metabolized, and it is exclusively eliminated in the urine unchanged. Lisinopril is dialyzable, and patients undergoing dialysis may require supplemental doses. The initial dosage should be reduced to 2.5 to 7.5 mg daily in patients with moderate to advanced chronic kidney disease (CKD).[22]

Perindopril is a nonsulfhydryl prodrug of the long-acting ACE inhibitor perindoprilat.[12,25] The usual daily dose is 4 to 8 mg (see Table 48.3). The response peaks at 3 to 7 hours. A single dose has a duration of action of 24 hours. Perindopril undergoes extensive first-pass hepatic metabolism into the active metabolite perindoprilat (see Table 48.4). Kidney excretion accounts for 75% of the clearance. The dosage should be reduced by 75% in patients with creatinine clearance <50 and by 50% for patients with creatinine clearance <10 mL/min/1.73 m^2, respectively.[22,25]

Quinapril hydrochloride is a nonsulfhydryl prodrug of the ACE inhibitor quinaprilat.[18,25,26] The initial dose is 10 mg, and the usual daily dose is 20 to 80 mg, which should be adjusted at 2-week intervals (see Table 48.3). Twice-daily therapy may provide a more sustained BP reduction. The onset of action occurs in 1 hour, and the peak response occurs in 2 hours and lasts for 24 hours. Quinapril is extensively metabolized in the liver into the active metabolite, quinaprilat (see Table 48.4). Kidney excretion by way of filtration and active tubular secretion accounts for 50% of the clearance. Quinapril is not dialyzable. The dosage should be reduced by 25% to 50% in patients with ESKD.[22]

Ramipril is a potent, nonsulfhydryl prodrug of the ACE inhibitor ramiprilat.[12,18] Ramipril capsules are available in 1.25, 2.5, 5, and 10 mg. The initial daily dose is 2.5 mg (see Table 48.3). The usual daily dose is 2.5 to 20 mg, and it can be titrated by doubling the current dose at 2- to 4-week intervals. Ramipril is well absorbed from the gastrointestinal

Table 48.3 Pharmacodynamic Properties of Angiotensin-Converting Enzyme Inhibitors

Generic Name (Trade Name)	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Alacepril (Cetapril)	12.5	12.5–100	100	qd	3	24
Benazepril (Lotensin)	10	20–40	40	qd	2–6	24
Captopril (Capoten)	25	12.5–50	150	tid	1–2	3–8
Cilazapril (Dynorm)	2.5	2.5–10	10	qd, bid	6	8–12
Enalapril (Vasotec)	5	10–40	40	qd, bid	3–4	12–24
Fosinopril (Monopril)	5	5–40	40	qd, bid	2–7	24
Imidapril (Tanatril)	5	10–40	40	qd	5–6	24
Lisinopril (Zestril, Prinivil)	10	20–40	40	qd	6	24
Perindopril (Aceon)	4	4–8	8	qd	3–7	24
Quinapril (Accupril)	10	20–80	80	qd	2	24
Ramipril (Altace)	2.5	2.5–20	40	qd, bid	2	24
Spirapril (Renormax)	6	6	6	qd	3–6	24
Trandolapril (Mavik)	1	2–4	8	qd	2–12	24
Zofenopril (Bifril)	15	30–60	60	qd	—	—

Table 48.4 Pharmacokinetic Properties of Angiotensin-Converting Enzyme Inhibitors

Drug	Absorption (%)	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Alacepril	—	70	—	1	1.9	L	U (70%)	Captopril
Benazepril	35	>37	No	2–6	22	L, K	F, U	Benazeprilat
Captopril	60–75	75	Yes	1	2	L	U	Inactive
Cilazapril	—	57–76	No	1–2	30–50	L	U (52%)	Cilazaprilat
Enalapril	55–75	73	—	3–4	11–35	L	F, U	Enalaprilat
Fosinopril	36	36	—	1	12	L, K, I	F, U	Fosinoprilat
Imidapril	—	40	—	3–10	10–19	L	U	Imidaprilat
Lisinopril	25	6–60	—	1	12	K	U	Enalaprilat
Quinapril	60	50	Yes	1	25	L	U (50%)	Quinaprilat
Perindopril	—	75	Yes	1.5	3–10	L	F, U (75%)	Perindoprilat
Ramipril	50–60	60	—	1–2	13–17	L	F, U	Ramiprilat
Spirapril	—	50	Yes	1	33–41	L	F (60%), U (40%)	Spiraprilat
Trandolapril	70	10	No	2–12	16–24	L	F (66%), U	`Trandolaprilat
Zofenopril	>80	96	Yes	5	5	K	F (26%), U (69%)	Zofenoprilat

F, Feces; *I,* intestine; *K,* kidney; *L,* liver; *U,* urine.

tract; peak concentrations are achieved in 1 to 2 hours (see Table 48.4). The peak response occurs in 2 hours and lasts for 24 hours. Ramipril is extensively metabolized in the liver into the active metabolite ramiprilat. The elimination half-life of the active compound is 13 to 17 hours, and it is prolonged in patients with kidney failure to approximately 50 hours. The dosage should be reduced by 50% to 75% in patients with a creatinine clearance of <50 mL/min/1.73 m^2.[18,22]

Trandolapril is a nonsulfhydryl ethyl ester prodrug of the ACE inhibitor trandolaprilat.[12,18] It is available in tablets of 1, 2, and 4 mg or in combination with verapamil. The usual starting dosage is 1 mg daily (see Table 48.3). Trandolapril is only 10% bioavailable, and its absorption is not affected by food (see Table 48.4).[27] Trandolapril undergoes extensive first-pass hepatic metabolism into trandolaprilat. The peak serum concentrations of trandolaprilat occur within 2 to 12 hours; the duration of action is 24 hours, but it may be as long as 6 weeks. The recommended starting dose in patients with a creatinine clearance of <30 mL/min/1.73 m^2 is 0.5 mg.

Phosphinyl angiotensin-converting enzyme inhibitor

Fosinopril sodium is a nonsulfhydryl prodrug of fosinoprilat, a long-acting ACE inhibitor.[18,28] The usual daily dose is 5 to 40 mg (see Table 48.3). Its maximal effects may not occur until 4 weeks. The initial response occurs in 1 hour, the peak response occurs in 2 to 7 hours, and the duration of response is 24 hours, which is prolonged in patients with ESKD (see Tables 48.3 and 48.4). The elimination half-life of fosinoprilat is 12 hours. All metabolites are excreted in the urine and feces. Hepatic biliary clearance increases significantly as kidney function declines. Thus the dosage should be reduced by 25% in patients with ESKD.[22]

CLASS KIDNEY EFFECTS

ACE inhibitors help slow CKD progression through several different hemodynamic and nonhemodynamic effects (Table 48.5). In patients with elevated BP, ACE inhibitors can restore the pressure-natriuresis relationship, thereby maintaining sodium balance at a lower arterial BP.[29] The

Table 48.5 Renal Protective Mechanisms of Angiotensin-Converting Enzyme Inhibitors

Decrease renal blood flow.
Decrease arterial pressure.
Decrease filtration fraction.
Decrease renal vascular resistance.
Decrease aldosterone production.
Decrease proteinuria.
Inhibit tubule sodium resorption.
Restore pressure-natriuresis relationship.
Improve altered lipid profile.
Reduce scarring and fibrosis.
Attenuate oxidative stress and reduce free radicals.

response is exaggerated in the setting of dietary sodium restriction. The mechanism responsible for this effect is the direct inhibition of proximal, and possibly distal, tubule sodium reabsorption.[30] The increased kidney excretory capacity plays a major role in the long-term antihypertensive activity of the drugs. Clinically, the increase in sodium excretion is transitory because the reduced arterial pressure allows the sodium excretion to return to normal. However, the maintenance of normal sodium excretion at lower arterial pressures correlates with increased excretion in the setting of hypertension.[29] After several days, inhibition of Ang II and aldosterone contributes to the natriuresis.[31] The long-term effects on water excretion are less certain. ACE inhibitors initially induce an increase in free water clearance, but there are no long-term changes in total body weight, plasma, or extracellular fluid volume. The decrease in aldosterone caused by ACE inhibition also correlates with decreased potassium excretion,[31] particularly in patients with impaired kidney function.

Intrarenal inhibition of ACE in the vascular wall and kidney vessels contributes to the antihypertensive activity of ACE inhibitors. Mice lacking only intrarenal ACE are protected against hypertension.[32] ACE inhibitor–induced changes in BP correlate with the degree of inhibition of RAAS in plasma and tissues.[33] The ACE inhibitors with the greatest tissue specificity are associated with prolonged activity at the tissue level, even after serum ACE levels have returned to normal.[33] Consequently, they are more efficacious with once-daily dosing.[33] Other potential kidney-protective effects that have been noted in experimental models include attenuation of oxidative stress,[34] scavenging of free radicals, and attenuation of lipid peroxidation.[35]

ACE inhibitors decrease circulating concentrations of Ang II and increase circulating concentrations of angiotensin (1–7) (a potential vasodilator) and plasma renin. The effects of ACE inhibitors on angiotensin peptide levels depend on the responsiveness of renin secretion.[36,37] When renin levels show little increase in response to ACE inhibition, the levels of Ang II and its metabolites decrease markedly, with little change in the levels of Ang I. Large increases in renin levels in response to ACE inhibition increase the levels of Ang I and its metabolites. The increased levels of Ang I can produce higher levels of Ang II through uninhibited ACE and other pathways, thereby blunting the effect of reduced Ang II. This phenomenon is termed "ACE escape" and may contribute to reduced ACE inhibitor efficacy when used over a long term.[38]

Most of the vasoconstrictive action of Ang II is confined to the efferent arteriole. ACE inhibitors preferentially dilate the efferent arteriole by reducing the systemic and intrarenal levels of Ang II. The effect is a reduction in intraglomerular capillary pressure. Many patients with impaired kidney function exhibit a reversible fall in GFR with ACE inhibitor therapy that is not detrimental. The GFR declines initially because of hemodynamic changes, but the long-term reduction in perfusion pressure is kidney protective. Even in patients receiving hemodialysis, ACE inhibitor therapy significantly preserves residual kidney function and helps maintain urine output.[39] An initial elevation in the serum creatinine level of up to 30% above baseline with ACE inhibitor initiation that stabilizes within the first 2 months of therapy has been considered acceptable, and it is no reason to discontinue therapy with these medications. In fact, a review of 12 randomized trials showed that in patients with CKD receiving ACE inhibitors, a stable rise in the serum creatinine of <30% was associated with long-term preservation of kidney function.[40,41]

Patients with severe bilateral renal artery stenosis,[42] unilateral renal artery stenosis of a solitary kidney, severe hypertensive nephrosclerosis, volume depletion, congestive heart failure, or cirrhosis[43] are at higher risk for deterioration in kidney function with ACE inhibitor therapy.[44,45] In these states of reduced kidney perfusion related to low effective arterial circulating volume or reduced perfusion because of narrowing of arterial inflow, the maintenance of kidney blood flow and GFR is highly dependent on increased efferent arteriolar vasoconstriction mediated by Ang II. Interruption of the increased tone causes a critical reduction in perfusion pressure and can lead to dramatic reductions in GFR and urinary flow, worsening of kidney ischemia and, in rare cases, anuria. The hemodynamic effect is typically reversible with cessation of therapy.[46] Studies of atherosclerotic renovascular disease have indicated that the risks of ACE inhibitors and other RAAS blockers (e.g., azotemia, hyperkalemia, and angioedema) can be balanced by their benefits when patients are carefully monitored.[47-49]

CLASS EFFICACY AND SAFETY

ACE inhibitors are recommended in patients with mild, moderate, and severe hypertension, regardless of age, sex, or ancestry.[50] They are effective in patients with diabetes, obesity, and/or following kidney transplantation.[51] All ACE inhibitors decrease urinary protein excretion[52-54] in normotensive and hypertensive patients with kidney disease of various etiologies. Individual response rates vary from an increase of 31% to a decrease of 100%, and they are strongly influenced by drug dosage and dietary sodium intake.[53,55] Importantly, multiple randomized controlled trials (RCTs) have shown that ACE inhibitors slow the progression of kidney disease,[56-58] even in advanced CKD.[59] Moreover, one RCT[60] showed that withdrawing ACE inhibitors and angiotensin receptor blockers (ARBs) in advanced CKD did not result in significant differences in the rate of GFR decline, nor in the requirement for initiation of kidney replacement therapy.

In most studies, ACE inhibitors elicit an adequate BP-lowering response in 40% to 60% of patients.[61] An immediate fall in BP occurs in 70% of patients. The enhanced efficacy of ACE inhibitors in the presence of salt restriction is paralleled by the additive effects of diuretic therapy.[62] The addition of low-dose hydrochlorothiazide (HCTZ) enhances the efficacy more than 80%, normalizing BP in another 20% to 25% of patients.[62] Adding a thiazide agent is generally more effective than increasing the dosage of the ACE inhibitor.[62,63]

Neither the duration nor the degree of BP lowering is predicted by the effect on blood ACE or Ang II levels, and all ACE inhibitors appear to have comparable efficacy. The response may be due in part to interindividual variability of the ACE genotype.[64,65] The activity of ACE is partially dependent on the presence or absence of a 287–base pair element in intron 16, and this insertion–deletion (I/D) polymorphism of a human Al repetitive DNA element accounts for 47% of the total phenotypic variation in plasma ACE. Deletion–deletion cases have the highest ACE concentrations, I/D cases have intermediate ACE concentrations and insertion–insertion cases have the lowest. Genotype also influences tissue ACE activity, but the clinical implications remain incompletely understood.

ACE inhibitors may cause fetal or neonatal injury or death when used during the second and third trimesters of pregnancy[66] because angiotensin appears to be required for normal fetal growth and development.[67] Use of ACE inhibitors during the first trimester might be associated with congenital malformations, but these malformations may be related to maternal factors. A systemic review showed that among 118 cases of ACE inhibitor exposure and 68 cases of ARB exposure, neonatal complications were more frequent following exposure to ARBs. Forty-eight percent of newborns exposed to ACE inhibitors exhibited features thought to be related to RAAS inhibitor exposure, in contrast to 87% of newborns exposed to ARBs ($P < 0.0001$).[68] Among 26 newborns affected by RAAS inhibitor exposure with >6 months of follow-up, 6 (23%) developed kidney failure, 4 (15%) had hypertension, and 3 (12%) experienced neurodevelopmental delay.

A prospective, observational, controlled cohort study of ACE inhibitors and ARB exposure during the first trimester was reported in 2012.[69] Participants were enrolled from an unselected sample of women contacting a teratogen information service. There were two comparison groups, women with hypertension treated with other antihypertensives (including α-methyldopa or calcium channel blockers [CCBs]) and healthy controls. In the ACE–ARB and disease-matched groups, the offspring exhibited significantly lower birth weights and gestational ages than those of the healthy controls ($P < 0.001$ for both variables). A significantly higher rate of miscarriage was noted in the ACE–ARB group ($P < 0.001$). These results suggest that ACE inhibitors and ARBs are not major human teratogens during the first trimester. There was, however, a higher rate of spontaneous abortions in the ACE–ARB group.

When a patient becomes pregnant during treatment with an ACE inhibitor or ARB, the drug should be discontinued immediately, and alternative antihypertensive therapy should be started. Pregnancy termination is at the discretion of the patient and treatment team, but physicians and patients can derive some reassurance from the small studies quoted earlier if the drug is discontinued during the first trimester. Recognizing that the benefits of ACE inhibitors are largely long term, women attempting conception should transition to an alternative antihypertensive agent with a proven safety profile in pregnancy (see Chapter 58).

ACE inhibitors are transferred into breast milk, but the drug levels in milk are low. Captopril and enalapril have been reviewed by the American Academy of Pediatrics[70] and the National Institute for Health and Care Excellence (NICE) guidelines for the management of hypertension in pregnancy from the United Kingdom,[71] and they are compatible with lactation. However, newborns may be more susceptible to the hemodynamic effects of these drugs (e.g., hypotension) and sequelae (e.g., oliguria and seizures).

Overall, ACE inhibitors are well tolerated and have relatively neutral or beneficial metabolic effects. ACE inhibitors are associated with 8% to 11% reductions in levels of low-density lipoprotein (LDL) cholesterol and triglycerides and 5% increases in levels of high-density lipoprotein (HDL) cholesterol.[72] They do not cause perturbations of serum sodium or uric acid levels. ACE inhibitors reduce the levels of plasminogen activator inhibitor-1 and may improve fibrinolysis.[73]

The effects of ACE inhibitors on glucose metabolism are favorable.[74] They may improve glucose tolerance by augmenting the insulin secretory response to glucose[74] and may also help ameliorate obesity and hyperinsulinemia.[75] The use of ACE inhibitors has been clinically associated with a 25% to 30% reduction in the risk of developing diabetes.[76] Several large clinical trials have evaluated the clinical relevance of this finding.[77] Many of the ACE inhibitors require dosage adjustment in the presence of impaired kidney function (Table 48.6).

The most common adverse effect (AE) of ACE inhibitors is a dry, hacking, nonproductive cough, which has been reported in up to 20% of patients.[78] The ACE inhibitor–induced cough is thought to be secondary to hypersensitivity to bradykinins, which are increased by ACE,[79] increased levels of prostaglandins,[80] accumulation of substance P,[78] a potent bronchoconstrictor,[81] or polymorphisms in the neurokinin-2 receptor gene.[82] The cough can begin initially or many months after the start of therapy.[83] It is more common in women, African Americans,[84] and Asians[85,86] and it may spontaneously disappear.[83] It may be more common in patients with bronchial hyperreactivity, but ACE inhibitors are safe to use in asthmatic patients.[87] Cessation of ACE inhibitors and switching to an ARB is an effective way to manage ACE inhibitor–induced cough. Nonsteroidal antiinflammatory drugs (NSAIDs), oral iron supplements, sodium cromoglycate,[88] and picotamide have also been reported to reduce ACE inhibitor–induced cough.[89]

Angioedema is a rare but potentially life-threatening complication of ACE inhibitor therapy. It occurs in 0.1% to 0.7% of patients within hours of the first dose of ACE inhibitor or after prolonged use.[90-92] The absolute risk of ACE inhibitor–induced angioedema is low, but with large numbers of prescriptions written annually, many patients are at risk for this disorder.[93] ACE inhibitor–induced angioedema accounts for one-third of all cases of angioedema seen in emergency departments.

ACE inhibitor–induced angioedema occurs five times more frequently in individuals of African ancestry. ACE inhibitor–induced angioedema involves several

Table 48.6 Dose Modifications of Antihypertensive Drugs Required for Reduced Kidney Function

	Estimated Glomerular Filtration Rate (Creatinine Clearance; mL/min/1.73 m²)			
Drug	**>50**	**10–15**	**<10**	**Dialysis**[a]
Angiotensin-Converting Enzyme Inhibitors				
Benazepril	No change	50%	25%	Negligible
Captopril	No change	50%	25%	H: 50%
Cilazapril	No change	50%	25%	H: 50%
Enalapril	No change	50%	25%	H: 50%
Fosinopril	No change	No change	75%	—
Imidapril	No change	No change	—	—
Lisinopril	No change	50%	25%	H: 50%
Perindopril	No change	75%	50%	—
Quinapril	No change	50%	25%	—
Ramipril	No change	50%	25%	—
Trandolapril	No change	50%	25%	—
Zofenopril	No change	—	—	—
Angiotensin Receptor Blockers				
Azilsartan				
Candesartan	No change	No change	No change	Negligible
Eprosartan	No change	No change	50%	Negligible
Irbesartan	No change	No change	—	Negligible
Losartan	No change	No change	No change	Negligible
Olmesartan	No change	—	—	—
Telmisartan	No change	No change	No change	Negligible
Valsartan	No change	No change	No change	—
β-Blockers				
Acebutolol	No change	50%	30%-50%	H: 50%
Atenolol	No change	50%	25%	H: 50%
Nadolol	No change	50%	25%	H: 50%
Betaxolol	No change	No change	50%	H: 50%
Bisoprolol	No change	50%	25%	Negligible
Carvedilol	No change	No change	No change	—
Celiprolol	No change	50%	Avoid	—
Metoprolol	No change	No change	No change	—
Nebivolol	No change	50%	—	—
Penbutolol	No change	No change	50%	Negligible
Pindolol	No change	No change	50%	Negligible
Calcium Channel Blockers				
Amlodipine	No change	No change	No change	Negligible
Diltiazem	No change	No change	No change	Negligible
Felodipine	No change	No change	No change	Negligible
Isradipine	No change	No change	No change	Negligible
Lacidipine	No change	No change	No change	Negligible
Lercanidipine	Dosage adjustment in renal failure unknown			
Manidipine	No change	No change	No change	Negligible
Nicardipine	No change	No change	No change	Negligible
Nifedipine	No change	No change	No change	Negligible
Nisoldipine	No change	No change	No change	Negligible
Verapamil	No change	No change	No change	Negligible
Central α_2-Adrenergic or I_1 Imidazole Receptor Agonists				
Clonidine	No change	50%	25%	Negligible
α-Methyldopa	No change	No change	50%	H: 50%
Moxonidine	No change	50%	—	—
Rilmenidine	No change	50%	25%	—

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Table 48.6 Dose Modifications of Antihypertensive Drugs Required for Reduced Kidney Function (Cont'd)

	Estimated Glomerular Filtration Rate (Creatinine Clearance; mL/min/1.73 m²)			
Drug	**>50**	**10–15**	**<10**	**Dialysis[a]**
Direct-Acting Vasodilators				
Hydralazine	No change	No change	75%[b]	Negligible
Minoxidil	No change	50%	50%	H and P: 50%
Peripheral Adrenergic-Neuronal Blocking Agents				
Guanethidine	No change	No change	—	—
Guanadrel	No change	50%	25% (avoid)	—
Renin Inhibitor				
Aliskiren	No change	No change	Not studied	Not studied
Tyrosine Hydroxylase Inhibitor				
Metyrosine	No change	50%	25%	—
Selective Aldosterone Receptor Antagonist				
Eplerenone	Dosage adjustment in renal failure unknown Caution regarding hyperkalemia			

[a]Replacement dose at end of dialysis as percentage of dose prescribed for patient with glomerular filtration rate <10 mL/min.
[b]Slow acetylators.
Percentage of usual dose given.
H, Hemodialysis; *P,* peritoneal dialysis; —, not applicable.

components, including tissue accumulation of bradykinin and inhibition of C1 esterase activity.[90,94] Susceptible individuals typically have defects in non-ACE, nonkininase I vasoactive pathways of bradykinin degradation; possess the XPNPEP2 gene variant;[95-102] have elevated des-Arg9-BK;[97] or are taking dipeptidyl peptidase inhibitors (e.g., sitagliptin, saxagliptin, and linagliptin) to treat diabetes.[90,98-100] Some patients also have defective degradation of substance P, thereby increasing vascular permeability.[101]

Clinical features including asymmetric swelling confined to the face, subcutaneous or submucous membranes, and lips usually resolve with discontinuation of the therapy, but obstructive sleep apnea may be exacerbated.[102] If the ACE inhibitor is not discontinued, the episode usually resolves, but the frequency and severity of future episodes will escalate.[103,104] Angioedema of the small intestine and acute appendicitis have also been reported.[105-108]

Involvement of the glottis and larynx requiring airway management occurs in 10% of all cases and may result in laryngeal obstruction and death.[109] Administration of epinephrine, histamine-2 blockers, glucocorticoids, and/or fresh-frozen plasma is indicated.[110] Icatibant, a selective bradykinin B2 receptor antagonist, is approved for the treatment of hereditary angioedema. However, a meta-analysis[111] of three RCTs involving 179 patients showed no significant benefit of icatibant in time to achieve complete resolution of ACE inhibitor–induced angioedema compared with placebo or other conventional treatments (pooled mean difference 7.8 hours, 95% confidence interval [CI] –25.2 to 9.6 hours). ACE inhibitors are contraindicated in patients with known hypersensitivity.

First-dose hypotension, with a reduction in BP of up to 30%, has been reported with all ACE inhibitors in up to 2.5% of patients. In the Studies of Left Ventricular Dysfunction (SOLVD trial), hypotension was observed in 14.8% of patients versus 7.1% of individuals who received a placebo ($P < 0.0001$).[112] In the Ongoing Telmisartan Alone and in Combination with Ramipril Global Endpoint Trial (ONTARGET), of the 8579 patients who received ramipril, only 1.7% and 0.2% permanently stopped therapy because of hypotension and syncope, respectively.[90]

Hypotension occurs more commonly in patients with effective arterial volume depletion, patients with high-renin hypertension, and those with heart failure with reduced ejection fraction (HFrEF).[112,113] Hypotension is usually well tolerated, although occasionally it is associated with syncope. In older patients, ACE inhibitor therapy more frequently causes nocturnal hypotension.[114] The accompanying increase in the plasma norepinephrine concentration may explain the low incidence of orthostatic symptoms.[114] In patients at high risk of orthostatic symptoms, therapy should be initiated at lower dosages, preferably after holding diuretics. Rebound hypertension has not been reported with discontinuation of ACE inhibitors.

AEs related to the chemical structure are more frequently seen with the sulfhydryl-containing captopril than with the other agents. Dysgeusia appears to be related to the binding of zinc by the ACE inhibitors.[115] Approximately 2% to 4% of patients experience a diminution or loss of taste sensation that is associated with a metallic taste. It is usually self-limited and resolves in 2 to 3 months, even with continued therapy. However, it may be severe enough to interfere with nutrition and cause weight loss.[116]

Cutaneous reactions manifest as a nonallergic, pruritic, maculopapular eruption that appears during the first few weeks of therapy. These reactions may be associated with a fever or arthralgias and may disappear, even with continuation of the ACE inhibitor.[117]

Leukopenia and anemia have been reported with ACE inhibitor therapy. Among patients with normal or near-normal kidney function, ACE inhibitors may reduce hemoglobin concentrations in a dose-dependent manner.[118] ACE inhibitors have been demonstrated to interfere with the response to erythropoietin. Patients receiving hemodialysis and kidney transplant recipients receiving erythropoietin frequently require higher dosages to maintain hemoglobin concentrations.[119] Consequently, ACE inhibitors can be used effectively to reduce posttransplantation erythrocytosis[120] but appear to have little effect on erythropoiesis in patients receiving hemodialysis.[121] Neutropenia (<1000 neutrophils/mm^3) with myeloid hypoplasia occurs almost exclusively in patients with impaired kidney function, immunosuppression, collagen vascular disease, or autoimmune disease.[12] Neutropenia occurs within 3 months of initiation of therapy and generally resolves 2 weeks after therapy is discontinued.[12,117]

Now mostly of historical interest, anaphylactoid reactions ranging from mild pruritus to bronchospasm and cardiopulmonary collapse have been reported in patients treated with ACE inhibitors who undergo dialysis with equipment that uses high-flux polyacrylonitrile, cellulose acetate, or cuprophane membranes or who undergo apheresis with equipment that uses dextran sulfate membranes.[122] The frequency of reactions is unknown, but they occur within the first few minutes of treatment. Dialyzers manufactured with these materials are rarely used in modern dialysis practice; nevertheless, exposure of the blood to any of these compounds should be avoided when patients are receiving ACE inhibitors. The use of ACE inhibitors in patients on plasmapheresis or an exchange protocol is safe as long as effective arterial volume is managed appropriately.

Few significant drug interactions occur with ACE inhibitors. Studies have shown that aspirin dosages of 100 mg/day or less do not negate the effects of ACE inhibitors.[123] Ang II stimulates the production of vasodilatory prostaglandins. Aspirin inhibits the production of vasodilators and antithrombotic prostaglandins. Theoretically, either agent may antagonize the effectiveness of the other. Concomitant use of ACE inhibitors and calcineurin inhibitors may exacerbate kidney hypoperfusion.[12] Use of the mammalian target of rapamycin (mTOR) inhibitors sirolimus or everolimus in transplant recipients decreases the metabolism of bradykinins and predisposes them to angioedema with ACE inhibitors.[124]

Hyperkalemia of more than 5.5 mmol/L was observed in 3.3% of patients taking ramipril in the ONTARGET study.[125] Production of Ang II systemically and locally in the zona glomerulosa of the adrenal gland, which is blocked by ACE inhibitors, will reduce subsequent aldosterone synthesis and urinary potassium excretion. In a Veterans Administration Medical Center case-control study of 1818 patients using ACE inhibitors, 194 (11%) developed hyperkalemia.[126] The results of laboratory studies indicated that a serum urea nitrogen concentration higher than 6.4 mmol/L and a serum creatinine concentration higher than 136 μmol/L, as well as congestive heart failure and long-acting ACE inhibitors, were independently associated with hyperkalemia; concurrent use of a loop diuretic or thiazide agent was associated with reduced risk. After 1 year of follow-up, 15 (10%) of 146 patients who remained on a regimen of an ACE inhibitor developed severe hyperkalemia (serum potassium >6.0 mmol/L). A serum urea nitrogen concentration higher than 8.9 mmol/L (25 mg/dL) and age older than 70 years were independently associated with subsequent severe hyperkalemia.

Hyperkalemia has been effectively and safely treated with patiromer sorbitex calcium and sodium zirconium cyclosilicate in an outpatient setting.[127,128] These drugs add to the outpatient pharmacopoeia for hyperkalemia management that until now has been limited to sodium and calcium polystyrene sulfonate. Correction of metabolic acidosis, if present, and the use of loop diuretics, thiazide, or thiazide-like agents or sodium-glucose transport protein 2 (SGLT2) inhibitors may also reduce the incidence and severity of hyperkalemia.[129]

ANGIOTENSIN II TYPE 1 RECEPTOR ANTAGONISTS

CLASS MECHANISMS OF ACTION

The ARBs allow more specific and complete blockade of the RAAS than the ACE inhibitors because they circumvent all pathways that lead to the formation of Ang II (see Fig. 48.1). For example, Ang I is metabolized by not only ACE to form Ang II but also chymase, cathepsin G, tissue plasminogen activator, and other enzymes.[130] Ang II can be formed at sites other than those in the systemic circulation, such as the brain, kidney, and heart. Furthermore, long-term ACE inhibitor therapy is associated with a return of Ang II levels to baseline, which possibly contributes to reduced efficacy. The ARBs selectively antagonize Ang II directly at the Ang II type 1 (AT_1) receptor, regardless of the source of production. Because Ang II plays a crucial multifactorial role in maintaining and regulating BP, blockade of the AT_1 receptor with ARBs is a powerful tool for targeting multiple pathways that contribute to hypertension.

Like ACE inhibitors, ARBs directly block the vasoconstrictive action of Ang II and cause a decrease in peripheral vascular resistance.[131] Interruption of the binding of Ang II at the tissue level also leads to other effects (beyond vasodilation) that contribute to the antihypertensive effect. Additional mechanisms include the following: 1. augmentation of kidney blood flow and reduction of aldosterone release to induce natriuresis and attenuate the compensatory increase in sodium retention that accompanies a fall in BP; 2. direct depression of tubular sodium reabsorption;[132,133] 3. improvement of nitric oxide–mediated endothelial function;[134] 4. reversal of vascular hypertrophy;[134] 5. blunting of sympathetic nervous system (SNS) activity and presynaptic norepinephrine release; 6. inhibition of postjunctional pressor responses to norepinephrine or Ang II; 7. inhibition of central Ang II–mediated sympathoexcitation and vasopressin release;[135-137] 8. inhibition of centrally controlled baroreceptor reflexes;[138] 9. inhibition of central nervous system (CNS) norepinephrine synthesis; 10. inhibition of thirst;[139] and 11. possible inhibition of RAAS-mediated action on endothelin-1.[140] The antihypertensive action of ARBs is dependent on activation of the RAAS and is associated with clinically insignificant increases in circulating levels of Ang II.[131] ARBs also increase bradykinin levels by antagonizing Ang II at its type 1 receptor and diverting Ang II to its counterregulatory type 2 receptor, which potentiates vasodilation.[141] ARBs also increase the level of other angiotensin peptides, including angiotensin-(1-7), Ang II, and angiotensin IV, which can act on their respective receptors to modulate vasoconstriction, kidney blood flow, and vascular hypertrophy.[142-147]

CLASS MEMBERS

The ARB class is composed of peptide and nonpeptide analogs that vary in structure, mechanism of receptor inhibition, metabolism, and potency. Many drugs were developed by modifying losartan, the first biologically active ARB oral agent. These drugs are categorized according to the substitution of carboxylic and other moieties into several groups—the biphenyl tetrazoles (derivatives of losartan), nonbiphenyl tetrazoles, and nonheterocyclic compounds. They are also classified according to their ability to antagonize Ang II.[148] The competitive (surmountable) antagonists shift the dose-response curve for Ang II–mediated contraction to the right without depressing the maximal response to Ang II. The noncompetitive (insurmountable) antagonists also depress the maximal response to Ang II. The variable effects of ARBs are mediated by differences in the interaction with allosteric binding sites on the receptor, dissociation of the drug–receptor complex, removal of the agonists from tissues, or the ability to modulate the amount of internalized receptors[148] (Tables 48.7–48.8).

Biphenyl Tetrazole and Oxadiazole Derivatives

Azilsartan medoxomil is a selective AT_1 receptor blocker that has demonstrated a potent 24-hour sustained antihypertensive effect. At the approved dosage, it reduces systolic BP (SBP) by 12 to 15 mm Hg and diastolic BP (DBP) by 7 to 8 mm Hg.[149-151] Azilsartan medoxomil is a prodrug that is hydrolyzed to azilsartan in the gastrointestinal tract during absorption. It possesses a unique moiety (5-oxo-1,2,4-oxadiazole) in place of the tetrazole ring that offers a strong inverse antagonism at the AT_1 receptor.[152,153] This bond is chemically stronger than that of its predecessors, which may explain its added potency when compared with other members of its class or ACE inhibitors.[152] The estimated oral bioavailability is about 60%; peak plasma concentration is reached after 1.5 to 3 hours following ingestion.[154] Food does not affect the bioavailability of the drug. More than 99% of azilsartan is bound to albumin. The initial starting dose is 20 mg daily, and it is available in 20-, 40-, and 80-mg tablets. The terminal half-life is 9 hours, and approximately 55% of the parent compound is excreted by the kidney.[154] In a study to assess the pharmacokinetics

Table 48.7 Pharmacodynamic Properties of Angiotensin Receptor Blockers

Generic Name (Trade Name)	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Azilsartan (Edarbi)	40	40–80	80	qd	1.5–3 peak response	≥24
Candesartan (Atacand)	16	8–32	32	qd	6-8	24
Eprosartan (Teveten)	200	400–800	800	qd, bid	4	24
Irbesartan (Avapro)	150	150–300	300	qd	4-6, 14	24
Losartan (Cozaar)	25	50–100	100	qd/bid	6	12–24
Olmesartan (Benicar)	20	20–40	40	qd	1-2	24
Telmisartan (Micardis)	40	40–80	80	qd	3	24
Valsartan (Diovan)	80	80–160	320	qd	4-6	24

bid, Two times a day; *qd,* once a day.

Table 48.8 Pharmacokinetic Properties of Angiotensin Receptor Blockers

Drug	Absorption (%)	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Candesartan	—	15	No	2–4	9	I, L, K	F (67%), U (33%)	None
Eprosartan	>80	13	Yes	4	6	L	F (70%), U (7%)	Inactive
Irbesartan	>80	60–80	No	1.5–2	10–14	L, K	F (65%), U (20%)	Inactive
Losartan	>80	25	No	1	4–9	L, K	F (60%), U (40%)	Active
Olmesartan	—	26	No	1	13	I	F (50%), U (50%)	Active
Telmisartan	—	42	Yes	0.5–1	24	L	F	Inactive
Valsartan	>80	25	Yes	2–4	6–9	L, K	F (83%), U (13%)	Inactive

F, Feces; *I,* intestine; *K,* kidney; *L,* liver; *U,* urine.

of kidney disease of azilsartan, no dosage adjustment was advised for kidney disease or hemodialysis.[155] In a report, 17 patients receiving hemodialysis who switched to azilsartan showed a decrease in SBP from 150.9 ± 16.2 to 131.3 ± 21.7 mm Hg ($P < .008$) and a decrease in DBP from 84.1 ± 6.3 to 74.9 ± 8.3 mm Hg.[156]

Candesartan cilexetil is an esterified prodrug imidazole that is rapidly and completely converted into the active 7-carboxylic acid candesartan (CV-11974) in the intestinal wall.[12] Candesartan is a selective, nonpeptide, noncompetitive ARB with the second highest receptor-binding affinity and a slow detachment rate from the receptor. Consequently, the effects are long lasting and unlikely to be overcome by the upregulation of Ang II that commonly accompanies AT_1 receptor blockade. The initial dose is 16 mg daily, and the usual daily dose is 8 to 32 mg in one or two divided doses. The antihypertensive response occurs initially in 2 to 4 hours, peaks at 6 to 8 hours, and lasts for 24 hours (see Tables 48.7 and 48.8).[148,157] Radioreceptor assays demonstrate the presence of candesartan at the receptor site for longer than predicted periods (from plasma half-life analysis), which correlates with the clinical observation of a sustained effect beyond 24 hours.[157,158] Maximal response is achieved in 4 weeks. The terminal half-life of candesartan is approximately 9 hours, and it is not affected by kidney failure. No unchanged parent compound is detected in the serum or urine. Candesartan is not dialyzable.

Irbesartan is a nonpeptide-specific imidazolinone derivative of losartan that acts as a noncompetitive AT_1 receptor blocker with a high receptor-binding affinity.[12,159] The initial dose is 150 mg daily, and the usual daily dose is 150 to 300 mg. The initial response occurs in 2 hours. The peak response is bimodal; in hypertensive patients, peak responses occur in 4 to 6 hours and 14 hours, corresponding to the peak increases in plasma renin activity and Ang II levels, respectively.[159] With continuous dosing, the maximal effect may not be seen for up to 6 weeks. The duration of action is 24 hours. Irbesartan is not dialyzable.

Losartan potassium is the prototype ARB. The tetrazole moiety on the biphenyl ring accounts for its activity in oral form and duration of action. It was the first oral active agent, and it is a nonpeptide competitive, selective AT_1 receptor inhibitor, with moderate receptor-binding affinity.[12,160] The usual starting dosage is 50 mg daily (see Table 48.7). Dosage adjustments should be made at weekly intervals. The antihypertensive efficacy may be improved with divided doses. The usual daily dose is 50 to 100 mg. The potassium content of the 25-, 50-, and 100-mg tablets is 0.054, 0.108, and 0.216 mEq, respectively. The oral bioavailability of losartan is 25%, and it is unaffected by food (see Table 48.8). The initial response occurs in 1 hour, and the response peaks at 6 hours and lasts for 24 hours. Only 5% of losartan is recovered unchanged in the urine, which supports extensive metabolism and biliary secretion. Neither the parent drug nor metabolites are removed by dialysis.[22]

Olmesartan medoxomil is a nonpeptide selective ARB prodrug that is rapidly and completely bioactivated by hydrolysis to olmesartan during absorption from the gastrointestinal tract.[161] The initial dosage is 20 mg daily, and the usual dosage is 20 to 40 mg daily (see Table 48.7). The peak plasma concentration is reached in 1 hour (see Table 48.8). The BP-lowering effect lasts for 24 hours and peaks at 2 weeks. Olmesartan is eliminated in a biphasic manner, with a terminal half-life of 13 hours. Olmesartan is not dialyzable.

Nonbiphenyl tetrazole derivatives

Telmisartan incorporates a carboxylic acid as the biphenyl acidic group. Telmisartan is a nonpeptide, noncompetitive ARB, with high specificity and receptor affinity.[12,162] The usual starting dosage is 40 mg daily, and the usual daily dose is 40 to 80 mg. The initial response occurs in 3 hours, and it is dose dependent (see Table 48.8). The duration of action is 24 hours but may last up to 7 days after discontinuing the drug. Women typically achieve plasma levels two to three times higher than those of men, but this result is not associated with differences in BP response. Less than 3% of the drug is metabolized in the liver into inactive compounds. The elimination half-life is 24 hours. Telmisartan is not dialyzable, and dosage adjustment is not necessary in patients with kidney disease.

Nonheterocyclic derivatives

Valsartan is a nonheterocyclic ARB in which the imidazole of losartan is replaced by an acetylated amino acid.[12] Valsartan is a noncompetitive antagonist with high specificity and receptor-binding affinity. The initial starting dosage is 80 mg daily (see Table 48.7), and the usual dosage is 80 to 160 mg daily but can be increased to 320 mg daily. The maximal BP response is achieved after 4 weeks of therapy. The initial response occurs in 2 hours, peaks at 4 to 6 hours, and lasts 24 hours. Valsartan does not undergo significant metabolism. The elimination half-life is 6 to 9 hours, and it is not affected by kidney failure (see Table 48.8).

CLASS KIDNEY EFFECTS

Intrarenal Ang II receptors are widely distributed in the afferent and efferent arterioles, glomerular mesangial cells, inner stripe of the outer medulla, and medullary interstitial cells,[163] as well as on the luminal and basolateral membranes of the proximal and distal tubule cells, collecting ducts, podocytes, and macula densa cells.[164] Most receptors are of the AT_1 subclass. Circulating and predominantly locally produced Ang II interacts with the receptors; the complex is internalized and Ang II is released into the intracellular compartment, where it exerts its effects. Studies have suggested that most kidney interstitial Ang II is formed at sites not readily accessible to ACE inhibition or is formed by non-ACE pathways.

ARBs antagonize the binding of Ang II and cause a number of intrarenal changes. The overall kidney hemodynamic responses of AT_1 receptor blockade are variable, depending on the counteracting influences of the decrease in arterial pressure.[132,165] Decreases in systemic arterial pressure by ARBs may be associated with compensatory activation of the intrarenal SNS. This effect is more pronounced in sodium-depleted states because activation of the renin–angiotensin system helps maintain arterial and kidney pressure. By contrast, direct intrarenal infusions of ARBs cause an increase in sodium excretion.[166] The enhanced sodium excretion has been shown to be caused by direct inhibition of sodium reabsorption by the proximal tubules, but it may also be caused by hemodynamic changes in medullary blood flow and tubule absorption in distal nephron segments. Because Ang II blockade enhances the ability of the kidneys to excrete sodium, sodium balance can be maintained at lower arterial pressures. Ang II blockade also reduces tubuloglomerular feedback sensitivity by decreasing macula densa transport of sodium chloride to the afferent arteriole.[167] This leads to increased delivery of sodium chloride to the distal segments for excretion, without compensatory changes in GFR.

In addition to the natriuretic and diuretic actions, short-term administration of some ARBs such as losartan has been observed to induce reversible kaliuresis in salt-depleted normotensive patients in the absence of changes in GFR.[168] However, long-term Ang II receptor blockade does not cause appreciable changes in urinary electrolyte excretion or volume. Another property unique to losartan is induction of uricosuria.[169] This effect is not observed with ACE inhibitors or other ARBs, including the active metabolite of losartan, and does not appear to be related to inhibition of the RAAS.[169] Losartan has a greater affinity for the urate-anion exchanger than other antagonists, and it inhibits urate reabsorption in the proximal tubule.[170] The uricosuria is associated with a concomitant decrease in serum urate concentrations in normal individuals, hypertensive individuals, and patients with kidney disease including kidney transplant recipients.[170] The effect occurs within 4 hours of drug administration and is dose dependent. Long-term administration reduces serum urate concentrations by approximately 0.4 mmol/L.[171] Concerns that increased uric acid supersaturation might perpetuate kidney uric acid deposition have not been borne out clinically, perhaps because losartan simultaneously increases urinary pH, which protects against crystal nucleation.[172]

Hypertensive patients treated with ARBs exhibit decreases in BP, increases in kidney blood flow, and decreases in filtration fraction and kidney vascular resistance.[173] These effects are probably a result of combined decreases in preglomerular and postglomerular resistances. It has been suggested that elevated intrarenal Ang II levels in the presence of AT_1 receptor blockade stimulate AT_2 receptors, which can increase the preglomerular vasodilator actions of bradykinin, cyclic guanosine monophosphate, and nitric oxide.[174] As with ACE inhibitors, ARBs can cause an initial elevation in the serum creatinine due to these hemodynamic changes and should not be a reason to stop ARBs because, like ACE inhibitors, ARBs slow CKD progression. The lowering of efferent arteriole resistance reduces intraglomerular hydrostatic pressure, which attenuates the progression of kidney injury, and increases kidney sodium excretory capacity. In concert with the reduction in systemic arterial pressure, these actions provide more kidney protection than other classes of antihypertensive agents, despite equivalent reductions in BP.[56,175,176]

In healthy and hypertensive patients, ARBs produce dose-dependent increases in circulating Ang II levels and plasma renin activity.[177] The increases occur at the peak plasma drug levels and persist for up to 24 hours; they remain elevated with long-term administration. Decreases in plasma concentrations of aldosterone have been reported, but they are variable.[178] In normal individuals, decreases in aldosterone coincide with the peak interval of ARB activity; in hypertensive patients consuming a fixed-sodium diet, there are no significant changes in plasma aldosterone concentrations relative to baseline.[168] ARBs suppress the Ang II–mediated adrenal cortical release of aldosterone, but these effects appear to be quantitatively less important than the intrarenal suppression of Ang II action. Long-term AT_1 receptor blockade does not appear to induce aldosterone escape.[179]

Urinary protein excretion is significantly decreased with administration of ARBs and parallels findings with ACE inhibitor therapy.[54] Antiproteinuric effects have been described in patients with and without diabetes mellitus, as well as in kidney transplant recipients.[54,180] The antiproteinuric effect has a slow onset, and the dose-response curves differ from those of the antihypertensive effects, in which the maximal effect occurs at 3 to 4 weeks. Whether the antiproteinuric effects are equivalent to or better than those of ACE inhibitors remains to be determined, but it is evident that the suppression of albuminuria is equivalent at all stages of CKD.[181] Combining ACE inhibitors and ARBs for renoprotection and proteinuria reduction is not recommended owing to the significant development of hyperkalemia, hypotension, and kidney hypoperfusion, particularly in patients with underlying CKD.[125,182-184] Like ACE inhibitors, ARBs have nonhemodynamic effects that may contribute to renoprotection, including antiproliferative actions on the vasculature and mesangium, inhibition of transforming growth factor-β (TGF-β),[25,185] inhibition of atherogenesis[186] and vascular deterioration,[187] improved superoxide production and nitric oxide bioavailability,[188] reductions of collagen formation, reduction of mesangial matrix production, improved vascular wall remodeling, decreased vasoconstrictor effects of endothelin-1, improved endothelial function,[187] reductions of oxidative stress and inflammation,[189] modulation of peroxisome proliferator–activated receptor γ activity, and protection from calcineurin inhibitor injury. ARBs also reduce salt sensitivity by restoring kidney nitric oxide synthesis.[190]

CLASS EFFICACY AND SAFETY

All ARBs lower BP effectively and safely in patients with mild, moderate, and severe hypertension, regardless of age, sex, or ancestry.[191-193] ARBs are indicated as first-line monotherapy or add-on therapy for hypertension and are comparable in efficacy to other agents.[194-196] They are safe and effective in patients with CKD, diabetes, heart failure, coronary artery disease, arrhythmias, and left ventricular hypertrophy (LVH)[197-200] and kidney transplant recipients. Some but not all studies have shown a weak lowering of the risk for new and recurrent atrial fibrillation because of the beneficial structural and electrical effects on the atria.[201,202] ARBs have been shown to diminish the rate of persistent atrial fibrillation in patients with preexisting recurrent atrial fibrillation.[203] The Prevention Regimen for Effectively Avoiding Second Strokes (PRoFESS) study showed no measurable benefit of ARBs for the prevention of recurrent stroke but may have been underpowered to show an effect in its well-treated patient population.[204]

In most patients, ARBs offer BP lowering comparable with that of all other antihypertensive drug classes, with an improved tolerability profile.[205] They provide effective control over a 24-hour period and are generally suitable for once-daily dosing (see Table 48.7).[206,207] ARBs do not affect the normal circadian BP variation.[208] The long onset of action (4–6 weeks) avoids the first-dose hypotension and rebound hypertension that are commonly observed with some other drugs. There is a dose-dependent response with newer agents, but losartan and valsartan have a relatively flat dose-response curve.[209] Azilsartan, candesartan, irbesartan, and olmesartan may have the greatest efficacy, with a longer duration of action because of their noncompetitive binding, and telmisartan may have an added advantage by the inhibition of SNS activation through an antioxidant effect in experimental models.[210,211]

ARBs may reversibly decrease kidney function and elevate serum potassium concentrations; serum chemistries should be checked after initiation or increase in dosage of these drugs. The overall incidence of hyperkalemia from ARBs is 3.3%, similar to that of ACE inhibitors.[125] Participants in the Candesartan in Heart Failure—Assessment of Reduction in Mortality and Morbidity (CHARM; n = 7599) program were randomized to standard heart failure therapy plus candesartan or placebo, with recommended monitoring of serum potassium and creatinine levels.[212] The authors assessed the incidence and predictors of hyperkalemia over the median 3.2 years of follow-up. Candesartan increased the risk of incident hyperkalemia compared with placebo 5.2% versus 1.8% (difference, 3.4%; $P < 0.0001$). Risk factors for hyperkalemia in treated patients with symptomatic heart failure include advanced age, male sex, baseline hyperkalemia, impaired kidney function, diabetes, or combined RAAS blockade. As with ACE inhibitors, hyperkalemia associated with ARB use can be medically managed.[127,128]

ARBs should be stopped immediately at the onset of pregnancy with the first missed menstrual period in the first trimester, as discussed previously in the context of ACE inhibitor use, because ARBs may cause fetal or neonatal death and congenital abnormalities when used during the second and third trimesters of pregnancy.[69] Breastfeeding is not contraindicated, according to Kidney Disease Improving Global Outcomes (KDIGO) and NICE guidelines; however, low drug levels are detected in breast milk.[70,71]

Overall, ARBs have neutral metabolic effects and are superior to most other hypertensive classes with respect to tolerability. ARBs do not cause hypernatremia or hyponatremia, and hyperkalemia is relatively uncommon. In the ONTARGET study, a serum potassium concentration higher than 5.5 mmol/L was observed in 3.3% of patients who resigned from the study and is comparable with that observed with ACE inhibitor therapy.[125] ARBs have no effect on serum lipids in hypertensive patients but may improve the abnormal lipoprotein profile of patients with proteinuric kidney disease and reduce obesity-related morbidity.[213,214] ARBs have favorable effects on serum glucose concentrations and insulin sensitivity.[215-217] Clinical trials comparing ARB-based therapy with treatment with other antihypertensive agents in patients with hypertension with and without LVH demonstrated a 25% reduced risk of the development of diabetes in the ARB-treated group.[218,219] The mechanism for this effect has not been defined.[220,221] Increased levels of liver transaminases are occasionally reported, but the effects are usually transient, even with continued therapy.[12]

Clinically relevant AEs are not observed more frequently than in placebo-treated patients. Because ARBs do not interfere with kinin metabolism, cough is rare, which is a major clinical advantage.[125] The incidence of cough in patients with a history of ACE inhibitor–induced cough is no greater than in those receiving placebo.[222] Similarly, the incidence of angioedema and facial swelling is no greater than that with placebo, but such swelling can occur.[223] ARBs are typically associated with a more potent antiinflammatory response than ACE inhibitors.[224] The most frequent AEs are headache (14%), dizziness (2.4%), and fatigue (2%), which occur at rates lower than those with placebo.[225] ARB therapy does not worsen sexual activity and may even improve it.[226] Like ACE inhibitors, ARBs may cause minor decreases in the hemoglobin concentration; they may also lower the hemoglobin effectively in posttransplantation erythrocytosis.[227] Rarely, there have been associated cutaneous eruptions.

A spruelike enteropathy associated with olmesartan therapy has been reported.[228,229] In a systematic review of 54 patients,[229] the clinical presentation was characterized by diarrhea (95%) and weight loss (89%). Less common symptoms were fatigue, nausea, vomiting, and abdominal pain. The patients had been taking olmesartan for 6 months to 7 years. A laboratory examination showed normochromic, normocytic anemia (45%) and hypoalbuminemia (39%). HLA-DQ2 or HLA-DQ8 was observed in more than 70%. Antibody testing for celiac disease was negative. Duodenal villous atrophy in varying degrees was described in all the reported patients, and they all showed resolution of diarrhea. The U.S. Food and Drug Administration (FDA) issued a drug safety communication on July 13, 2013 (http://www.fda.gov/Drugs/DrugSafety/ucm359477.htm).

ARBs have not been shown to increase the risk of cancer. Although one meta-analysis suggested a slightly increased risk of cancer with ARBs relative to other antihypertensive agents,[230] there were questions about the study's methodology and exclusion of certain key RCTs that would have changed the results had they been included. Subsequent meta-analyses[231-234] and cohort studies[235-238] demonstrated no association of ARBs with cancers of any type in large populations, and one study suggested that ARBs may actually lower the incidence.[235] A subsequent systematic review of observational and interventional studies suggested that the use of ACE inhibitors and ARBs may improve cancer outcomes.[239]

β-ADRENERGIC RECEPTOR ANTAGONISTS

CLASS MECHANISMS OF ACTION

β-Adrenergic receptor antagonists (β-blockers) exert their antihypertensive effects by attenuating sympathetic stimulation through the competitive antagonism of catecholamines at the β-adrenergic receptor.[240] However, the precise mechanism of the antihypertensive effect of β-blockers remains incompletely understood. β_1-Adrenergic receptor blockade has generally been considered responsible for the BP-lowering effect; however, β_2-receptor blockade has an independent antihypertensive effect.[241] Inhibition of β_1-adrenergic receptors in the juxtaglomerular cells in the kidney may inhibit renin release. A direct action on the CNS, with a reduction in CNS sympathetic outflow, may also be involved. Attenuation of cardiac pressor stimuli related to β-blockade may result in baroreceptor resetting. In addition, adrenergic neuron output may be blocked because of the inhibition of β_2-adrenergic receptors at the vascular wall.

β_1-Selective blockers may have a slightly more potent antihypertensive effect than nonselective agents. This differential effect may be in the range of 2 to 3 mm Hg. It may be that β_2-blockade in some fashion blunts the antihypertensive effects of β_1-blockade.[242] The β_2 partial agonist activity may mediate peripheral vasodilator effects that could contribute to the antihypertensive action. A β_1-selective antagonist with partial agonist activity at the β_1 receptor may result in less hypotensive effect. The magnitude and clinical significance of these differences are unclear.

In addition to β-adrenergic antagonist properties, certain drugs have antihypertensive effects that are mediated through different mechanisms (Table 48.9), including α_1-adrenergic antagonist activity and effects on nitric oxide–dependent vasodilator action. Partial agonist activity is a property of certain β-blockers that results from a small degree of direct stimulation of the β-adrenergic receptor by the drug, while simultaneously blocking the same receptor to access by stimulating catecholamines.[242-244] Whether the presence of partial agonist activity is advantageous or disadvantageous remains unclear. Drugs with partial agonist activity slow the resting heart rate less than drugs that lack this pharmacologic effect.[245] The exercise-induced increase in heart rate is similarly blocked by both groups of drugs.[245] However, β-blockers with nonselective partial agonist activity may reduce peripheral vascular resistance and cause less atrioventricular conduction depression than drugs without partial agonist activity. The specificity of partial agonist activity for β_1 or β_2 receptors may also have a role in the antihypertensive response to a given drug.

β-Blockers may be nonspecific and block β_1- and β_2-adrenergic receptors, or they may be relatively specific for β_1-adrenergic receptors. β_1-adrenergic receptors are found predominantly in heart, adipose, and brain tissue, whereas β_2 receptors predominate in the lung, liver, smooth muscle, and skeletal muscle. Many tissues, however, have both β_1 and β_2 receptors, including the heart, and it is important to realize that the concept of a cardioselective drug is only relative.

β-Blockers differ significantly in gastrointestinal absorption, first-pass hepatic metabolism, protein binding, lipid solubility, penetration into the CNS, and hepatic or kidney clearance. β-Blockers that are eliminated primarily by hepatic metabolism have a relatively short plasma half-life; however, the duration of the clinical pharmacologic effect does not correlate well with the plasma half-life in many of these drugs. Water-soluble drugs that are eliminated by the kidney may have longer half-lives. Bioavailability varies greatly across the class, as does the degree to which individual agents are dialyzed.[246]

Nonselective β-Adrenergic Antagonists

Nadolol is a nonselective β-blocker without partial agonist activity (see Table 48.9). The average adult dosage is 40 to 80 mg daily, with a maximum daily dose of 320 mg (Tables 48.10 and 48.11). Nadolol is not appreciably metabolized, and elimination occurs predominantly in the urine and feces. Dosage adjustment is indicated in patients with CKD. Dosage intervals should be increased to 24 to 36 hours, 24 to 48 hours, and 40 to 60 hours in patients with creatinine clearances of 30 to 50, 10 to 30, and <10 mL/min/1.73 m^2, respectively.[22] Dosage adjustment is not necessary for patients with hepatic insufficiency. Hemodialysis reduces the serum concentration of nadolol, but specific recommendations for dosage during dialysis are not available.

Propranolol is a noncardioselective β-blocker that has no partial adrenergic activity. The usual daily dosage range is 80 to 320 mg. The drug may be administered in a single daily dose if a long-acting preparation is used. The drug is metabolized by the liver. The major metabolite, 4-hydroxypropranolol, has β-blocking activity. Kidney excretion is <1%. Dosage adjustment in patients with CKD is not necessary.[22] Patients with liver disease may require variable dosage adjustments and more frequent monitoring.

Penbutolol is a nonselective β-blocker[247] with low partial agonist activity. Usually, dosages are 20 to 40 mg given as a single dose or divided twice daily. Hepatic metabolism to inactive metabolites occurs with subsequent kidney elimination. The optimal antihypertensive effect is observed at an average of 14 days after initiation of therapy. Dosage adjustments for patients with CKD are not recommended,

Table 48.9 Pharmacologic Properties of β-Adrenergic Antagonists

Generic Name (Trade Name)	β_1 Selectivity	Partial Agonist Activity	Membrane-Stabilizing Activity	α-Adrenergic Antagonist Activity
Acebutolol (Sectral)	+	+	+	
Atenolol (Tenormin)	+			
Betaxolol (Kerlone)	+		+	
Bisoprolol (Zebeta)	+			
Carvedilol (Coreg)			+	+
Celiprolol (none in the United States)	+			+
Labetalol (Trandate)		+		+
Metoprolol (Lopressor)	+			
Nadolol (Corgard)				
Penbutolol (Levatol)		+		
Pindolol (Visken)		+	+	
Propranolol (Inderal)			+	
Nebivolol (Bystolic)	+			

Table 48.10 Pharmacokinetic properties of β-adrenergic antagonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion
Acebutolol	90	No	2–3	3–8	L	U
Atenolol	40–60	Yes	—	14–16	—	U, F
Betaxolol	78–90	No	2–6	12–22	L	U
Bisoprolol	90	—	2.3	9.6	L	U
Carvedilol	25–30	Yes	1.5	6–10	L	U, F
Metoprolol	50	—	1.5–2	3–7	L	U
Nadolol	20–40	No	2–4	20–24	—	U, F
Penbutolol	100	No	—	17–24	L	U
Pindolol	95	No	2	3–11	L	U (40%)
Propranolol	16–60	Yes	—	3–4	L	—
Timolol	50–90	—	—	2–4	L	U (20%)

F, Feces; *L,* liver; *U,* urine.

Table 48.11 Pharmacodynamic Properties of β-Adrenergic Antagonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Nadolol	40	40–80	320	qd	—	—
Propranolol	40	80–320	640	bid	—	—
Timolol	10	20–40	60	bid	—	—
Penbutolol	10	20–40	80	qd, bid	2	20–24
Pindolol	5	10–40	60	qd, bid	—	24
Atenolol	25	50–100	200	qd	3	24
Metoprolol	12.5–50	100–200	400	qd, bid	1	3–6
Betaxolol	10	10–40	40	qd	3	23–25
Bisoprolol	5	2.5–20	20	qd	2-4	24
Acebutolol	400	400-800	1200	qd	3	24

bid, Two times a day; *qd,* once a day.

but adjustments may be required for patients with hepatic insufficiency.[22]

Pindolol is a nonselective β-blocker with high partial agonist activity. The usual adult oral dosage is 5 mg twice daily, with incremental doses of 10 mg every 3 to 4 weeks. The maximum daily recommended dose is 60 mg. Approximately 40% of a dose of pindolol is excreted unchanged in the urine; 60% is metabolized in the liver. The drug half-life increases modestly in patients with impaired kidney function. Dose adjustments do not appear to be necessary.[22] Dose adjustments may be necessary in patients with cirrhosis and advanced CKD.

β_1-selective adrenergic antagonists

Acebutolol is a β_1-selective blocker with low partial agonist activity. Dosages of 400 to 1200 mg daily are effective in treating hypertension. The drug is metabolized to diacetolol, an active metabolite, with the parent compound being excreted by the kidneys and in bile. Diacetolol is excreted predominantly by the kidneys. Dosage reduction of 50% to 75% is recommended for patients with advanced CKD.[22]

Atenolol is a long-acting β_1-selective blocker with no partial agonist activity. The usual dosage is 50 to 100 mg daily. Approximately 50% of the drug is eliminated by the kidneys, and 50% is excreted in the feces. Dosages of more than 100 mg daily are unlikely to produce additional benefits. The time required to achieve the optimal antihypertensive effect is 1 to 2 weeks. In patients with moderate CKD, the dosing interval should be increased to 48 hours, and in patients with advanced CKD, dosing intervals should be increased to 96 hours to prevent excessive accumulation and subsequent bradycardia.[22] Atenolol is not significantly metabolized by the liver, and no dosage adjustment is necessary in patients with liver disease. Atenolol is removed by dialysis, and a maintenance dose should be given after a dialysis treatment.

Betaxolol is a long-acting β_1-selective blocker[248] with no partial agonist activity. The usual oral dosage for hypertension is 10 to 40 mg daily. Therapy is typically started at a dosage of 10 mg daily. Most patients respond to 20 mg daily. The time to achieve the optimal antihypertensive effect is approximately 1 to 2 weeks. Betaxolol is metabolized predominantly in the liver, with metabolites excreted by the kidney. Approximately 15% of the dose is recovered unchanged in the urine. CKD results in a decrease in betaxolol clearance, and titration should begin at 5 mg daily in these patients.

Bisoprolol is a long-acting β_1-selective blocker[249] with no partial agonist activity. The usual oral dosage is 2.5 to 20 mg given daily. Hepatic metabolism occurs with the kidney excretion of metabolites; however, 50% of the drug is excreted by the kidney unchanged. In patients with CKD, the initial oral dosage should be 2.5 mg daily, with careful monitoring of dose titration. The maximum recommended dosage of bisoprolol in patients with CKD is 10 mg/day. Similar dosage reduction is also required for patients with hepatic insufficiency. It is modestly removed during hemodialysis.[246]

Metoprolol is a β_1-selective blocker with no partial agonist activity. Extensive hepatic metabolism occurs primarily by the cytochrome P450 (CYP) 2D6 system (CYP2D6), and 3% to 10% of the drug is excreted unchanged in the urine. Metoprolol pharmacokinetics is heavily influenced by the CYP2D6 genotype and metabolizer phenotype, with up to a 15-fold difference in clearance between ultrarapid and poor metabolizers.[250] The initial oral dosage is 12.5 to 50 mg once or twice daily, increasing to 100 to 200 mg twice daily. Sustained-release (SR) preparations may be substituted as a once-daily dose. Metoprolol is extensively removed by hemodialysis.[246]

Nonselective β-adrenergic antagonists with α-adrenergic antagonism or other mechanisms of antihypertensive action

Carvedilol is a nonselective β-blocker with peripheral α_1-blocker activity (Tables 48.12 and 48.13)[251,252] and no partial agonist activity. The drug is approximately equipotent in blocking β_1- and β_2-adrenergic receptors. The ratio of α_1- to β_1-blocking activity is estimated to be 1:7.6. There is evidence that the therapeutic actions of carvedilol may depend in part on the endogenous production of nitric oxide, which may improve endothelial dysfunction in hypertensive patients. For the management of hypertension, an initial oral dosage of 6.25 mg twice daily is recommended and may be increased to 12.5 to 25 mg twice daily, if needed. Dose adjustments are not required for patients with CKD. Carvedilol is extensively metabolized in the liver, and dose reductions are suggested for patients with hepatic insufficiency. Carvedilol is not removed by hemodialysis.[246]

Celiprolol is a β-blocker with several unique properties.[253,254] It is a β_1-selective blocker with α_2-receptor blocking activity (see Tables 48.12 and 48.13). Celiprolol also causes vasodilation through β_2-receptor stimulation and possibly nitric oxide, with a subsequent decrease in systemic vascular resistance. In contrast to other β-blockers, celiprolol does not appear to induce bronchospasm or have negative inotropic effects. It does have moderate partial agonist activity. The initial dosage of celiprolol is 200 mg daily and can be increased to 400 mg daily. Kidney excretion is 35% to 42%. A 50% dosage reduction is suggested in patients with a creatinine clearance of 15 to 40 mL/min/1.73 m^2. Celiprolol is not recommended for patients with a creatinine clearance <15 mL/min/1.73 m^2. It is not currently available in the United States.

Labetalol is a nonselective β-blocker[255] with weak partial agonist activity and α_1-receptor blocking activity (see Tables 48.12 and 48.13). The drug is approximately equipotent in blocking β_1- and β_2-adrenergic receptors. With oral administration, the ratio of α_1- to β-blocking potency is approximately 1:3. With intravenous administration, the β-blocking potency is more prominent. The usual initial dosages for treatment of hypertension are 100 mg orally twice

Table 48.12 Pharmacokinetic Properties of β-Adrenergic Antagonists With Vasodilatory Properties

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Labetalol	25–40	Yes	1–2	5–8	L	U (50%–60%)	—
Carvedilol	25–35	No	1–1.5	6-8	L	F	—
Celiprolol	30–70	Yes	—	5-6	—	F, U	—
Nebivolol	12–96	No	2.4-3.1	8-27	L	—	—

F, Feces; *L,* liver; *U,* urine.

Table 48.13 Pharmacodynamic Properties of β-Adrenergic Antagonists with Vasodilatory Properties

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Labetalol	100	200–800	1200–2400	bid	3	8–12
Carvedilol	6.25	12.5–25	50	bid	4–7	24
Celiprolol	200	200–400	400	qd	—	—
Nebivolol	5	5	40	qd	6	24

bid, Two times a day; *qd,* once a day.

daily, increasing gradually to a maintenance dosage of 200 to 400 mg twice or thrice daily. The drug is metabolized in the liver, with 50% to 60% of a dose excreted in the urine and the remainder in the bile. Dose adjustment is not required in CKD.[22] Chronic liver disease has been demonstrated to decrease the first-pass metabolism of labetalol, and dosage reduction is required in these patients.

Nebivolol is a long-acting, β_1-selective blocker (see Tables 48.12 and 48.13).[256-260] The compound is a 1:1 racemic mixture of two enantiomers: D-nebivolol and L-nebivolol. The actions of nebivolol, which are unique and unlike those of other β-blocking agents, are attributable to the individual effects of the isomers. When administered alone, the L-isomer does not produce significant effects on BP, but its presence enhances the antihypertensive effects of the D-isomer. The L-isomer may potentiate the effects of endothelium-derived nitric oxide to induce decreases in BP and peripheral vascular resistance. These effects may improve endothelial dysfunction and potentially influence cardiovascular risks.[261,262] The L-isomer may also inhibit norepinephrine actions at the presynaptic β receptors. The initial oral dosage is 5 mg daily. The drug is metabolized in the liver; rapid and slow metabolizers have been identified. The half-life of nebivolol is 8 hours in rapid metabolizers and 27 hours in slow metabolizers. Reduced initial dosages are recommended for patients with CKD.

CLASS KIDNEY EFFECTS

α- and β-Adrenergic receptors in the kidney mediate vasoconstriction and vasodilation, as well as renin secretion. β-Blockers may influence kidney blood flow and GFR through their effects on cardiac output and BP in addition to direct effects on intrarenal adrenergic receptors. β-Adrenergic receptors have been localized to the juxtaglomerular apparatus in autoradiographic studies.[263] β_2 Receptors predominate in the kidney. The degree of specificity of β-adrenergic blockers for β_1 and β_2 receptors might be expected to influence the effect on kidney function, as might the degree of intrinsic partial agonist activity. In general, the short-term administration of a β-adrenergic blocker usually results in a reduction of GFR and effective kidney plasma flow.[264] This effect is independent of whether the drug has β_1 selectivity or intrinsic partial agonist activity. Nebivolol, carvedilol, and celiprolol, however, have vasodilatory properties and have been shown to increase GFR and kidney plasma flow.[265] Nebivolol dilates glomerular afferent and efferent arterioles by a nitric oxide–dependent mechanism, in contrast to metoprolol, which had no similar effect.[266] This effect may be mediated by the increased synthesis of vasodilatory nitric oxide. Nadolol, when administered intravenously, has been shown in some studies to increase kidney plasma flow and glomerular filtration, whereas oral administration may result in decreased blood flow and GFR. β_1-Selective drugs, when administered orally, tend to produce smaller reductions in GFR and kidney plasma flow. The long-term use of propranolol has been characterized by a 10% to 20% decrease in kidney plasma flow and GFR. The degree of reduction in GFR and kidney plasma flow is modest and probably not of clinical significance in most cases. Labetalol, with its combined α and β blockade, has shown little effect on kidney hemodynamics. The fractional excretion of sodium has been observed to decrease by up to 20% to 40% in some studies of the acute kidney effects of β blockade.[267]

CLASS EFFICACY AND SAFETY

β-Blockers provide effective therapy for the management of mild-to-moderate hypertension, but their use as first-line therapy was not recommended[268-270] in the 2017 American College of Cardiology (ACC)/AHA guidelines based on results from a contemporaneous systematic review and network meta-analysis. In that review, β-blockers were found to be less effective than calcium channel blockers (CCBs) or thiazide or thiazide-like agents for reducing stroke and cardiovascular risk.[271] No significant differences in outcomes were observed in these class-to-class comparisons by age, sex, race, and diabetes mellitus status. For patients with CKD who may need multiple antihypertensive agents to effectively manage BP, the combination of β-blockers with nondihydropyridine CCBs can cause excessive heart rate lowering (bradycardia) and should be avoided.

β-Blockers are recommended for patients with specific comorbid conditions.[240,272-274] For example, the use of β-blockers after an acute myocardial infarction (MI) has been shown to reduce morbidity and mortality,[275] although one randomized trial called into question the benefit of these agents post-MI after early coronary angiography among lower-risk patients with preserved left ventricular ejection fraction.[276] Other studies have shown a 20% reduction in total mortality and a 32% to 50% reduction in sudden death with β-blocker therapy in patients who have experienced an MI.[277] For treatment of hypertension in patients with a history of MI (at least in the past year), β-blockers may still be the drugs of choice.[50,278,279]

Patients with coexisting heart failure and hypertension are another group that benefit from treatment with β-blockers.[280-282] The Cardiac Insufficiency Bisoprolol Study II demonstrated a 20% reduction in mortality in patients with moderate heart failure randomly assigned to therapy with a β-blocker.[283] Hospitalizations for heart failure and sudden cardiac death were also significantly lower in the β-blocker group. Similar benefits were observed in randomized trials of metoprolol succinate and carvedilol.[284] Over the long term, treatment with β-adrenergic blockers improves exercise tolerance, left ventricular geometry, and left ventricular structure and reduces myocardial oxygen demand. The magnitude of heart rate reduction with β-blockers, but not the dose, is significantly associated with the survival benefit in heart failure.[285] β-Blockers may differ in effects on cardiovascular outcomes. A meta-analysis has suggested that the vasodilatory β-blocker carvedilol has a greater benefit on all-cause mortality in HFrEF compared with β_1-selective β-blockers, although this distinction has not been observed in all studies.[285-287] Genetic polymorphisms affecting the β_1-adrenergic receptor, the α_{2C}-adrenergic receptor and the G protein–coupled receptor kinase have been suggested to modify heart failure risk and the response to β-blocker therapy.[288]

Agents with β_1-selectivity or intrinsic sympathetic activity have a therapeutic advantage over nonselective β-adrenergic antagonists in the treatment of patients with bronchospastic airway disease, chronic obstructive pulmonary disease, peripheral vascular disease, and diabetes mellitus.[243,289,290] Bronchoconstriction is mediated in part by β_2-adrenergic receptors in the airways. β-Blockade with nonselective agents can lead to increased airway resistance. This increase is less likely to occur with β_1-selective agents. β_1-Selectivity is relative, however, and may be less apparent at higher dosages. In general, patients with severe bronchospastic airway disease should not receive β-blockers. In patients with mild-to-moderate disease, β_1-selective agents may be used cautiously; it has been proposed that they have beneficial effects on airway hyperresponsiveness.[291]

Symptoms of peripheral artery disease may be exacerbated by β-blocker therapy.[292] Cold extremities and absent pulses have been described in patients with severe disease. Raynaud phenomenon has been reported with nonselective β-blockade.[293] Blockade of β_2-receptor–mediated skeletal muscle vasodilation, as well as decreased cardiac output, may contribute to vascular insufficiency.[294] However, a meta-analysis showed that treatment with β-blockers does not worsen intermittent claudication or walking capacity in patients with mild-to-moderate peripheral artery disease.[295] Current treatment guidelines do not recommend against the use of β-blockers in patients with peripheral artery disease, evidence is limited, and controversy remains.[296]

The CNS symptoms of sedation, sleep disturbance, depression, and visual hallucinations have been reported with β-blockers. In a review of 15 randomized trials involving more than 35,000 patients, β-blockers were not associated with a significant increase in the risk of reported depressive symptoms (6/1000 patient-years; 95% CI, −7 to 19). β-Blockers were associated with a small but significant annual increase in reported fatigue (18/1000 patient-years; 95% CI, 5–30). These symptoms may be more common with lipid-soluble β-blockers and less common with nebivolol.[297]

β-Blockers are associated with weight gain and an increased risk of new-onset diabetes mellitus,[298,299] perhaps by mediating increases in glycogenolysis and gluconeogenesis from amino acids and glycerol and inhibiting glucose uptake in the periphery. β-Blockers differ in terms of their effects on glucose metabolism. Nonvasodilating β-blockers such as metoprolol decrease insulin sensitivity and are associated with a worsening of glycemic control, whereas nebivolol does not.[300,301] In patients with established diabetes mellitus, β-blockers can blunt the effects of epinephrine secretion resulting from hypoglycemia and lead to hypoglycemia unawareness.[302,303]

Nonselective β-blockers and, to a lesser degree, β_1-selective agents have been associated with a rise in the serum potassium level.[304] Suppression of aldosterone and inhibition of β_2-linked, sodium–potassium membrane transport in skeletal muscle have been proposed as possible mechanisms.[305,306] This effect may be of most relevance to patients with CKD and in patients treated with ACE inhibitors or ARBs, particularly when used in conjunction with MRAs for resistant hypertension.

β-Blockers can affect plasma lipids.[307] Long-term use of β-blockers has been associated with an increase in triglyceride levels and a decrease in the level of HDL cholesterol. β-Blockers with increased β_1 selectivity or with partial agonist activity appear to have less effect on the lipid profile. Nonselective β-blockers without partial agonist activity may decrease the HDL cholesterol level by up to 20%; an increase in triglyceride levels of up to 50% has been reported. The effects of β-blockade on lipid metabolism are due primarily to the modulation of lipoprotein lipase activity. Very-low-density lipoprotein (VLDL) cholesterol and triglyceride metabolism is reduced in the setting of unopposed β-adrenergic stimulation of lipoprotein lipase activity. Decreased VLDL metabolism results in decreases in HDL cholesterol levels.

β-Blockers were associated with a small but significant annual increase in reported sexual dysfunction (5/1000 patients; 95% CI, 2–8). None of the AEs differed by lipid solubility.[308]

Abrupt withdrawal of β-blockers may be associated with rebound hypertension and worsening angina in patients with coronary artery disease.[309] MI has been reported. These withdrawal symptoms may be caused by increased sympathetic activity, which could reflect adrenergic receptor upregulation during long-term sympathetic blockade. Gradual tapering of β-blockers decreases the risk of withdrawal. Withdrawal symptoms have been reported more commonly with abrupt discontinuation of relatively short-acting drugs.[310]

CALCIUM CHANNEL BLOCKERS

CLASS MECHANISMS OF ACTION

Initially introduced in the 1960s as antianginal agents, dihydropyridine CCBs are now advocated as first-line therapy for hypertension.[50,311] The pharmacologic effects of these drugs are related to their ability to attenuate cellular calcium uptake.[311-315] CCBs inhibit the entry of calcium or its mobilization from intracellular stores. Calcium channels have binding sites for activators and antagonists. The voltage-dependent, L-type calcium channel is a multimeric complex composed of α_1-, α_2-, ω-, β-, and γ-subunits.[316] These channels have different binding sites for the various CCBs and are regulated by voltage-dependent and receptor-dependent events involving protein phosphorylation and G-protein coupling resulting from, for example, β-adrenergic stimulation.[317] Each class of CCB is quantitatively and qualitatively unique; the CCB compounds possess different sensitivities and selectivities for binding pharmacologic receptors and the slow calcium channel in various vascular tissues. Even within the dihydropyridine class, there is considerable pharmacologic variability.[318] This differential selectivity of action has important clinical implications for the use of these drugs and explains why the CCBs vary considerably in their effects on regional circulatory beds, sinus and AV nodal function, and myocardial contractility. The selectivity further explains the diversity of indications for clinical use, ancillary effects, and side effects.[319]

CCBs uniformly lower peripheral vascular resistance in patients regardless of age, sex, ancestry, salt sensitivity, or comorbid conditions. There are at least three mechanisms through which CCBs lower BP. First, CCBs reduce peripheral vascular resistance by attenuating the calcium-dependent contractions of vascular smooth muscle. Contraction of vascular smooth muscle depends on the total cytosolic calcium concentration, which in turn is regulated by two distinct mechanisms. Depolarization of vascular smooth muscle tissue depends on the inward flux of calcium through voltage-sensitive L-type and T-type calcium channels. Hypertensive patients have an abnormal influx of calcium, which promotes increased peripheral vascular resistance.[312] Calcium is released from the sarcoplasmic reticulum in response to extracellular calcium influx via a nonvoltage-dependent pathway. Cytosolic calcium binds to calmodulin, initiating a sequence of cellular events that promote the interaction between actin and myosin and results in smooth muscle contraction. Therefore the importance of calcium channels lies in their pivotal role in linking cell membrane electrical activity to biologic responses. Calcium influxes through L-type channels from extracellular sources and intracellular sources are both attenuated by CCBs.[311,320]

Second, CCBs decrease vascular responsiveness to Ang II and the synthesis and secretion of aldosterone.[314] CCBs also interfere with α_2-adrenergic receptor–mediated vasoconstriction and possibly α_1-adrenergic receptor–mediated vasoconstriction.[313,321] The maximal vasodilatory response, as measured by forearm blood flow, appears to be inversely related to the patient's plasma renin activity and Ang II concentration. Thus it is possible that there is a greater influence of the calcium influx–dependent vasoconstriction in patients with low-renin hypertension.

Finally, CCBs may induce mild diuresis. Dihydropyridines, in particular, reduce preglomerular resistance and maintain or increase the GFR because of their preferential vasodilatory action on the kidney afferent arteriole.[315] The vasorelaxant properties of nifedipine and verapamil appear to be nitric oxide independent, whereas those of amlodipine are partly nitric oxide dependent.[322,323] This effect of amlodipine is thought to be mediated by the inhibition of local ACEs and increases in vasodilatory bradykinins. Subsequently, decreased tubular sodium reabsorption and improved kidney blood flow and natriuresis are observed. The sodium excretion rate tends to correlate with the reduction in BP.

CLASS MEMBERS

Despite their shared mechanism of action, the CCBs are a heterogeneous group of compounds. They differ with respect to pharmacologic profile, chemical structure, pharmacokinetic profile, tissue specificity, receptor binding, clinical indications, and side-effect profile (Tables 48.14 and 48.15). Two primary subtypes are distinguished on the basis of their behavior: dihydropyridines and nondihydropyridines. The nondihydropyridines are further divided into two classes—benzothiazepines (diltiazem) and diphenylalkylamines (verapamil). Their distinctly different pharmacologic effects are summarized in Tables 48.14 and 48.15.

Although all CCBs vasodilate coronary and peripheral arteries, the dihydropyridines are the most potent. Because medications in this subclass of CCBs are membrane-active drugs, they exert a greater effect on the peripheral vessels than on myocardial cells, which depend less heavily on the external calcium influx.[311] Their potent vasodilatory action prompts a rapid compensatory increase in sympathetic nervous activity, as mediated by baroreceptor reflexes creating a neutral or positive inotropic stimulus.[324] Longer-acting dihydropyridines, however, do not appear to activate the SNS.[325] By contrast, the nondihydropyridines are moderately potent arterial vasodilators but directly decrease AV nodal conduction and have negative inotropic and chronotropic effects, which are not abrogated by the reflex increase in sympathetic tone. Because of their negative inotropic action, their use is contraindicated in patients with HFrEF. As expected, these drugs are more effective

Table 48.14 Pharmacokinetic Properties of Calcium Channel Blockers

Drug	Oral Absorption (%)	First-Pass Effect	Bioavailability (%)	Peak Blood Level	Elimination Half-Life (h)	Metabolism and Excretion	Protein Binding (%)	Active Metabolites
Amlodipine	>90	M	88	6–12 h	30–50	L/U	>95	Yes
Diltiazem	98	50%	40	2–3 h	4–6	L, F, U	77–93	Yes
Diltiazem SR	>80	50%	35	6–11 h	5–7	L, F, U	77–93	Yes
Diltiazem CD	95	E	35	12 h	5–8	L, F, U	77–93	Yes
Diltiazem XR	95	E	41	4–6 h	5–10	L, F, U	95	Yes
Diltiazem ER	93	E	40–60	4–6 h	10	L, F, U	95	Yes
Felodipine	>90	E	13–18	2.5–5 h	11–16	L/U	>95	No
Isradipine	>90	E	15–25	2–3 h	8	L, F, U	>95	No
Isradipine CR	>90	E	15–25	7–18 h	—	L, F, U	>95	No
Lacidipine	>90	E			12–19	L, F, U	>95	No
Lercandipine	>90	E	-	1.5–3	-	L, F, U	>99	No
Manidipine		E		2–3.5 h	5–8	L, F, U	99	No
Nicardipine	>90	E	35	0.5–2 h	8.6	L, F, U	>95	No
Nicardipine SR	>90	E	35	1–4 h	—	L, F, U	>95	No
Nifedipine	>90	20%–30%	60	<30 min	2	L, U	98	Yes
Nifedipine GITS	>90	25%–35%	86	6 h	—	L, U	98	Yes
Nifedipine ER	>90	25%–35%	86	2.5–5 h	7	L, U	98	Yes
Nisoldipine	>85	E	4–8	6–12 h	10–22	L, F, U	99	No
Verapamil	>90	70%–80%	20–35	1–2 h	2.8–7.4	L, F, U	85–95	Yes
Verapamil SR	>90	70%–80%	20–35	5–6 h	4–12	L, F, U	85–95	Yes
Verapamil SR pellet	>90	70%–80%	20–35	7–9 h	12	L, F, U	85–95	Yes
CODAS Verapamil	>90	70%–80%	20–35	11 h	—	L, F, U	85–95	Yes

CD, Controlled-diffusion; *CODAS,* chronotherapeutic oral drug absorption system; *CR,* controlled release; *E,* extensive; *ER,* extensive release; *F,* feces; *GITS,* gastrointestinal therapeutic system; *L,* liver; *M,* minimal; *SR,* sustained release; *U,* urine; *XR,* extended release.

Table 48.15 Pharmacodynamic Properties of Calcium Channel Blockers

Generic Name (Trade Name)	First Dose (mg)	Usual Daily Dosage (mg)	Maximum Daily Dose (mg)	Peak Response (h)	Duration of Response (h)
Diltiazem (Cardizem)	60	60–120 tid, qid	480	2.5–4	8
Diltiazem SR (Cardizem SR)	180	120–240 bid	480	6	12
Diltiazem CD (Cardizem CD)	180	240–280 qd	480	—	24
Diltiazem XR (Dilacor XR)	180	180–480 qd	480	3–6	24
Diltiazem ER (Tiazac)	180	180–480 qd	480	4–6	24
Amlodipine (Norvasc)	5	5–10 qd	10	30–50	24
Felodipine (Plendil ER)	2.5	2.5 qd	10	2–5	24
Isradipine (DynaCirc)	2.5	2.5–5 bid	20	2–3	12
Isradipine CR (DynaCirc CR)	5	5–20 qd	20	2	7–18
Lacdipine	4	4–6 qd	6	1–2	12–24
Lercandipine	10	10–20 qd	40		24
Manidipine	10	10–20 qd	40	2–3.5	24
Nicardipine (Cardene)	20	20–40 tid	120	0.5–2	8
Nicardipine SR (Cardene SR)	30	30–60 bid	120	1.4	12
Nifedipine (Procardia, Adalat)	10	10–30 tid, qid	120	0.1	4–6
Nifedipine GITS (Procardia XL)	30	30–90 qd	120	4–6	24
Nifedipine ER (Adalat CC)	30	30–90 qd	120	2–6	24
Nisoldipine (Sular)	20	20–40 qd	60	—	24
Verapamil (Calan, Isoptin)	80	80–120 tid	480	6–8	8
Verapamil SR (Calan SR, Isoptin SR)	120	120–240 bid	480	—	12–24
Verapamil SR Pellet (Verelan)	120	240–480 qd	480	—	24
Verapamil COER-24 (Covera-HS)	180	180–480 qhs	480	>4–5	24

bid, Two times a day; *CD,* Controlled-diffusion; *COER,* controlled-onset extended-release; *CR,* controlled release; *ER,* extended release; *GITS,* gastrointestinal therapeutic system; *qd,* once a day; *qhs,* every bedtime; *qid,* four times a day; *SR,* sustained release; *tid,* three times a day; *XR,* extended release.

at reducing stress-induced cardiovascular responses than dihydropyridines.[326]

A clinically useful classification system for CCBs categorizes them by their duration of action into short-acting and long-acting agents (see Tables 48.14 and 48.15). This schema is helpful because the short-acting agents are no longer recommended for the management of hypertension because of their rapid duration of action and risk of orthostatic hypotension and associated complications, as well as stimulation of the SNS, which may predispose patients to angina, MI, and stroke.[327] The long-acting drugs are commonly divided into three generations. First-generation agents, such as nifedipine, have shorter half-lives and require multiple daily doses. Second-generation agents have been modified into extended-release (ER) formulations, requiring once-daily dosing. The third-generation agents have intrinsically longer plasma or receptor half-lives, possibly related to their greater lipophilicity.[328]

Benzothiazepines

Diltiazem hydrochloride is the prototype of the benzothiazepine CCBs. Diltiazem is 98% absorbed from the gastrointestinal tract, but because of extensive first-pass hepatic metabolism, its bioavailability is only 40% compared with intravenous dosing[12] (see Tables 48.14 and 48.15). In vivo, the competitively inhibited liver CYP2D6 isoenzyme is the most important metabolic pathway and probably accounts for the substantial proportion of drug interactions that occur with diltiazem.[329] The rates of elimination are lower in older persons and those with chronic liver disease but unchanged in patients with CKD.

Oral forms of diltiazem have been modified to improve delivery and currently include tablets, SR capsules, controlled-diffusion capsules, Geomatrix ER capsules, ER capsules, and buccoadhesive formulations.[330-332] The usual starting dosage for the drug in tablet form is 180 mg/day in three divided doses, and the drug may be titrated to a total dosage of 480 mg/day (see Table 48.15).

Diphenylalkylamine

Verapamil hydrochloride, the oldest CCB, is the prototype diphenylalkylamine derivative. Verapamil inhibits membrane transport of calcium in myocardial cells, particularly the AV node, and smooth muscle cells, which renders it antiarrhythmic, antihypertensive, and a negative inotrope. The drug is available for oral administration as film-coated tablets containing 40, 80, or 120 mg of racemic verapamil hydrochloride.[12] The usual daily dose is 80 to 120 mg three times daily (see Table 48.15). The elimination half-life increases with long-term administration and in older patients with CKD (see Table 48.14).

The SR caplets are available in scored 120-, 180-, and 240-mg forms. The usual antihypertensive dose is equivalent to the total daily dose of immediate-release tablets and can be given as 240 to 480 mg/day. An adequate antihypertensive response may be improved by divided twice-daily dosing.

The SR pellet–filled verapamil capsules are gel coated with an onset of action of 7 to 9 hours that is not affected by food.

The peak concentrations are approximately 65% of those of immediate-release tablets, but the trough concentrations are 30% higher. The usual daily dose is 240 to 480 mg.

The controlled-onset, ER, and chronotherapeutic oral drug absorption system (CODAS) tablets have unique pharmacologic properties and deliver verapamil 4 to 5 hours after ingestion. A delay coating is inserted between the outer semipermeable membrane and active inner drug core. As the delay coating expands in the gastrointestinal tract, the pressure causes drug from the inner core to be released through laser-drilled holes in the outer membrane, making this formulation ideal for nighttime dosing by providing maximal plasma levels in the early morning hours, from 6 AM to noon, and minimizing nighttime diurnal BP variations.[333] A buccal gel formulation of verapamil that provides SR of the drug up to 6 hours has been reported.[334]

Of the 13 known metabolites of verapamil, norverapamil is the only one with cardiovascular activity; it has 20% of the potency of the parent compound. Kidney excretion accounts for 70% of clearance and occurs within 5 days. The remainder is excreted in the feces. Clearance decreases with increasing age and decreasing weight.[335] With long-term administration, there is a significant increase in bioavailability, possibly as a result of saturation of hepatic enzymes. Dose adjustment is necessary in patients with liver disease but not CKD. However, verapamil should be used with caution in patients who ingest large amounts of grapefruit juice or patients taking other AV nodal blocking agents.[336]

Dihydropyridines

Nifedipine is a dihydropyridine CCB that causes decreased peripheral arterial resistance, with no clinically significant depression of myocardial function. Because of the reflex sympathetic stimulation triggered by vasodilation, nifedipine has no tendency to prolong AV conduction or sinus node recovery or slow the sinus rate. Clinically, there is usually a small increase in heart rate and cardiac index. The labeling for immediate-release nifedipine capsules has been revised to recommend against using this dosage form for the management of hypertension[12,337] due in part to its association with increased mortality in older persons compared with the use of other antihypertensive agents, including other CCBs.[338] Also, in some patients the hypotensive effect is profound and has resulted in MI, stroke, and death.[337] This effect appears to be more pronounced in patients also taking β-blockers.[339] Consequently, its use should be reserved to settings in which multiple alternative agents have failed and should be avoided altogether in the acute setting. The usual adult dosage is 10 to 30 mg three times daily, and the dose can be titrated weekly (see Table 48.15). Nifedipine is extensively metabolized in the liver and excreted in the urine. Because nifedipine is 98% protein bound, the dosage should be adjusted in patients with hepatic insufficiency or severe malnutrition.

The ER tablets of nifedipine are available in 30-, 60-, and 90-mg doses. The ER form should not be bitten or divided. The time-to-peak concentration is 6 hours, and plasma levels remain steady for 24 hours. The bioavailability of the ER tablet is 86% compared with that of immediate-release forms, and tolerance does not develop.[340] Of the metabolites, 80% are excreted in the urine and the rest in the feces. The usual adult maintenance dosage is 30 to 90 mg daily.

Amlodipine besylate is unique among the dihydropyridine CCBs. It appears to bind to dihydropyridine and nondihydropyridine sites to produce peripheral arterial vasodilation without significant activation of the SNS.[22] The parent compound has substantially slower but more complete absorption than others in the class (see Table 48.14). After ingestion, amlodipine is almost completely absorbed, peak plasma concentrations are achieved in 6 to 12 hours, and the clinical response can be detected at 24 hours. The mean peak serum concentrations are linear, age independent, and achieved after 7 to 8 days of continuous dosing.[341] The elimination half-life is long, ranging from 30 to 50 hours, and is prolonged in older adults. The long half-life permits once-daily dosing; the hypotensive response may last up to 5 days,[342] 90% of amlodipine is metabolized in the liver, and 10% is excreted unchanged. The metabolites are excreted primarily in the urine, but no dosage adjustment is necessary with impaired kidney function. The minimum effective dose is 2.5 mg, particularly in older patients. Most patients require a dosage of 5 to 10 mg daily.

Felodipine is a dihydropyridine CCB that is administered in ER tablets of 2.5, 5, and 10 mg (see Table 48.15).[12] Felodipine is almost completely absorbed from the gastrointestinal tract, with a time to peak concentration of 2 to 5 hours (see Table 48.14). There is extensive first-pass hepatic metabolism. Bioavailability is influenced by food. Large meals and the flavonoids in grapefruit juice increase the bioavailability by approximately 50%.[343] The overall half-life is 11 to 16 hours. Felodipine is metabolized in the liver to inactive metabolites, most of which are excreted in the urine. The usual daily dose is 2.5 to 10 mg, and titration can be instituted at 2-week intervals. The dosage should be adjusted for liver disease but not for CKD.

Isradipine is a dihydropyridine CCB effective alone or in combination with other antihypertensive agents for the management of mild-to-moderate hypertension[12] (see Tables 48.14 and 48.15). Isradipine is rapidly and almost completely absorbed after oral administration. Extensive first-pass hepatic metabolism reduces bioavailability to <25%. The hypotensive effect peaks at 2 to 3 hours for the regular release form. The drug is active for 12 hours; however, the full antihypertensive response does not occur until 14 days. The usual dosage is 2.5 to 5 mg two to three times daily. The onset of action of the SR formulation is achieved in 2 hours and lasts for 7 to 18 hours. The usual daily dose of the controlled-release tablet is 5 to 20 mg. Isradipine is extensively protein bound. The elimination half-life is biphasic, with a terminal half-life of 8 hours. Dosage adjustment is unnecessary in liver disease or CKD.

Lacidipine is a second-generation dihydropyridine CCB available in tablet form. It is reported to be unusually potent and long acting, possibly because it diffuses deeper into lipid bilayer membranes. A unique attribute of this drug is its apparently greater vascular selectivity, but the clinical relevance of this remains unclear.[344] The usual dosage is 4 to 6 mg daily, and the dose should be titrated at 2- to 4-week intervals.[345] The duration of action is 12 to 24 hours. The elimination half-life is 12 to 19 hours. The parent compound is converted 100% by the liver into inactive fragments that are excreted primarily in the feces (70%)

and kidney. Dosage adjustment is necessary in older persons and in patients with liver disease but not CKD.

Lercanidipine is a dihydropyridine CCB whose molecular design imparts greater solubility within the arterial cellular membrane bilayer, conferring a tenfold higher vascular selectivity than that of amlodipine.[345] In contrast to amlodipine, lercanidipine has a relatively short half-life but a long-lasting effect at the receptor and membrane levels and is associated with significantly less peripheral edema.[345] The drug is administered at a starting dose of 10 mg and increased to 20 mg daily as needed. It has a gradual onset of action, and its effects last for 24 hours.[346] Lercanidipine also appears to dilate the efferent kidney arteriole.[347]

Manidipine is a third-generation dihydropyridine CCB structurally related to nifedipine.[348,349] The usual adult dosage is 10 to 20 mg daily. Dosage should be adjusted at 2-week intervals. Manidipine is highly protein bound and extensively metabolized in the liver. Metabolism is impaired by grapefruit juice;[350] 63% of the drug is excreted in the feces. The peak plasma concentration occurs after 2 to 3.5 hours, with an elimination half-life of 5 to 8 hours. Dose adjustment is not necessary in CKD. Manidipine may be less likely to cause significant ankle edema than amlodipine.[351]

Nicardipine hydrochloride is a dihydropyridine CCB available as 20- and 40-mg immediate-release gelatin capsules or 30-, 45-, and 60-mg SR capsules.[12] The usual dosage is 20 to 40 mg three times daily for the immediate-release form and 30 to 60 mg twice daily for the SR preparation. When conversion is made to the SR form, the previous daily total of immediate-release drugs should be administered on a twice-daily regimen. Titration should be instituted at least 3 days after administration. Nicardipine is well absorbed orally but has only 35% systemic bioavailability because of its extensive first-pass hepatic metabolism. The time to peak concentration is 30 minutes to 2 hours for immediate-release capsules and 1 to 4 hours for SR forms. The elimination half-life is 8.6 hours. Nicardipine is 100% oxidized in the liver to inactive pyridine metabolites. There is no evidence of microsomal enzyme induction. Metabolites are excreted primarily in the urine and feces. The parent compound is not dialyzable. Dosage adjustments are necessary with liver disease but not CKD.

Nisoldipine is a dihydropyridine CCB that is formulated as ER tablets of 10, 20, 30, and 40 mg.[12] The initial starting dose is 20 mg, and the usual maintenance dose is 20 to 40 mg given once daily, which can be titrated at weekly intervals. The bioavailability of nisoldipine is low and variable (4%–8%). The coat core design affords a full 24-hour effect after oral administration. The drug reaches therapeutic concentrations in 6 to 12 hours, and absorption is slowed by high-fat meals. The elimination half-life ranges from 10 to 22 hours. Nisoldipine is metabolized in the liver and intestine. Variable hepatic blood flow induced by the drug probably contributes to its pharmacokinetic variability. Most of the metabolites are excreted in the urine and the remainder in the feces. Dose adjustments are necessary with liver disease but not CKD.

CLASS KIDNEY EFFECTS

All CCBs exert natriuretic and diuretic effects.[352,353] Experimental studies and studies in humans with hypertension have indicated that the increase in sodium excretion is, in part, independent of vasodilatory action or changes in GFR, kidney blood flow, or filtration fraction. This effect is probably the result of changes in kidney sodium handling that can potentiate the antihypertensive vascular effect. In normal persons, CCBs acutely increase sodium excretion, frequently in the absence of changes in BP. In hypertensive persons, the short-term administration of CCBs uniformly increases sodium excretion 1.1- to 3.4-fold; the magnitude of the increase is not related to the decrease in BP.[352]

The natriuretic effect appears to persist in the long term. Long-term administration of CCBs to hypertensive patients results in a cumulative sodium deficit that is abruptly reversed with the discontinuation of the drug. Natriuresis occurs 3 to 6 hours after the morning dose.[353] The net negative sodium balance levels off after the first 2 to 3 days of administration but persists for the duration of therapy.[354] There are no significant changes in long-term body weight, serum concentrations of potassium, urea nitrogen, catecholamines, or GFR. Moreover, stimulation of renin release and aldosterone does not occur to an appreciable degree. It has been postulated that the natriuresis induced by CCBs increases distal sodium delivery to the macula densa, suppressing renin release. Because Ang II mediates aldosterone synthesis by way of cytosolic calcium messengers, CCBs blunt this response as well.[355]

The mechanism whereby CCBs induce natriuresis appears to be direct inhibition of kidney tubular sodium and water absorption. Dihydropyridines increase urinary flow rate and sodium excretion without changing the filtered water and sodium load. Studies have suggested that CCBs may diminish sodium uptake at the amiloride-sensitive sodium channels.[356] Inhibition of water reabsorption occurs distally to the late distal tubule. Proximal tubular sodium reabsorption may be inhibited by higher dosages. One possible mediator of this effect is atrial natriuretic peptide. In human studies, CCBs augment atrial natriuretic peptide release and potentiate its action at the level of the kidney. Other potential mediators are under investigation. How much the natriuretic effects contribute to the antihypertensive response is unknown, but unlike effects of other vasodilators, the changes attenuate the expected adaptive changes in sodium handling.

The kidney hemodynamic effects of CCBs are variable and depend primarily on which vasoconstrictors modulate the kidney vascular tone.[357] Experimentally, CCBs improve GFR in the presence of the vasoconstrictors norepinephrine and Ang II, as well as others, by preferentially attenuating afferent arteriolar resistance.[358] The efferent arteriole appears to be refractory to these vasodilatory effects. Patients with primary hypertension appear to be more sensitive to the kidney hemodynamic effects of CCBs than normotensive patients, and this effect is more pronounced with more advanced CKD.[359] Short-term administration of CCBs results in little change in, or augmentation of, GFR and kidney plasma flow, no change in the filtration fraction, and reduction of kidney vascular resistance. Long-term administration is not associated with significant changes in kidney hemodynamics. The response is maximal in the presence of Ang II, which selectively causes postglomerular vasoconstriction. Clinically significant changes are counteracted by the reduction in kidney perfusion pressure coincident with a reduction of BP.

The long-term effects of CCBs on kidney function are variable.[360,361] In hypertensive patients, the effects on kidney hemodynamics vary. Some patients exhibit no change in GFR, whereas others have an exaggerated increase in GFR and kidney plasma flow.[357] Even normotensive patients with a family history of hypertension have an exaggerated hemodynamic response.[362]

The effects of CCBs on proteinuria also vary with respect to the specific drug and the degree of BP reduction achieved.[363] Some dihydropyridines increase protein excretion by up to 40%. It is not clear whether this increase is a result of hemodynamic vasodilation at the afferent arteriole, resulting in increased glomerular capillary pressure (because CCBs directly impair kidney autoregulation), changes in glomerular basement membrane permeability, or increased intrarenal Ang II. By contrast, felodipine, diltiazem, and verapamil do not appear to have this effect and may lower protein excretion, possibly by also decreasing the efferent arteriolar tone and glomerular pressure.[364] Clinical implications of these differential effects remain to be determined.

Large clinical trials underscore this controversy. In African Americans with hypertension and mild-to-moderate CKD, treatment with an ACE inhibitor demonstrated superior renoprotective effects compared with amlodipine.[365] This effect was independent of BP reduction and was more evident in proteinuric patients; it was also suggestive in patients with <300 mg of protein/day at baseline. Hypertensive patients with diabetic nephropathy also fared worse with amlodipine than with ARB therapy.[175] Patients experienced higher rates of progression of CKD and all-cause mortality in the amlodipine- and placebo-treated groups. The latter effect was independent of the achieved BP. However, it should be emphasized that coadministration of a dihydropyridine and a RAAS inhibitor does not abrogate the protective effect of RAAS inhibitors on kidney function.[56,366] It has been postulated that the selective dilation of the afferent arteriole favors an increase in glomerular capillary pressure that perpetuates progression.

CLASS EFFICACY AND SAFETY

All long-acting CCBs are considered among first-line antihypertensive agents and appear to be equally efficacious and safe vis-à-vis blood pressure lowering.[367,368] In contrast to other vasodilators, CCBs attenuate the reflex increase of neurohormonal activity that accompanies a reduction in BP, and in the long term, they inhibit or do not change the sympathetic activity.[369,370] The longer-acting agents such as amlodipine produce sustained SBP reductions of 10 to 14 mmHg in placebo-controlled RCTs when used as monotherapy, with no appreciable development of tolerance.[371] The CCBs are effective in young, middle-aged, and older patients with white coat hypertension and mild, moderate, or severe hypertension.[372-375] Their efficacy may be determined by genetic polymorphisms.[376] CCBs are equally efficacious in men and women, in patients with a high or low plasma renin activity regardless of dietary salt intake.[377] Effects of CCBs are diminished in smokers.[378] CCBs are effective and safe in patients with hypertension and coronary artery disease,[379] as well as in patients with ESKD.[380] CCBs also reduce adverse cardiovascular events and slow the progression of atherosclerosis in normotensive patients with coronary artery disease.[381]

The use of CCBs is contraindicated in patients with HFrEF (except, perhaps, amlodipine or felodipine). These drugs should not be used as first-line antihypertensive agents in patients with heart failure, a history of MI, or unstable angina.[365]

Among the different categories, dihydropyridines appear to be the most powerful for reducing BP but may also be associated with more pronounced activation of baroreceptor reflexes.[382] Dihydropyridines induce a more prominent shift in the sympathovagal balance that favors sympathetic predominance compared with nondihydropyridines.[326] In general, however, compared with other vasodilators, CCBs attenuate the reflex increase in sympathetic activity—increased heart rate, cardiac index, plasma norepinephrine levels, and renin activity.

The nondihydropyridines verapamil and, to a lesser extent, diltiazem exert greater effects on the heart and have less vasoselectivity. These drugs typically reduce the heart rate, slow AV conduction, and depress contractility (Table 48.16). Generally, they should not be used together with β-blockers because of increased risk for heart block.

CCBs are not associated with significant impairments in glycemic control or sexual dysfunction.[367] The rapid antihypertensive action of CCBs may encourage patient adherence. Orthostatic changes tend not to occur because venoconstriction remains intact. AEs are usually transient and are the direct result of vasodilation. Hypotension is most common with intravenous administration. The most common AE of the dihydropyridines is peripheral edema;[383] it is dose related and thought to be the result of uncompensated precapillary vasodilation, which causes increased intracapillary hydrostatic pressure. The edema is not responsive to diuretics but improves or resolves with the addition of an ACE inhibitor or ARB, which preferentially vasodilates postcapillary beds and reduces intracapillary

Table 48.16 Hemodynamic Effects of Calcium Channel Blockers

Class	Arteriolar Dilation	Coronary Dilation	Cardiac Afterload	Cardiac Contractility	Myocardial O_2 Demand	Cardiac Output	AV Conduction	SA Automaticity	Heart Rate: Short Term/ Long Term	Activation of Baroreceptor Reflexes
Dihydropyridines	↑↑↑	↑↑↑	↓↓	↔	↓	↓ or ↔	↔	↔	↑/↑	↑ or ↔
Diltiazem	↑↑	↑↑↑	↓	↓	↓	↔	↓	↓↓	↓/↓ or ↔	↔
Verapamil	↑↑	↑↑	↓	↓↓	↓	↔	↓↓	↓	↓/↓ or ↔	↔

AV, Atrioventricular; *SA,* sinoatrial.

hydrostatic pressure.[384] Other AEs related to vasodilation include headache, nausea, dizziness, and flushing, and occur more commonly in women. The nondihydropyridines verapamil and isradipine more commonly cause constipation and nausea. The gastrointestinal effects are directly related to the inhibition of calcium-dependent smooth muscle contraction–reduced peristalsis and relaxation of the lower esophageal sphincter. Another common AE of dihydropyridines is gingival hyperplasia, which is exacerbated in patients who are also taking cyclosporine. Dihydropyridines lead to the accumulation of gingival inflammatory B-cell infiltrates as stimulated by bacterial plaque, immunoglobulins, and folic acid, which causes the growth of the gingiva.[385] This growth can be controlled with regular periodontal treatment and reversed with discontinuation of the drug.[386]

CCBs are notable among antihypertensive agents because of their metabolic neutrality. Because the calcium influx across β-cell membranes helps regulate insulin release,[387] CCBs might predispose to low insulin levels. At typical therapeutic levels, CCBs have no effect on serum glucose concentrations, insulin secretion, or insulin sensitivity in persons with and without diabetes mellitus. The use of CCBs was not significantly associated with incident diabetes compared with other antihypertensive agents in a meta-analysis of RCTs.[388] The association with diabetes was lowest for ACE inhibitors and ARBs, followed by CCBs, β-blockers, and diuretics.[388] Furthermore, it was recently demonstrated that CCBs can even prevent diabetes and increase β-cell survival in vitro. The mechanism behind the β-cell mass destruction is the human islet cell protein TXNIP (thioredoxin-interacting protein), which is upregulated by hyperglycemia. Orally administered verapamil resulted in a reduction of TXNIP expression and β-cell apoptosis, enhanced endogenous insulin levels, and rescued mice from streptozotocin-induced diabetes. Verapamil also promoted β-cell survival and improved glucose homeostasis and insulin sensitivity in ob/ob mice.[389] CCBs do not increase triglyceride or LDL cholesterol and do not reduce HDL cholesterol. CCBs do not precipitate hyponatremia, hyperkalemia, hypokalemia, or hyperuricemia. Therefore they are suitable agents for patients with dysmetabolic syndromes or diabetes.

Properties beyond their antihypertensive actions make the CCBs particularly useful in certain clinical situations. CCBs lower arterial pressure and also have variable effects on cardiac function. All CCBs are vasodilators and increase coronary blood flow. With the exception of the short-acting dihydropyridines, most CCBs reduce heart rate, improve myocardial oxygen demand, improve ventricular filling, diminish ventricular arrhythmias, reduce myocardial ischemia, and conserve contractility,[390,391] making them suitable for patients with angina or diastolic dysfunction.[379] In short-term use, CCBs improve diastolic relaxation; when administered over a long term, they reduce left ventricular wall thickness,[392] may prevent the development of hypertrophy and may improve arterial compliance.[393-395] These effects may be crucial in hypertensive patients because LVH is one of the strongest risk predictors for cardiovascular morbidity and mortality.[396] Verapamil may also be used for secondary cardioprotection to reduce reinfarction rates in patients who are intolerant of β-blockers (unless they have concomitant heart failure)[397] and in patients with chronic headaches.[398] The BP-independent inhibition of atherogenesis by CCBs may be another indication to use a CCB, particularly in high-risk patients, such as those with diabetes and ESKD.[399,400]

The use of CCBs may prevent or slow the decline of dementia. In the Systolic Hypertension in Europe (SYST-EUR) trial,[401] nitrendipine use was associated with a lower risk of incident dementia, a finding also seen in some other observational studies.[402,403] However, a meta-analysis of data from 31,090 individual participants from 6 observational studies found no significant benefit of CCB over other antihypertensive classes on dementia risk.[404]

Drug interactions are not uncommon (Table 48.17). Concurrent use of a CCB and amiodarone exacerbates sick sinus syndrome and AV block. Diltiazem, verapamil, and dihydropyridines have been shown to increase the levels of cyclosporine (including the microemulsion formulation), tacrolimus,[405] and sirolimus.[406] This interaction may be clinically useful for reducing the dosage and cost associated with immunosuppressive therapy. Frequent monitoring of serum concentrations of calcineurin inhibitor is recommended. Diltiazem is a potent inhibitor of CYP3A4, which is responsible for the metabolism of methylprednisolone. Coadministration of diltiazem and methylprednisolone resulted in a more than 2.5-fold increase in the steroid blood level and enhanced adrenal suppressive responses.[407] Coadministration of diltiazem also increased nifedipine levels by 100% to 200%.[408] This combination has additive antihypertensive efficacy and appears to be safe.[409] Concomitant administration of CCBs with the digitalis glycosides resulted in up to a 50% increase in serum digoxin concentrations because of reduced kidney clearance of digoxin, an effect that appears to be dose dependent.[410] Insofar as dihydropyridine CCBs partially suppress aldosterone synthesis, they provide an attractive alternative for patients who cannot tolerate blockade of the RAAS.[411]

Several issues regarding the inherent safety of CCBs have come under scrutiny. CCBs may be associated with an increased risk of gastrointestinal hemorrhage, particularly in older persons.[412] Diltiazem inhibits platelet aggregation in vitro,[413] but the clinical relevance of this finding has not been substantiated. A systematic review and meta-analysis of 17 studies showed a marginally higher pooled risk of GI bleeding among CCB users (RR 1.17, CI 1.01–1.36), but this increased risk was observed primarily in observational studies that did not adjust for prior history of GI bleeding.[414] Nonetheless, it is prudent to use caution when coadministering CCBs with NSAIDs because NSAIDs may exacerbate the risk of bleeding and may antagonize the antihypertensive effects of CCBs.[415,416] There have been concerns regarding a possible relation between the long-term use of CCBs and breast cancer.[417] A population case–control study of breast cancer showed that the use of CCBs for more than 10 years was associated with ductal breast cancer (odds ratio [OR], 2.4; 95% CI, 1.2–4/9; $P = .04$) and lobular breast cancer (OR, 2.6; 95% CI, 1.3–5.3; $P = .01$), respectively. However, two recent large population-based observational studies demonstrated no association between CCB use and breast cancer.[418,419]

Short-acting CCBs have been associated with a small increased risk of MI in meta-analyses[337,420] when compared with other agents. It has been speculated that the

Table 48.17 Drug–Drug Interactions With Calcium Channel Blockers

Calcium Channel Blocker	Interacting Drug	Result
Verapamil	Digoxin	Digoxin level ↑ by 50%–90%
Diltiazem	Digoxin	Digoxin level ↑ by 40%
Verapamil	β-Blockers	AV nodal blockade, hypotension, bradycardia, asystole
Verapamil, diltiazem, dihydropyridines	Cyclosporine-tacrolimus and sirolimus	Increase immunosuppressive drug levels due to decreased metabolism
Verapamil, diltiazem	Cimetidine	Verapamil and diltiazem levels ↑ by decreased metabolism
Verapamil	Rifampin–phenytoin	Verapamil level ↓ by enzyme induction
Dihydropyridines	Amiodarone	Exacerbation of sick sinus syndrome and AV nodal blockade
Dihydropyridines	α-Blockers	Excessive hypotension
Dihydropyridines	Propranolol	Increases propranolol level
Dihydropyridines	Cimetidine	Increased area under the curve and plasma level of calcium channel blocker
Nicardipine	Cyclosporine	Cyclosporine level ↑ by 40%–50%
Amlodipine	Cyclosporine	Cyclosporine level ↑ by 10%
Felodipine	Flavonoids	Bioavailability ↑ by 50%
Diltiazem	Methylprednisolone	Methylprednisone ↑ 2.5-fold
Nifedipine	Diltiazem	Nifedipine level ↑ 100%–200%

AV, Atrioventricular.

disadvantageous activation of the RAAS and SNS induced by the short-acting agents may predispose to myocardial ischemia. Currently, there is no evidence to prove the existence of additional beneficial or detrimental effects of CCBs on coronary disease events, including fatal or nonfatal MIs and other deaths from coronary heart disease. Because of the potential risk, however, as well as for simplicity and improved patient adherence, longer-acting agents should be considered over short-acting CCBs for the management of hypertension.[421,422]

The use of CCBs has also been associated with the development of chyloperitoneum among patients receiving peritoneal dialysis.[421,422] A rare complication, chyloperitoneum seems to be most frequently described with the use of the dihydropyridines manidipine and lercanidipine. Cessation of the CCB generally leads to complete resolution. The mechanism of chyloperitoneum via CCBs in PD is not clear but possibly related to inhibition of lymphatic pump contraction.

CENTRAL ADRENERGIC AGONISTS

CLASS MECHANISMS OF ACTION

Central adrenergic agonists act by crossing the blood-brain barrier and have a direct agonist effect on α_2-adrenergic receptors located in the midbrain and brainstem.[423-425] Binding to the I_1 imidazoline receptors in the brain may also play a role in the inhibition of central sympathetic output.[5,426-431] Drugs in this class bind to the α-adrenergic or I_1 imidazoline receptors with some degree of specificity (Table 48.18). Moxonidine and rilmenidine have a 30-fold greater specificity for the I_1 imidazoline receptor than the α_2 receptor. Clonidine, by contrast, exhibits a fourfold greater specificity for the I_1 imidazoline receptor than for the α_2 receptor.

AEs are thought to be largely related to α_2-receptor binding. Moxonidine and rilmenidine have reduced central side effects because of the lower activity at the α_2 receptor relative to other agents.[431,432] In addition to decreasing the total sympathetic outflow, binding to these receptors results in increased vagal activity. A reduction in catecholamine release and turnover, as evidenced by decreased biochemical markers of noradrenergic activity, such as plasma concentrations of norepinephrine, correlated with the magnitude of BP lowering.

Table 48.18 Receptor Binding of Centrally Acting Antihypertensives

Drug	Receptor
Clonidine	α2, I_1
α-Methyldopa	α2
Guanfacine	α2
Rilmenidine	I_1 > α2
Moxonidine	I_1 > α2

I_1, Imidazole receptor; *α2,* α2-adrenergic receptor.

Stimulation of both receptor types is probably mediated through the same neuronal pathways.[432] The classic α_2-receptor agonists, such as clonidine and α-methyldopa (acting through its active metabolite, α-methylnorepinephrine), result in vasodilation in the resistance vessels and thus a reduction in peripheral vascular resistance. Despite vasodilator action, reflex tachycardia generally does not occur, probably as a result of peripheral sympathetic inhibition.

The selective I_1 receptor agonists moxonidine and rilmenidine are predominantly arterial vasodilators that lead to a reduction in peripheral vascular resistance.[427] Moxonidine is associated with a reduction in plasma renin activity. The central α_2-adrenergic agonists may

Table 48.19 Pharmacokinetic Properties of Central Adrenergic Agonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Clonidine (Catapres)	50	—	—	6–23	L	F (30%–50%) U (24%)	Methyldopa-*o*-sulfite
Guanfacine (Tonex)	80	—	1–4	17	L	U (40%–75%)	—
α-Methyldopa (Aldomet)	65–96	—	1.5–5	6–23	L	F (22%) U (65%)	—
Rilmenidine (Hyperium)	80–90	No	2	2–3	L	U (90%)	—
Moxonidine (Physiotens)	100	No	0.5–3	2	L	U (90%)	—

F, Feces; *L,* liver; *U,* urine.

Table 48.20 Pharmacodynamic Properties of Central Adrenergic Agonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Clonidine	0.1	0.3–0.9	2.4	bid, tid	2–4	6–10
α-Methyldopa	250	250–500	3000	bid, tid, qid	6–9	24–48
Guanfacine	1	1–3	3	qd	6	24
Moxonidine	0.1–0.2	0.2–0.3	0.6	bid	1.5–4	48–72
Rilmenidine	1	1–2	2	qd, bid	1–2	10–12

bid, Two times a day; *qd,* once a day; *qid,* four times a day.

also stimulate peripheral α_2-adrenergic receptors. This effect predominates at high drug concentrations. These receptors mediate vasoconstriction, which may result in a paradoxical increase in BP.[424] Overall, these drugs generally result in a decrease in peripheral vascular resistance, slowing of the heart rate, and either no change or a mild decrease in cardiac output.[432,433] The pharmacokinetic and pharmacodynamic properties of these drugs are shown in Tables 48.19 and 48.20.

CLASS MEMBERS

Clonidine is a central-acting α-adrenergic agonist.[81,428,434] The usual oral dosage is 0.1 mg two to three times daily, adjusted as needed in 0.1- to 0.2-mg increments. The usual maintenance dosage is 0.3 to 0.9 mg daily in two to three divided doses. Total doses more than 1.2 mg daily are usually not associated with a greater effect. The onset of activity is 30 to 60 minutes after an oral dose. The peak antihypertensive activity occurs within 2 to 4 hours. The duration of the antihypertensive effect is 6 to 10 hours. The half-life of the absorbed drug is 6 to 23 hours. Hepatic metabolism to inactive metabolites is followed by kidney excretion. Transdermal patches are available and may be applied on a once-weekly basis. The drug half-life with the transdermal patch is approximately 20 hours after removal of the patch. With a transdermal patch, steady-state drug levels are reached within approximately 3 days. Dose adjustment is not needed for patients with any degree of CKD including ESKD. Approximately 5% of clonidine body stores are removed after a 5-hour hemodialysis session.

Guanfacine is a centrally acting antihypertensive drug with actions similar to those of clonidine.[435] Effective dosages are 1 to 3 mg daily. Peak levels are noted between 1 and 4 hours. The drug half-life is approximately 17 hours. The drug is 70% protein bound. It is metabolized in the liver, with kidney excretion of 40% to 75% as an unchanged drug. Limited data are available on dosing in CKD, but dosage adjustments do not appear warranted.

α-methyldopa (or methyldopa) is a methyl-substituted amino acid that is active after conversion to an active metabolite. This active metabolite, α-methylnorepinephrine, accumulates in the CNS and is selective for α_2-adrenergic receptors. The initial dosage of α-methyldopa in hypertension is 250 mg two to three times daily. This dose may be increased at intervals of not less than 2 days until a therapeutic response is achieved. The usual maintenance dosage is 500 mg to 2 g daily in two to four doses. The maximum recommended daily dose is 3 g. An initial response occurs within 3 to 6 hours after dosing. The peak response occurs at 6 to 9 hours. The drug is approximately 50% metabolized by the liver. The drug half-life is increased in patients with CKD. Excretion in the urine is largely in the form of an inactive metabolite. The dosing interval should be increased to every 12 to 24 hours in patients with advanced CKD. Approximately 60% of α-methyldopa is removed with hemodialysis. A supplemental dose is recommended after dialysis treatment.

Moxonidine is a central I_1 imidazole and α_2-receptor agonist.[81,436,437] Serum concentration peaks are reached within 30 to 180 minutes; 90% of the dose is excreted through the urine within 24 hours, and 50% of this is as unchanged drug. The average half-life is 2 hours. For the management of hypertension, the starting dosage is 0.2 to 0.4 mg/day. The dose may be increased after several weeks to 0.2 to 0.3 mg twice daily. The maximum daily dose is 0.6 mg. Selectivity for the I_1 imidazoline receptor results in fewer central AEs, such as dry mouth and sedation, compared with those of clonidine. Drug clearance is delayed in patients with impaired kidney function. Single doses of 0.2 mg and a maximum daily dosage of 0.4 mg should not be exceeded in patients with CKD.

Rilmenidine is a centrally acting imidazole receptor and α_2-adrenergic receptor agonist.[81,430,438-442] Rilmenidine binds preferentially to central I_1 imidazoline receptors in the brainstem. At higher doses, rilmenidine can bind and activate central α_2-adrenergic receptors. Antihypertensive effects occur within 1 hour after a single 1-mg dose. The duration of action is 10 to 12 hours. The concentration after oral dosing peaks at approximately 2 hours. Steady-state plasma levels are reached by day 3. Rilmenidine is eliminated primarily unchanged in the urine. The usual oral dosage is 1 mg once or twice daily. Dose reductions are required for patients with impaired kidney function. In patients with advanced CKD, the dose should be decreased to 1 mg every other day.

CLASS KIDNEY EFFECTS

Central α_2- and I_1 imidazoline receptor agonists have little, if any, clinically important effect on kidney plasma flow, GFR, or the RAAS. The fractional excretion of sodium is unchanged. Body fluid composition and weight are not altered. These agents may result in decreased kidney vascular resistance, as mediated by a decrease in preglomerular capillary resistance related to decreased levels of circulating catecholamines.

CLASS EFFICACY AND SAFETY

The antihypertensive efficacy of this class of drugs has been confirmed in large numbers of patients and although not considered first-line, they provide effective monotherapy for hypertension.[424] Moxonidine and rilmenidine have been associated with decreased plasma glucose concentrations and may improve insulin sensitivity. These drugs may also decrease total cholesterol, LDL, and triglyceride levels[426,443,444] and may play a role in the management of metabolic syndrome. They may also be of benefit in patients with congestive heart failure. Treatment with rilmenidine and moxonidine reverses LVH and improves arterial compliance. This effect was associated with a reduction in plasma concentrations of atrial natriuretic peptide.

Stimulation of α_2-adrenergic receptors in the CNS induces several AEs of these drugs, including sedation and drowsiness. The most common AE related to α_2-adrenergic activation is dry mouth caused by a decrease in salivary flow. This decrease is attributed to centrally mediated inhibition of cholinergic transmission. Clonidine in high doses may precipitate a paradoxic hypertensive response related to the stimulation of postsynaptic vascular α_2-adrenergic receptors.[424] α-Methyldopa use has been associated with a positive result on the direct Coombs test in patients with and without hemolytic anemia.[424] Because of a long history of safe use during pregnancy, α-methyldopa remains a common therapeutic agent for hypertensive disorders of pregnancy and for essential hypertension during pregnancy.[423,425] Despite its position as one of the first-line agents for management of hypertension in pregnancy, α-methyldopa is no longer commercially available in the United States.[445,446] The α_2-adrenergic agonists are associated with sexual dysfunction and may produce gynecomastia in men and galactorrhea in men and women.

Abrupt cessation of α_2-adrenergic blockers may result in rebound hypertension, which occurs 18 to 36 hours after the cessation of short-acting agents.[447] Patients may experience tachycardia, tremor, anxiety, headache, nausea, and vomiting. This syndrome may be related to downregulation of the α_2-adrenergic receptors in the CNS in the setting of long-term use of these agents. These agents have a higher specificity for the I_1 receptor and appear to produce significantly fewer CNS effects, such as dry mouth and drowsiness. Rebound hypertension secondary to abrupt withdrawal has not been associated with moxonidine or rilmenidine.

CENTRAL AND PERIPHERAL ADRENERGIC-NEURONAL BLOCKING AGENT

MECHANISMS OF ACTION AND CLASS MEMBER

Reserpine, a *Rauwolfia* alkaloid, reduces BP by decreasing the activity of central and peripheral noradrenergic neurons. Reserpine blocks norepinephrine and dopamine uptake into the storage granules of noradrenergic neurons. The result is norepinephrine depletion. A similar effect is seen in central dopaminergic and serotoninergic neurons. At the dosages currently used to treat hypertension, the major effect of the use of reserpine is in the CNS. Reserpine results in a rapid reduction in cardiac output, heart rate, and peripheral vascular resistance. Enhanced vagal activity may also be involved. Tolerance to the antihypertensive effects of reserpine does not occur.

Reserpine is used at initial dosages of 0.1 to 0.25 mg daily.[448] Approximately 40% of an oral dose is absorbed. The half-life is 50 to 100 hours. Extensive hepatic metabolism occurs; 1% is recovered as unchanged compound in the urine. The maximal clinical effect is observed 2 to 3 weeks after initiation of therapy. No dosage adjustment is necessary for patients with kidney insufficiency. Dosage supplementation is not required after hemodialysis.

KIDNEY EFFECTS

The GFR and kidney plasma flow are not affected by reserpine therapy. Kidney vascular resistance may be reduced, perhaps mediated by decreased sympathetic stimulation of vascular α-adrenergic receptors. Significant effects on the RAAS have not been observed. Kidney handling of sodium and potassium is unchanged.

EFFICACY AND SAFETY

Reserpine provides effective therapy as a single agent or in combination with HCTZ or HCTZ and hydralazine.[448,449] This has been observed in numerous large and small trials, including the Veterans Administration Cooperative Study on Antihypertensive Agents, Hypertension Detection and Follow-up Program, and the Multiple Risk Factor Intervention Trial. Reserpine used in combination with a diuretic has shown comparable efficacy to combinations of β-blockers and diuretics. In these studies, the dose of reserpine was between 0.1 and 0.3 mg daily, which is many times lower than the doses used in the 1960s that led to reserpine's reputation as having a poor side-effect profile.

The most common AE of reserpine is nasal congestion, which is reported in 6% to 20% of patients. Unlike other AEs, nasal congestion does not appear to decrease at lower drug dosages and is thought to be related to the cholinergic effects of the drug. Increased gastric motility and gastric acid secretion can occur; however, the incidence of dyspepsia or peptic ulcer disease with reserpine therapy is not greater than that with other antihypertensive drug treatments. Inability to concentrate, sedation, sleep disturbance, and depression have been reported. Other AEs include weight gain, increased appetite, and sexual dysfunction.

DIRECT-ACTING VASODILATORS

CLASS MECHANISMS OF ACTION

Direct-acting vasodilators reduce BP by decreasing peripheral vascular resistance. These drugs act directly on vascular smooth muscle with the selective vasodilation of the arteriolar resistance vessels and have little or no effect on the venous capacitance vessels.[450] There is no effect on the functioning of carotid or aortic baroreceptors. The vasodilating effects are thought to involve inhibition of calcium uptake into the cells. Decreases in arterial pressure are associated with a decrease in peripheral resistance and a reflex increase in cardiac output. Sodium and water retention are promoted secondary to the stimulation of renin release and possibly by direct effects on kidney tubules. The arteriolar dilation produced by these drugs causes a decrease in cardiac afterload,[450] and the absence of venodilation leads to an increase in venous return to the heart, which produces an elevated preload. These combined effects result in increased cardiac output.[450] The pharmacokinetic and pharmacodynamic properties of these drugs are shown in Tables 48.21 and 48.22.

CLASS MEMBERS

The initial oral dose of hydralazine for hypertension should be 10 mg four times daily, increasing to 50 mg four times daily over several weeks. Patients may require doses of up to 300 mg/day. Dosing can be changed to twice daily for maintenance. The drug may also be used as an intravenous bolus injection or a continuous infusion. The elimination half-life is 1.5 to 8 hours and varies with the acetylation rate in the liver. Slow and fast acetylators have been described. The onset of action is approximately 1 hour. In patients with mild-to-moderate CKD, the dosing interval should be increased to every 8 hours. In patients with advanced CKD, the dosing interval should be increased to every 8 to 24 hours. No dose supplement is required after hemodialysis or peritoneal dialysis (see Table 48.6).

Minoxidil is more potent than hydralazine. For severe hypertension, the initial recommended starting dose is 2.5 mg as a single daily dose, increasing to 10 to 20 or 40 mg in single or divided doses. Minoxidil is usually used in conjunction with salt restriction and diuretics to prevent fluid retention. Concomitant therapy with a β-adrenergic blocking agent is often required to control tachycardia related to minoxidil use. The onset of the antihypertensive effect is within 30 to 60 minutes. The peak response occurs at 4 to 8 hours. The drug is 90% metabolized by the liver. The glucuronide metabolite has reduced pharmacologic effects but accumulates in patients with ESKD. Kidney excretion is 90%. Dose adjustments may be required for patients with advanced CKD, although the mean daily doses required to control BP are similar in patients with normal and impaired kidney function (see Table 48.6).

CLASS KIDNEY EFFECTS

Hydralazine and minoxidil both increase the juxtaglomerular cell secretion of renin, which is associated with increased Ang II and aldosterone levels. Long-term use is associated with the return of plasma aldosterone levels to baseline. Retention of salt and water may be attributed to direct drug effects on the proximal convoluted tubule. Kidney vascular resistance is decreased in association with

Table 48.21 Pharmacokinetic Properties of Direct-Acting Vasodilators

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Hydralazine (Apresoline)	20–50	No	1–2	1.5–8	L	U (3%–14%) F (3%–12%)	—
Minoxidil (Loniten)	90–100	—	1	4.2	L	U (90%) F (3%)	Glucuronide

F, Feces; *L,* liver; *U,* urine.

Table 48.22 Pharmacodynamic Properties of Direct-Acting Vasodilators

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Hydralazine	10	200–400	400	bid, qid	1	3–8
Minoxidil	2.5	10–20	40	qd, qid	4–8	10–12

bid, Two times a day; *qd,* once a day; *qid,* four times a day.

a relaxation of resistance vessels.[450] GFR and kidney plasma flow are preserved.

CLASS EFFICACY AND SAFETY

Hydralazine is commonly used in practice to reduce afterload in heart failure among patients with reduced left ventricular function. Hydralazine has been frequently used to treat pregnancy-associated hypertension in view of its relatively low teratogenicity. Its use can cause adrenergic activation and fluid retention.

Treatment with hydralazine has been associated with the development of drug-induced autoimmunity, including systemic lupus erythematosus and antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis. Hydralazine-induced lupus generally occurs early in therapy and can occur in 6% to 10% of patients receiving high doses of hydralazine; kidney involvement is rare.[451] It is seen most frequently in women and rarely in African Americans. Hydralazine-induced lupus is reversible when hydralazine is discontinued, but months may be required for complete clearing of symptoms.[452] Hydralazine-induced ANCA-associated vasculitis generally occurs with longer duration of treatment and involves high titers of myeloperoxidase-specific (MPO) ANCA antibodies. It manifests as rapidly progressive glomerulonephritis (RPGN). Treatment often involves not only cessation of hydralazine but also immunosuppression and, in some cases, plasmapheresis. In one case series[452] of 51 patients with hydralazine-induced ANCA GN and follow-up data available, 37% of patients reached the combined endpoint of ESKD or death, while the remaining 63% had favorable outcomes with mean serum creatinine at end of follow-up of 1.49 mg/dL.

Minoxidil is commonly reserved for severe or intractable hypertension.[453,454] Minoxidil is frequently a last-resort therapy in patients with CKD unresponsive to other therapies. It must nearly always be used in combination with a β-blocker and a loop diuretic to prevent tachycardia and fluid retention. Hypertrichosis is a common AE. Pericarditis and pericardial effusions have been described.[455] An increase in left ventricular mass has been reported, which may be related to adrenergic hyperactivity. Oral minoxidil has gained renewed popularity as an off-label treatment for androgenetic hair loss in women and men.[456]

MODERATELY SELECTIVE PERIPHERAL α_1-ADRENERGIC ANTAGONISTS

CLASS MECHANISMS OF ACTION

The nonselective agents phentolamine and phenoxybenzamine have an occasional role in hypertension management. Phentolamine is administered parenterally, and the longer-acting agent phenoxybenzamine has been used orally for the management of hypertension associated with pheochromocytoma.[457] Phenoxybenzamine is a moderately selective, peripheral α_1-adrenergic antagonist. Its specificity for the α_1-adrenergic receptor is 100 times greater than that for the α_2-adrenergic receptor.

CLASS MEMBERS

Phenoxybenzamine is a long-acting α-adrenergic blocking agent. This agent irreversibly and covalently binds to α receptors only. β Receptors and the parasympathetic system are not affected by phenoxybenzamine. The total peripheral resistance is decreased, and cardiac output increases with phenoxybenzamine. Phenoxybenzamine is also believed to inhibit the uptake of catecholamines into adrenergic nerve terminals and extraneural tissues. The usual oral dose of phenoxybenzamine for the treatment of pheochromocytoma is initially 10 mg twice daily, with the dose gradually increased every other day to dosages ranging between 20 and 40 mg two or three times daily. The final dosage should be determined by the BP response. Phenoxybenzamine may be administered with a β-blocking agent if tachycardia becomes excessive during therapy. The pressor effects of a pheochromocytoma must be controlled by α-blockade before β-blockers are initiated. With oral use, the pheochromocytoma symptoms decrease after several days. The oral bioavailability is 20% to 30%. The drug is extensively metabolized by the liver. Phenoxybenzamine should be administered cautiously to patients with kidney impairment. Specific dosage recommendations are not available.

Phentolamine is an α-adrenergic blocking agent that produces peripheral vasodilation and cardiac stimulation, with a resulting decrease in BP in most patients. The drug is used parenterally. The usual dose is 5 mg, repeated as needed. The onset of activity with intravenous dosing is immediate. The drug is not absorbed well orally; its half-life is 19 minutes. Phentolamine is metabolized by the liver, with 10% excreted in the urine as unchanged drug.

CLASS KIDNEY EFFECTS

Phenoxybenzamine has no clear effect on RAAS. Blood volume and body weight are not altered. Salt and water retention do not occur. GFR and effective kidney plasma flow would be expected to increase. Kidney vascular resistance probably decreases in proportion to the degree of blockade of α-adrenergic receptors.

CLASS EFFICACY AND SAFETY

Phenoxybenzamine is used primarily as an agent to counteract the excessive α-adrenergic tone associated with pheochromocytoma. Tachycardia may result from an α-adrenergic blockade, which unmasks β-adrenergic effects with epinephrine-secreting tumors. This may be controlled with concurrent use of a β-adrenergic antagonist. α-Adrenergic blockade must be initiated before β-adrenergic blockade to avoid paradoxic hypertension. AEs of phenoxybenzamine are sedation, weakness, nasal congestion, hypertension, and tachycardia.

PERIPHERAL α_1-ADRENERGIC ANTAGONISTS

CLASS MECHANISMS OF ACTION

Drugs of the peripheral α_1-adrenergic antagonist class, including doxazosin, prazosin, and terazosin, are selective antagonists of the postsynaptic α_1-adrenergic receptor. These drugs, which blunt the increases in arteriolar and venous tone mediated by norepinephrine released from sympathetic nerve terminals, act at the α_1-adrenergic receptor located postjunctionally in the blood vessel wall. The affinity of these drugs for the α_2 receptor is low. Because of the selective α_1 action, there is no interference with the negative feedback control mechanisms that are mediated by the prejunctional α_2 receptors. As a result, the reflex tachycardia associated with the blockade of the presynaptic α_2 receptor decreases substantially. The pharmacokinetic and pharmacodynamic properties of these drugs are shown in Tables 48.23 and 48.24.

CLASS MEMBERS

Doxazosin is a selective long-acting α_1-adrenergic antagonist. The initial antihypertensive dosage is 1 mg daily. This dose can be titrated up to a maximum of 16 mg daily. The maximal antihypertensive effect is seen 4 to 8 hours after a single dose. The drug is highly plasma protein bound and extensively metabolized. Most of the administered dose is excreted in the feces. The estimated half-life ranges from 9 to 22 hours. Doxazosin pharmacokinetics are not altered in patients with impaired kidney function. The drug should be used with caution in patients with advanced liver dysfunction.

Prazosin is a selective α_1-adrenergic antagonist structurally related to doxazosin. Oral dosing is 3 to 20 mg total daily in divided doses due to its relatively shorter half-life. Full therapeutic effects are seen within 4 to 8 weeks after initiation of therapy. Peak serum levels are reached 1 to 3 hours after an oral dose. The drug is highly protein bound with an elimination half-life of 2 to 4 hours. There is extensive hepatic metabolism followed by kidney excretion of a small amount of unchanged drug. Dose adjustment is not required for patients with CKD. Patients with significant liver disease may require dose adjustment and more frequent monitoring.

Terazosin is a selective, long-acting α_1-adrenergic antagonist that has structural similarities to prazosin and doxazosin. The initial dosage is 1 mg orally at bedtime, with titration to 5 mg nightly. Doses of 10 to 20 mg orally have been given. Peak serum levels after oral administration occur within 1 hour. The half-life is approximately 12 hours. Terazosin is extensively metabolized in the liver and eliminated primarily through the biliary tract. Pharmacokinetics are not affected by CKD, and dose adjustment is not required. Patients with severe hepatic insufficiency may require dose adjustments.

CLASS KIDNEY EFFECTS

GFR and kidney blood flow are maintained during long-term treatment. In some studies, there was a slight increase in kidney blood flow. Kidney vascular resistance may be reduced, perhaps mediated by a reduction in preglomerular capillary resistance related to inhibition of α_1-mediated vasoconstriction. Urinary protein excretion has been reported to be reduced. The RAAS is not significantly affected by specific α_1-adrenergic antagonists. Extracellular fluid volume has been reported to be increased, and fractional excretion may be decreased.

CLASS EFFICACY AND SAFETY

Comparative clinical studies of the efficacy of α_1-adrenergic blockers have shown that the antihypertensive responses are similar to those elicited by other antihypertensive drugs.[61] The availability of ER formulations of these drugs has improved tolerability. These drugs have been shown to increase insulin sensitivity[458] to exert potentially beneficial effects on lipid metabolism.[459-463] They can cause a modest reduction in total and LDL cholesterol and a small increase in HDL cholesterol levels. This metabolic benefit may be linked to the beneficial effect on insulin responsiveness, leading to increased peripheral glucose uptake.

There are also potential AEs of α_1-adrenergic receptor blockers. In the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT),

Table 48.23 Pharmacokinetic Properties of Peripheral α_1-Adrenergic Antagonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Doxazosin (Cardura)	62–69	No	2–5	9–22	L	F (63%–65%) U (1%–9%)	—
Prazosin (Minipress)	—	No	1–3	2–4	L	F	—
Terazosin (Hytrin)	90	Yes	1	12	L	F (45%–60%) U (10%)	—

F, Feces; *L,* liver; *U,* urine.

Table 48.24 Pharmacodynamic Properties of Peripheral α_1-Adrenergic Antagonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Doxazosin	1	8	16	qd, qid	4–8	24
Prazosin	1	3–20	20	bid, qid	0.5–1.5	10
Terazosin	1	5	20	qd, bid	3	24

bid, Two times a day; *qd,* once a day; *qid,* four times a day.

patients randomized to receive doxazosin as their initial antihypertensive drug were found to have poorer BP control than those receiving a chlorthalidone-based treatment.[464] No difference was seen in the primary outcomes of fatal coronary heart disease or nonfatal MI in patients, but patients randomized to doxazosin had higher rates of stroke and congestive heart failure.[465] Data from ALLHAT have largely relegated doxazosin and other peripheral α_1-adrenergic antagonists to second- or third-line agents in the management of hypertension.

α_1-Adrenergic receptor blockers can also cause a first-dose orthostatic hypotension effect, resulting in lightheadedness, palpitations, and occasionally syncope, related to the drugs' effect on the venous capacitance vessels, which results in venous dilation and inadequate venous return. It may occur when peak drug levels are reached 30 to 90 minutes after the first dose; it can be minimized by initiating therapy with a small dose taken at bedtime.[466] This effect can be exacerbated in patients with underlying autonomic insufficiency.

α_1-Adrenergic antagonists are also used for symptomatic management of prostatic hyperplasia. Prostatic smooth muscle has significant α_1-adrenal receptor expression. Blockade of these receptors results in smooth muscle relaxation within the prostate.[467,468] Terazosin has been marketed in combination with amlodipine for male patients with hypertension and lower urinary tract symptoms.[469]

DIRECT RENIN INHIBITORS

CLASS MECHANISM OF ACTION AND CLASS MEMBER

Aliskiren is an oral nonpeptide, low-molecular-weight renin inhibitor used for the management of hypertension (see Fig. 48.1).[470,471] Renin inhibition interferes with the first and rate-limiting step in the renin enzyme cascade, the interaction of renin with its substrate angiotensinogen. The renin blockade step is an attractive target for hypertension therapeutics, in large part because of the remarkable specificity of renin for its substrate.[472] This specificity reduces the likelihood of unwanted interactions and possible AEs. In addition, unlike ACE inhibitors or ARBs, which lead to a reactive increase in renin and associated angiotensin peptides, direct renin inhibition renders the RAAS quiescent. Although aliskiren has low inherent bioavailability (2.6%), it is potent in reducing BP and has an effective half-life of 40 hours.[473] The drug is not actively metabolized by the liver and is primarily excreted in the urine, with most of it as unchanged drug. CYP enzymes are not involved in the metabolism of aliskiren. Thus clinically significant CYP inhibition by aliskiren is unlikely.[474]

CLASS KIDNEY EFFECTS

Clinical trial data in humans indicated that aliskiren reduces proteinuria in conjunction with decreasing BP.[472] In a study involving 600 patients with hypertension, diabetes, and nephropathy, the administration of aliskiren, 300 mg/day, facilitated a 20% incremental reduction in proteinuria compared with placebo in patients being treated concurrently with losartan, 100 mg/day. The baseline BP was 135 mm Hg with losartan treatment before random assignment to aliskiren therapy and did not change with the addition of aliskiren.[475] Kidney vascular response curves for ACE inhibition or renin inhibition at the top of the dose-response curve indicated greater increase in kidney blood flow with the renin inhibitor, despite similar changes in BP.[472] However, despite these encouraging clinical observations, a subsequent clinical trial, ALTITUDE, did not demonstrate an incremental benefit of aliskiren when used with an ACE inhibitor or ARB in patients with diabetic kidney disease when examining cardiovascular and kidney endpoints.[184] From a safety standpoint, there were more AEs in the patients receiving both classes of RAAS-blocking drugs.

CLASS EFFICACY AND SAFETY

Clinical trials of aliskiren have demonstrated a dose-dependent efficacy in reducing BP. In an 8-week, double-blind, placebo-controlled trial, aliskiren was studied in dosages from 150 to 600 mg/day.[476] The placebo-corrected reduction in sitting SBP was approximately 10 to 11 mm Hg for both the 300- and 600-mg doses. Also evaluated in the same study was irbesartan, an ARB, at a dosage of 150 mg/day, which effectuated BP reduction comparable with that of aliskiren at a dosage of 150 mg/day. Other clinical studies have confirmed the antihypertensive efficacy of aliskiren compared with placebo or with other active therapies, such as ARB losartan. As expected, BP reduction was comparable with that produced by the active agents. The only observed difference was suppression of plasma renin activity and Ang I and Ang II levels with aliskiren but increased levels of Ang I and Ang II with ARB therapy.

The tolerability of aliskiren is comparable with that of placebo, with a relatively low incidence of AEs at all dosages tested in the 150- to 600-mg range.[476] Aliskiren treatment was comparable in tolerability to treatment with an ARB or placebo, with a low discontinuation rate and no statistical difference in the incidence of different types of AEs, except for some diarrhea at the 600-mg dose.

Aliskiren has also been studied in combination with HCTZ, CCBs, ARBs, and ACE inhibitors in the management of hypertension.[477-480] In one small clinical trial, all patients received aliskiren 150 mg once daily for 3 weeks. Patients who had a daytime BP above 130/80 mm Hg, as measured by ambulatory BP monitoring, were given HCTZ 25 mg daily, for an additional 3 weeks, resulting in an additional 10 mm Hg of SBP reduction. Not surprisingly, there was no difference in plasma renin activity between the patients receiving aliskiren with HCTZ and those receiving aliskiren alone. Adding aliskiren to amlodipine, valsartan, or ramipril provides incremental and statistically significant improvements in BP reduction. However, the ALTITUDE study raised safety concerns about using aliskiren with other RAAS-blocking drugs due specifically to declining kidney function and hyperkalemia.[183] The additional BP reduction may or may not be related to the neutralization of plasma renin activity that occurs with the addition of the renin inhibitor. Aliskiren use has been associated with the development of angioedema and is generally contraindicated among patients with a history of angioedema with other RAAS inhibitor use.[481]

SELECTIVE ALDOSTERONE RECEPTOR ANTAGONISTS

CLASS MECHANISM AND CLASS MEMBERS

Spironolactone and eplerenone are members of the class of MRAs. The mineralocorticoid receptor forms part of the steroid–thyroid–retinoid–orphan receptor family of nuclear transactivating factors.[482] When unbound, these receptors are in an inactive multiprotein complex of chaperones. On binding of aldosterone, the chaperones are released and the receptor hormone complex is translocated into the nucleus, where it binds to hormone response elements on DNA and interacts with transcription initiation complexes, which ultimately modulate gene expression.[483] In the kidney, mineralocorticoid receptors are located primarily in the epithelial cells of the distal nephron. These receptors bind physiologic glucocorticoids and mineralocorticoids with a similar affinity. Activation of mineralocorticoid receptors by aldosterone results in the activation of epithelial sodium channels, which leads to a rapid increase in sodium and water reabsorption and promotes the tubular secretion of potassium.[484,485] A persistent increase in sodium balance does not occur, even with continued stimulation of mineralocorticoid receptors by aldosterone. The mechanism of the aldosterone escape phenomenon has not been fully elucidated.

There is evidence indicating the presence of biologic activity of mineralocorticoid receptors in nonepithelial tissues.[486] These receptors have been identified in blood vessels of the heart and brain and may be involved in vascular injury and repair responses.[435,487] Aldosterone mediates fibrosis and collagen formation through the upregulation of Ang II receptor responsiveness.[434,486] Aldosterone increases sodium influx in vascular smooth muscle and inhibits norepinephrine uptake in vascular smooth muscle and myocardial cells.[488] Aldosterone also directly participates in vascular smooth muscle cell hypertrophy.

Spironolactone and eplerenone are both inhibitors of the mineralocorticoid receptor. Spironolactone is a prodrug that is metabolized in the liver and excreted in the urine. The half-life of the prodrug is 1.3 to 2 hours, whereas the half-life of the active metabolites ranges from 14 to 17 hours.

Eplerenone is approximately 24 times less potent than spironolactone but is more specific for blocking mineralocorticoid receptors with little agonist activity at estrogen and progesterone receptors.[489] Therefore eplerenone is associated with a lower incidence of gynecomastia, breast pain, and impotence in men and diminished libido and menstrual irregularities in women. The time to peak concentration is 1 to 2 hours. No significant accumulation occurs with multiple-dose administration, but up to 4 weeks may be required for the full antihypertensive effects to be evident. It appears to be well absorbed, but absolute (oral vs. intravenous) data are unavailable; specific data on protein binding and metabolism are also unavailable. The elimination half-life of eplerenone is 4 to 6 hours. Bioavailability is 69%, with approximately 50% protein binding. It is metabolized primarily by the hepatic CYP3A4 system to inactive metabolites excreted two-thirds in the urine and one-third in the feces.

Finerenone is a newer nonsteroidal MRA that delays kidney disease progression and reduces the risk of cardiovascular outcomes among patients with CKD and type 2 diabetes mellitus.[490] The effect of this agent on BP ranges from 2 to 3 mm Hg to 8 mm Hg compared with placebo, depending on the specific patient population.[491,492]

CLASS KIDNEY EFFECTS

Selective aldosterone receptor antagonism may have benefits for the kidney, independently of its effects on BP. Experimental and clinical studies have demonstrated that Ang II may be the primary mediator of the RAAS associated with the progression of kidney disease.[493,494] The relative importance of aldosterone in this cascade has been the subject of experimental and clinical studies. Hyperaldosteronism and adrenal hypertrophy are common observations in remnant kidney models and correlate with progressive loss of kidney function.[493] Hypertension, proteinuria, and structural injury are less prevalent in subtotally nephrectomized rats that have undergone adrenalectomy, even when administered large doses of replacement glucocorticoids.[495] Other investigators have demonstrated that aldosterone infusion can reverse the kidney protective effects of an ACE inhibitor in stroke-prone, spontaneously hypertensive rats.[496] Interestingly, in this model, the kidney injury induced by aldosterone was independent of increases in BP, suggesting a toxic tissue effect of aldosterone. Other experimental studies have indicated that aldosterone receptor antagonism can prevent the development of proteinuria.[496]

Even though selective aldosterone receptor antagonists have no observable effects on glomerular hemodynamics, therapy with these drugs may provide an incremental effect in protecting the kidney when added to ACE inhibitors or ARBs by inhibiting the effects of aldosterone that persist, despite treatment with the latter drugs. Studies in patients with diabetic nephropathy or other glomerular diseases have demonstrated that the addition of spironolactone to ACE inhibition markedly reduces albuminuria.[497,498]

CLASS EFFICACY AND SAFETY

Spironolactone and eplerenone can be effective for BP control, particularly in patients with resistant hypertension, while finereone's use in clinical practice is primarily related to reducing the risk of CKD progression and CV events in patients with diabetic kidney disease. In the PATHWAY-2 trial, a double-blind, placebo-controlled, crossover trial, 285 patients with resistant hypertension received spironolactone, doxazosin, bisoprolol, or placebo as the fourth agent.[499] During spironolactone therapy versus placebo, the participants' average home systolic BP decreased by 8.7 mm Hg (95% CI, 7.7–9.7 $P < 0.001$). Spironolactone was also superior to doxazosin and bisoprolol. The drug was generally well tolerated; the rates of discontinuations due to kidney impairment, hyperkalemia, and gynecomastia were not increased with spironolactone relative to other treatments and placebo. Collectively, these data suggest that spironolactone at 25 to 50 mg/day is an effective treatment option for individuals with resistant hypertension (uncontrolled on three antihypertensive medications including a diuretic or four or more antihypertensive medications independent of class).

Eplerenone is less potent than spironolactone and has a shorter half-life (3–4 hours), leading to reduced antihypertensive efficacy and a requirement for twice-daily dosing.[500] An open-label prospective study of 52 patients demonstrated the efficacy of the selective aldosterone blocker eplerenone at a dose of 50 to 100 mg daily in patients with resistant hypertension. After eplerenone treatment, the clinical BP was reduced by 18/8 mm Hg compared with baseline ($P < 0.0001$ for SBPs and DBPs), though there was no placebo control arm in this study.[501]

A meta-analysis of RCTs showed a positive effect of MRAs on changes in the cardiac structure and left ventricular function in patients with heart failure.[502] The role of MRAs in treating hypertensive obese patients with sleep apnea has been reported, representing approximately 20% of all individuals with sleep apnea. Adipocytes are directly responsible for some of the production of excess aldosterone.[503-506]

When using MRAs, patients must be monitored for hyperkalemia, particularly among those persons with CKD or if other RAAS inhibitors are also being used. Conventional MRAs are not recommended for use among patients with eGFR lower than 30 mL/min/1.73 m^2; hyperkalemia may be somewhat less common with finerenone than with eplerenone and particularly spironolactone, although finerenone, too, should be used with caution in patients with low or declining GFR.

TYROSINE HYDROXYLASE INHIBITOR

MECHANISMS OF ACTION

Metyrosine, the only drug in the tyrosine hydroxylase inhibitor class, blocks the rate-limiting step in the biosynthetic pathway of catecholamines. Metyrosine inhibits tyrosine hydroxylase, the enzyme responsible for the conversion of tyrosine to dihydroxyphenylalanine. This inhibition results in decreased levels of endogenous catecholamines. In patients with pheochromocytomas, metyrosine reduces catecholamine biosynthesis by up to 80%, resulting in a decrease in total peripheral vascular resistance. Heart rate and cardiac output increase because of vasodilation. It is used primarily for the management of pheochromocytoma.

CLASS MEMBER

The recommended initial dosage of metyrosine is 250 mg four times a day orally. The dose may be increased by 250 to 500 mg every day until a maximum of 4 g/day is given. Following oral absorption, metyrosine is eliminated primarily unchanged in the urine. The half-life is 7.2 hours. Dose reduction is appropriate in patients with CKD.

KIDNEY EFFECTS

Little information is available on the kidney effects of metyrosine. On the basis of its mechanism of action, which would counteract the kidney effects of excessive circulating catecholamines, kidney plasma flow, and glomerular filtration would be expected to increase and kidney vascular resistance would be expected to decrease.

EFFICACY AND SAFETY

Metyrosine is used in the preoperative or intraoperative management of pheochromocytoma. Hypertension and reflex tachycardia may result from vasodilation. These effects can be minimized by volume expansion. AEs include sedation, changes in sleep patterns, and extrapyramidal signs. Metyrosine crystals have been noted in the urine of patients receiving high dosages. Patients taking metyrosine should maintain a generous fluid intake; some patients have occasionally experienced diarrhea.

ENDOTHELIN RECEPTOR ANTAGONISTS

Endothelin is among the most potent endogenous vasoconstrictors known.[507] Endothelin also enhances mitogenesis and induces extracellular matrix formation. Endothelin is also thought to be involved in vascular remodeling and end-organ damage under several different cardiovascular conditions.[508] As a result, endothelin receptor antagonists have been studied as targets for treatment of hypertension.

CLASS MECHANISM OF ACTION AND CLASS MEMBERS

The two primary receptor sites, endothelin type A (ET-A) and endothelin type B (ET-B), can be selectively blocked with different chemicals, or both sites can be blocked simultaneously. Bosentan,[509] a mixed ET-A/ET-B receptor antagonist reduces BP but concerns about its safety and tolerability have heretofore limited its use largely to treatment of pulmonary hypertension. More recently, a multicenter randomized phase 3 study of the dual endothelin antagonist aprocitentan demonstrated dose-dependent BP lowering among patients with resistant hypertension compared with placebo.[510] However, there was more edema and there were more frequent fluid retention AEs in the aprocitentan group. Aprocitentan was approved by the U.S. FDA to treat resistant hypertension on March 19, 2024, marking the first new antihypertensive medication with a novel mechanism to be approved since aliskiren in 2007.

CLASS RENAL EFFECTS

Endothelin receptor antagonists have been used to study the role of endothelin in the development of acute CKD in different experimental models.[511] Clinical studies using endothelin receptor blockers have shown reductions in proteinuria. This effect is independent of the antiproteinuric effects of inhibition of the RAAS and may delay CKD progression.[512] What is unknown is whether the selective blockage of the ET-A receptor over the ET-B receptor will not only control BP but also mitigate kidney ischemia and reduce proteinuria.

NOVEL ANTIHYPERTENSIVE CLASSES

Several novel classes of antihypertensive medications are undergoing investigation. In a phase 1 study, zilebesiran, an investigational RNA interference therapeutic agent that reduces the projection of angiotensinogen by the liver, demonstrated dose-dependent decreases in 24-hour ambulatory BP.[513] The decrease in BP was sustained for up to 24 weeks after a single subcutaneous dose. Two phase 2 studies are underway (clinicaltrials.gov number NCT04936035 and NCT05103332). Ocedurenone, a nonsteroidal MRA, had shown promise for treating resistant hypertension among patients with stage 3b/4 CKD in a phase 2[514] trial, but a phase 3 trial of this drug failed to meet its primary endpoint.[515]

Several aldosterone synthase inhibitors are under development. A 12-week phase 2 trial showed that baxdrostat reduced systolic BP by 11.0 mm Hg (CI 5.5–16.4 mm Hg) more than placebo among patients with treatment-resistant hypertension.[515-517] An 8-week trial of 200 participants with uncontrolled hypertension on two or more antihypertensive medications showed that the least-squares mean difference in SBP between placebo and the aldosterone synthase inhibitor lorundrostat 100 mg daily was −7.8 mm Hg (90% CI −14.1 to 1.5 mm Hg; $P = 0.04$).

SELECTION OF ANTIHYPERTENSIVE DRUG THERAPY

CATEGORIZATION OF BLOOD PRESSURE AND DIAGNOSIS OF HYPERTENSION

In observational studies, BP is associated with cardiovascular disease (CVD) and kidney disease in a direct, log-linear fashion.[518] At any level of BP, risk is greatly influenced by the presence or absence of other CVD risk factors.[519] In clinical practice, most patients are older adults and at increased risk for CVD.[520,521] For pragmatic purposes, BP can be categorized, with the term hypertension being applied to the category at which antihypertensive drug treatment should be added to lifestyle counseling. In contemporary guidelines, an average SBP exceeding either 130 or 140 mm Hg or DBP exceeding either 80 or 90 mm Hg, or treatment with antihypertensive medication are commonly used to identify hypertension.[522]

The 2017 ACC/American Heart Association (AHA) BP guideline recommends use of the BP categories displayed in Table 48.25.[523] Hypertension should be diagnosed in adults who report taking antihypertensive medication and in those with an average SBP of 130 to 139 mm Hg or DBP of 80 to 89 mm Hg (stage 1 hypertension) or an SBP of ≥140 mm Hg or DBP of ≥90 mm Hg (stage 2 hypertension).[523] The ACC/AHA decision to lower the SBP/DBP thresholds for diagnosis of hypertension from the previously recommended SBP and DBP cut points of 140 mm Hg and 90 mm Hg[524] was based on BP risk estimation and the benefits of antihypertensive drug therapy in previously untreated and treated adults.[525-527] A presumptive diagnosis of hypertension based on office (clinic) BP measurements should be confirmed by out-of-office measurements. Almost one-quarter of adults with a high office BP have a nonhypertensive level of BP outside the office (white coat hypertension). A similar percentage of adults have nonhypertensive BPs in the office but out-of-office BPs that meet the criteria for diagnosis of hypertension (masked hypertension) and have a CVD risk that is similar to their counterparts with sustained (in and out of office) hypertension. Masked hypertension is most common in those with elevated BP and evidence of target-organ damage.

ACCURATE MEASUREMENT OF BLOOD PRESSURE

BP measurements are generally obtained to facilitate diagnosis and management by estimating an individual's usual average level of BP. Several factors can predictably (systematically) influence the level of BP, and their effects should be minimized by use of a standardized measurement method. In addition, BP varies in an unpredictable (random) fashion within and between visits, which can be minimized by averaging readings at repeated visits. The practice of evidence-based medicine assumes use of standardized BP measurements for diagnosis and management of high BP similar to the methods used in the landmark antihypertensive RCT and cohort studies. Detailed recommendations for accurate estimation of average BP are provided in the ACC/AHA BP Guideline[523] and elsewhere, but the core elements are outlined in Fig. 48.2.[528] Common reasons for inaccurate office BP readings are not proceeding the measurement by a period of quiet rest, failure to use a clinically validated BP measurement device (device validation websites include the U.S. Blood Pressure Validated Device site: https://www.validatebp.org and the European STRIDE BP site: https://www.stridebp.org), improper BP cuff choice or positioning, failure to train and certify the personnel performing the BP measurement by using one of the many courses available online (e.g., https://campus.paho.org/en/node/29166), improper preparation and positioning of the patient, and failure to confirm an elevated BP by averaging two or more readings on two or more occasions. Inaccurate BP measurements are common in clinical practice and usually result in overestimation of SBP by 5 to 7 mm Hg,[529] with consequent overestimation of hypertension prevalence by about 10% to 15%.[530] However, BP is underestimated in some patients and the errors in measurement vary by level of BP, practice setting, and time.[529] The only way to avoid these errors is to use an accepted standardized approach to measurement of BP.

Table 48.25 Categories of Blood Pressure in Adults

BP category	Systolic BP, mm Hg		Diastolic BP, mm Hg
Normal	<120	and	<80
Elevated	120–129	and	<80
Stage 1 Hypertension	130–139	or	80–89
Stage 2 Hypertension	≥140	or	≥90

Individuals with SBP and DBP in 2 categories should be designated to the higher BP category.

BP, Blood pressure (based on an average of ≥2 readings obtained on ≥2 occasions); *DBP,* diastolic blood pressure; *SBP,* systolic blood pressure.

Reprinted with permission from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: A report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:e13–e115. © 2017 American Heart Association, Inc.

RECOMMENDATIONS FOR TREATMENT AND FOLLOW-UP IN ADULTS, BY CATEGORY OF BLOOD PRESSURE

The ACC/AHA BP Guideline recommendations for treatment and follow-up in adults, by category of BP, are displayed in Fig. 48.3. Adults with normal or elevated BP should be encouraged to adhere to healthy lifestyles that result in a reduced BP (heart-healthy diet, weight loss, reduced dietary sodium and enhanced dietary potassium

Step 1 Facility and equipment	• Quiet room with a comfortable temperature. • Clinically validated BP measurement device; an automated device measuring BP at the brachial artery is recommended. • A range of cuff sizes to fit a range of upper-arm circumferences.
Step 2 Personnel performing BP measurement	• Trained health care professional should perform the BP measurement. Annual retraining is recommended.
Step 3 Prepare the patient	• The patient should be provided with instructions to abstain from caffeine, alcohol, nicotine, and exercise for at least 30 minutes prior to the BP measurement. • Eliminate discomfort such as a full bladder. • Prior to the BP measurement, there should be a short rest period (3–5 minutes) without provocation (including talking, or being talked to in-person or on the phone).
Step 4 The measurement procedure (see figure below)	• The health care professional should explain the procedure, including the number of BP measurements to be obtained. • Use the arm with the higher SBP readings during an initial visit, unless a new medical condition (e.g., arm ischemia) has developed in the interim in that arm. • ≥2 measurements should be obtained at least 30 seconds apart; the values should be averaged and recorded.

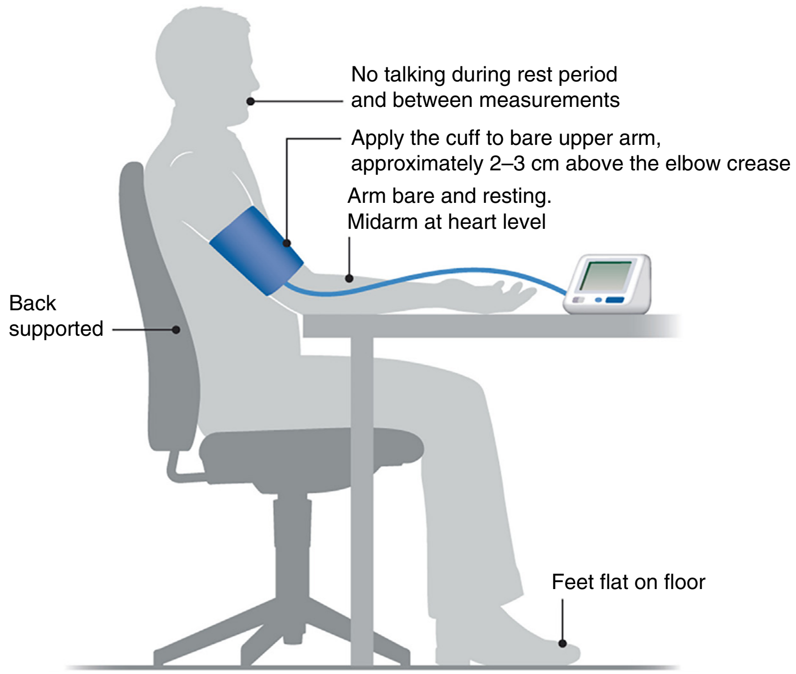

Fig. 48.2 Core elements of a standardized approach to office/clinic measurement of blood pressure. (Reproduced with permission from Cheung AK, Whelton PK, Muntner P, et al. *Am J Med.* 2023;136:438–455.)

intake, physical activity, and avoidance or moderation in alcohol consumption). Those with stage 2 hypertension should additionally be treated with antihypertensive medication. Most adults with stage 1 hypertension should be managed with lifestyle improvement, but antihypertensive medication should be added in those with CVD, or an apparently high risk of atherosclerotic CVD based on an ACC/AHA calculator estimate in adults 40 to 75 years. CVD risk is usually high in adults with hypertension who are ≥65 years or have diabetes mellitus or CKD,[521] and these phenotypic combinations can serve as a substitute indication for addition of antihypertensive medication.

SELECTION OF ANTIHYPERTENSIVE DRUG THERAPY

The primary purpose of treating hypertension is to prevent CVD, extend lifespan, and enhance quality of life. Provision of effective treatment is complicated by the asymptomatic and chronic nature of high BP, widespread inaccuracy of BP

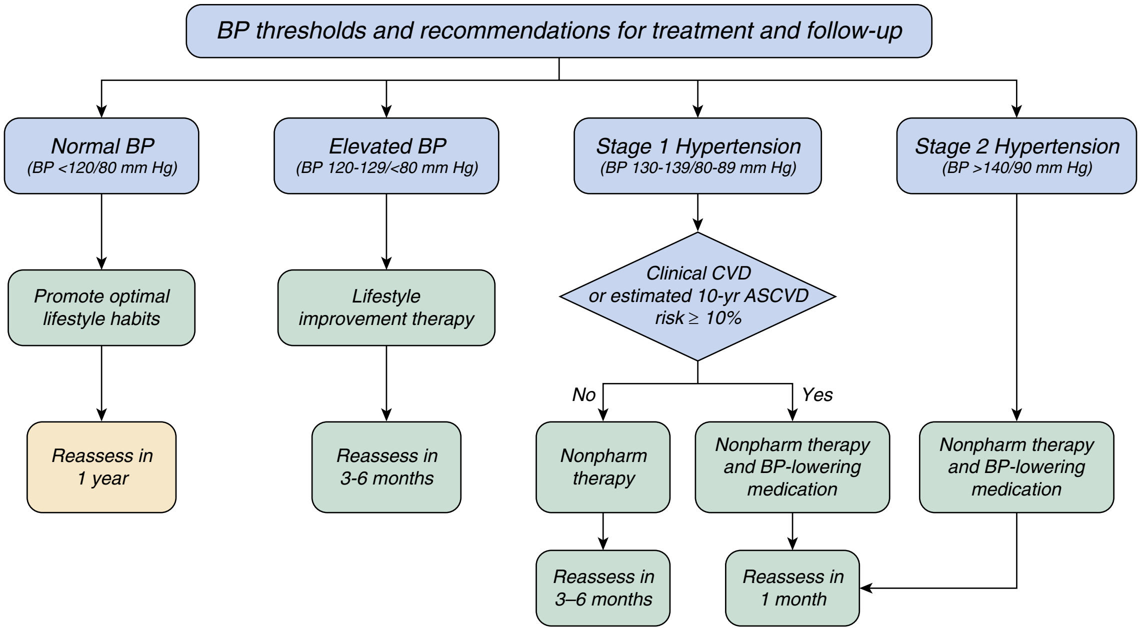

Fig. 48.3 American College of Cardiology/American Heart Association recommendations for treatment and frequency of follow-up in adults by category of blood pressure. (Reprinted with permission from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: A report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:e13–e115. © 2017 American Heart Association, Inc.)

measurement in clinical practice, and a tendency to undertreat patients (therapeutic inertia). When contemplating the addition of antihypertensive drug therapy to lifestyle counseling measures, the following considerations apply:

- Does antihypertensive drug therapy reduce the risk of CVD?
- Which antihypertensive drug classes should be used for treatment?
- Is BP lowering or choice of antihypertensive drug class more important?
- What are the best strategies for antihypertensive therapy implementation and adherence?
- What is the optimal BP target during antihypertensive drug therapy?

DOES ANTIHYPERTENSIVE DRUG THERAPY REDUCE THE RISK OF CVD?

Clinical trials have repeatedly demonstrated the efficacy of antihypertensive drug therapy for prevention of CVD in adults with hypertension who are at increased risk for CVD. In a meta-analysis of 123 RCT, antihypertensive drug therapy resulted in an average reduction of 20% for major CVD events, 27% for stroke, 28% for heart failure, 17% for coronary heart disease, and 13% for all-cause mortality[531] (Fig. 48.4). The treatment-related reduction in relative risk for CVD was similar in those with and without CVD at baseline and across five baseline categories of SBP from <130 to ≥160 mm Hg. There was no evidence for prevention of kidney disease, but the trials included in this meta-analysis excluded adults with CKD.

WHICH ANTIHYPERTENSIVE DRUGS SHOULD BE USED FOR TREATMENT?

The Antihypertensive and Lipid Lowering to Prevent Heart Attach Trial (ALLHAT) results provide the best head-to-head randomized comparison of first-step antihypertensive drug therapies prescribed at recommended doses. The trial failed to demonstrate superiority for the CCB amlodipine or ACE inhibitor lisinopril compared with the long-acting thiazide-type diuretic chlorthalidone for the trial's primary outcome (coronary heart disease), stroke, or death.[532] Chlorthalidone was superior for prevention of heart failure, especially compared with amlodipine. Meta-analyses have documented a reduction in CVD risk for agents from five antihypertensive drug classes (thiazides, β-blockers, CCB, ACE inhibitor, and ARB) compared with placebo.[533] In meta-analyses of head-to-head active treatment comparison that employed agents from the same five drug classes, β-blockers have been less effective for prevention of stroke compared with agents from the other four drug classes.[534,535] CCB were less effective for prevention of heart failure, especially compared with thiazides. Thiazides have generally been the best agents for "all-around" CVD prevention.

On the basis of ALLHAT results and drug comparison meta-analyses, the ACC/AHA BP guideline recommends use of thiazides, CCB, ACE inhibitor, and ARB for initial drug therapy of hypertension, with β-blockers reserved for those with a compelling indication for their use.[523] An overall approach to antihypertensive therapy in adults with uncomplicated hypertension is presented in Fig. 48.5. All adults with hypertension should be treated with nonpharmacologic therapy. Antihypertensive medication should also be employed in those with stage 2 hypertension or stage 1 hypertension when the risk for CVD is high (prior CVD, ACC/AHA risk calculator 10-year risk of ASCVD ≥10% in those 40–75 years, or the combination of hypertension with age ≥65 years, diabetes mellitus, or CKD).[521,523] Drug monotherapy may be sufficient in those with a starting SBP that is close to 130 mm Hg, but a combination of antihypertensive agents is usually necessary to achieve an average SBP/DBP <130/80 mm Hg. Logical combinations

	Studies	Intervention Events	Intervention Participants	Control Events	Control Participants	RR (95% CI) per 10 mm Hg reduction in systolic blood pressure
Major cardiovascular events	55	13,209	137,319	14,068	128,259	0.80 (0.77–0.83)
Coronary heart disease	56	4862	136,986	5301	128,548	0.83 (0.78–0.88)
Stroke	54	4635	136,682	5378	128,641	0.73 (0.68–0.77)
Heart failure	43	3284	115,411	3760	107,440	0.72 (0.67–0.78)
Renal failure	16	890	39,888	834	39,043	0.95 (0.84–1.07)
All-cause mortality	57	9775	138,298	9998	129,700	0.87 (0.84–0.91)

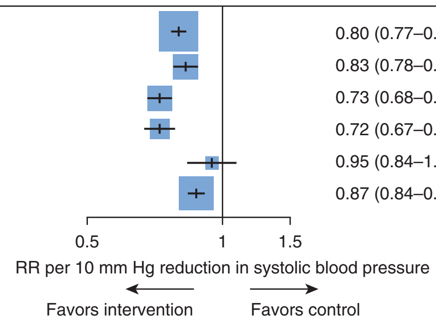

Fig. 48.4 Group meta-analysis of the efficacy of blood pressure lowering with antihypertensive medication for prevention of cardiovascular disease and all-cause mortality. The results are based on pooling of the findings in 123 randomized controlled trials, with an overall sample size of 613,815 persons, and have been standardized to depict the effect of a 10 mm Hg lowering of systolic blood pressure. (Reproduced with permission from Ettehad D, Emdin CA, Kiran A, et al. *Lancet.* 2016;387:957–967.)

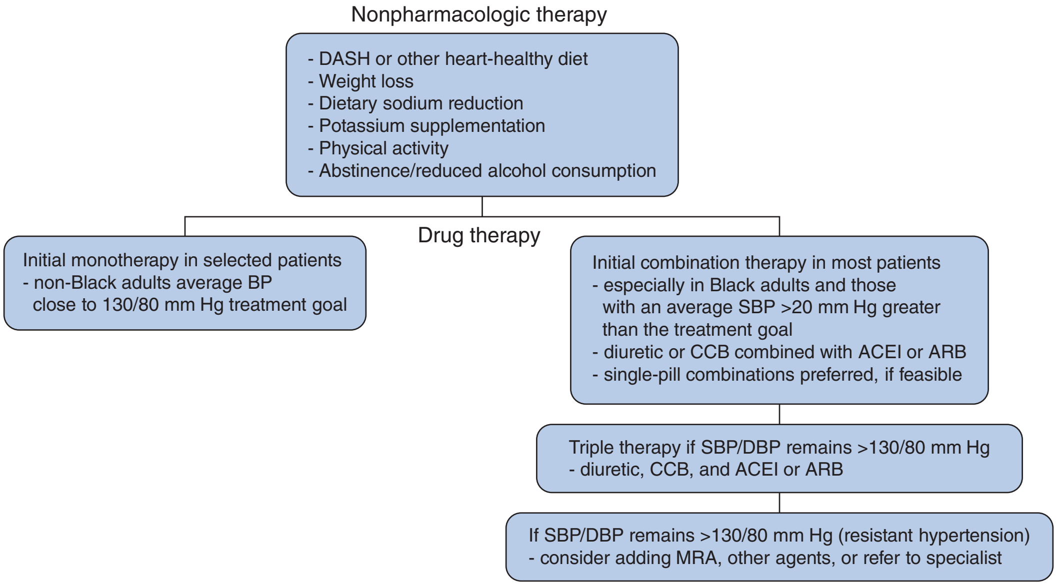

Fig. 48.5 Antihypertensive treatment in adults with uncomplicated hypertension. *ACEI,* Angiotensin-converting enzyme inhibitor; *ARB,* angiotensin receptor blocker; *BP,* blood pressure; *CCB,* calcium channel blocker; *DASH,* dietary approaches to stop hypertension; *DBP,* diastolic BP; *MRA,* mineralocorticoid receptor antagonists; *SBP,* systolic BP.

using agents with complementary mechanisms of action provide an effective treatment approach. Combinations of a thiazide or CCB with an ACE inhibitor or ARB are excellent choices for initial drug therapy, followed by use of the triple combination of a thiazide, CCB, and ACE inhibitor or ARB in those with an SBP/DBP that remains ≥130/80 mm Hg. In more complicated and difficult-to-manage patients, drug choice may require additional careful consideration and individualization or referral to a specialist.

The ACC/AHA BP guideline recommends chlorthalidone as the preferred diuretic for treatment of hypertension because it is longer acting, more potent for BP lowering compared with hydrochlorothiazide, and was the diuretic most used in the landmark antihypertensive drug trials. There was no difference in CVD events in a recent pragmatic RCT of hydrochlorothiazide versus chlorthalidone.[536] However, the results are difficult to interpret because the trial employed an open design and most of the trial participants were treated with low doses. A subsequent meta-analysis suggested chlorthalidone might be better for BP lowering but result in more hypokalemia (a finding in almost all of the studies) and provide no additional benefit related to MI, stroke, hospitalization for heart failure, or all-cause mortality.[537]

IS BP LOWERING OR CHOICE OF ANTIHYPERTENSIVE DRUG CLASS MORE IMPORTANT?

Apart from heart failure, CVD prevention in the ALLHAT was similar in those assigned to first-step drug therapy with thiazide, CCB, and ACE inhibitors. Likewise, meta-analyses generally suggest that most of the benefits derived from antihypertensive drug therapy with a diuretic, CCB, ACE inhibitor, or ARB result from BP lowering rather than a specific drug effect.

SINGLE-PILL COMBINATION THERAPY

Most antihypertensive drugs reduce SBP by about 8 to 10 mm Hg at full dosage.[538] Often, two to four once-daily drugs may be required to achieve BP control. In addition, other medications may be needed to control diabetes, dyslipidemia, angina, and/or other comorbidities. During the 1990s it was recognized that a low-dose thiazide or thiazide-type agent in combination with other antihypertensive drugs such as an ACE inhibitor or ARB nearly doubled the antihypertensive effect without adding toxicity to the regimen.[539] Fixed-dose combination therapy using a thiazide or CCB with an ACE inhibitor or ARB is well proven to be effective,[540,541] tends to reduce the incidence of side effects and is readily available in many countries. Most thiazide-containing fixed-dose combinations in the United States include hydrochlorothiazide but azilsartan, the most potent of drugs in the ARB class, is formulated in combination with chlorthalidone.[542] Triple-therapy fixed-dose combination tablets usually combine a CCB, ARB, and hydrochlorothiazide.[542,543] BP guidelines encourage the use of these formulations to enhance adherence. A list of commercially available fixed-dose formulations of antihypertensive agents in the United States, as of January 2023, was published by King and colleagues.[542] Only one triple-therapy and no quadruple fixed-dose combination agent was available in the United States, and merely 3.2% of the antihypertensive combinations used in Systolic Blood Pressure Intervention Trial (SPRINT) were available. At the time of this writing, three triple-therapy antihypertensive formulations were being marketed in the United States.

WHAT IS THE OPTIMAL BP TARGET DURING ANTIHYPERTENSIVE DRUG THERAPY?

Many RCTs have demonstrated better CVD outcomes in groups treated to a lower compared with a higher BP target, starting with the UKPDS trial reported in 1998[544] and being demonstrated most recently in the STEP, China Rural Hypertension Control Project (CRHCP), and the Effects of Intensive Systolic Blood Pressure Lowering Treatment in Reducing Risk of Vascular Events (ESPRIT) trials.[545-547] Thus the SPRINT is arguably the most influential and relevant of these trials, especially for U.S. clinicians.[526,527] The trial was stopped early due to a 25% lower risk of the primary CVD outcome (composite of MI, non-MI acute coronary syndrome, stroke, heart failure, CVD death) and a 27% lower risk of death due to any cause in the more intensive (SBP target <120 mm Hg) compared with the standard (SBP target <140 mm Hg) treatment group. Pooling of clinical trial results based on network,[548] individual participant data,[549] and group[535] meta-analyses have all documented benefits of more intensive antihypertensive therapy. The ACC/AHA guideline recommends an SBP/DBP treatment target of <130/80 mm Hg, including in community-dwelling older adults and patients with diabetes mellitus and/or CKD.[523] A meta-analysis confined to trials that randomized to different BP targets provided convincing evidence of CVD and all-cause mortality benefit in those randomized to an SBP <130 mm Hg versus ≥130 mm Hg. The meta-analysis also provided support for use of an SBP treatment target <120 mm Hg versus <140, but this was based on pooling information from only four trials.[550] Results from three ongoing SPRINT-like RCTs, two being conducted in patients with diabetics and one in stroke survivors, will help to clarify the value of an SBP <120 versus <140 mm Hg treatment target.[518]

Despite well-documented benefits of RCT, concerns have been raised related to more intensive antihypertensive therapy.[551] This reflects an understanding that there is some BP level at which organs will be underperfused but has primarily been based on reports of a J-shaped association between DBP and CVD, especially coronary heart disease[552] in some[553] but not all[554] observational analyses, including nonrandomized analyses of RCT data sets.[555] Some observers have drawn strong inferences based on J-curve reports, and some guidelines have recommended lower bounds for DBP during BP lowering based on the J-curve reports.[556] A central question is whether the J-curve phenomenon reflects a causal effect of lower DBP on CVD or results from CVD causing a low DBP. In SPRINT, baseline DBP had a U-shaped association with the hazard of the primary CVD outcome and all-cause mortality in both treatment groups, with a higher risk of CVD in those with the lowest DBP.[557] However, in randomized comparisons, the beneficial effect of the intensive SBP intervention on the primary outcome and all-cause mortality was not modified by baseline DBP. A causal mediation analysis of achieved low DBP in SPRINT also supported this observation.[558] Likewise, Mendelian randomization studies of antihypertensive therapy have failed to demonstrate a J- or U-shaped association between DBP and CVD.[555] Thus the ACC/AHA guideline SBP goal of <130 mm Hg is supported by the best available evidence.

The promptness of achieving BP control has long been considered beneficial because initial therapy is often materially unaltered in clinical practice due to therapeutic inertia.[559] RCTs and observational studies have indicated that initiating treatment with a two-drug antihypertensive combination results in more rapid achievement of target BP goals and better adherence compared with monotherapy.[560,561] Recognizing this, a 3-year event-based RCT that compared treatment based on a stepped-care approach with usual care yielded an SBP difference of 23.1 mm Hg and a 33% reduction in the primary outcome (CVD composite of MI, stroke, heart failure, and/or CVD death).[547] Thus while initial combination therapy is generally the preferred strategy, any strategy that leads to effective BP lowering is acceptable.

SPECIAL CONSIDERATIONS FOR CHOICE OF ANTIHYPERTENSIVE MEDICATIONS

TREATMENT OF HYPERTENSION IN OLDER ADULTS

Vascular stiffening is common in older adults and often leads to isolated systolic hypertension with wide pulse pressure.[562] Management of hypertension in older adults may require several antihypertensive agents, consideration of the metabolic and excretory routes of medications being considered, and attention to the potential for drug–drug interactions and the burden of following a complex regimen that includes taking multiple medications. Intensive antihypertensive therapy has been beneficial in RCT and can be safely achieved in most older adults who are ambulatory and free-living by treatment with an average of three antihypertensive agents,[545,563-565] including in those who are frail or have slow gait speed.[563,564] In addition to CVD and all-cause mortality benefits, intensive antihypertensive treatment in the SPRINT and CRHCP participants reduced the risk of cognitive impairment.[565,566] In those who are institutionalized, have many comorbidities, or have a relatively limited life expectancy, individualization of care is sensible.

Sitting and upright BP measurements should be obtained during medication initiation and titration. In the Systolic Hypertension in the Elderly Program (SHEP), approximately 10% experienced a decrease in SBP >20 mm Hg after standing for 1 minute, but only about half of them had persistent postural hypotension after 3 minutes.[567] Symptomatic hypotension, which is more important, is much less common. A meta-analysis of intensive treatment trials documented a reduction rather than worsening of postural hypotension.[568] Diuretics, CCBs, ACE inhibitors, or ARBs tend to vasodilate during long-term therapy and are good treatment choices in persons who are prone to postural hypotension. All these agents are better tolerated at lower doses and are quite effective in combination even in the presence of a high-salt diet, perhaps owing to their natriuretic effects.[569] Heart failure is increasingly common at older age, especially in the seventh and eighth decades of life, making inclusion of a diuretic especially desirable in older adults. Inclusion of an α-blocker may be useful in older men with benign prostatic hyperplasia because it facilitates prostatic urethral relaxation and improves urination. Older patients are also more likely than younger patients to have pseudohypertension[570] and hypertrophic cardiomyopathy[571] with impaired diastolic function, which may impair cardiac output.

TREATMENT BASED ON SEX

Women have a lower likelihood of coronary disease than men during the first half of life but after menopause or in the presence of diabetes risks are similar. Despite reported differences in the pathophysiologic mechanisms for high BP in men and women,[572] they benefit equally from the cardioprotective effects of antihypertensive therapy.[526,573] However, sex differences may help to guide the selection of antihypertensive therapy.[574] Women should avoid ACE inhibitors and ARB therapy during pregnancy because of the potential for teratogenic effects. In addition, CCB may delay labor. There is a well-established track record for safety using α-methyldopa, hydralazine, and β-blocker therapy in pregnancy. Women experience higher rates of osteoporosis than men, and thiazide agents have been shown to enhance mineral density,[575] presumably by reducing calciuria, and may reduce fracture risk.[576-578] Women report more cough with ACE inhibitor and more pedal edema with CCB than men.[579]

RACE AND ETHNICITY

Compared with other major race/ethnicity groups, African Americans tend to have a higher average BP and a higher prevalence of hypertension, present with hypertension at an earlier age, have more BP-related complications at any level of BP, and have evidence of target-organ damage at an earlier age.[580] Beyond differences in traditional risk factors for high BP, many social determinants of health such as socioeconomic, neighborhood, and geographic factors contribute to the high prevalence of hypertension, poor control rates, and disproportionately high frequency of BP-related CVD in African Americans compared with other racial/ethnic groups.

Racial differences in response to antihypertensive medications have been demonstrated repeatedly.[569,581] Genetic differences in kidney sodium handling and salt sensitivity have been proposed as a pathophysiologic mechanism, but differences have been noted independent of dietary salt or potassium intake.[582,583] Thiazides and CCB are especially effective antihypertensive agents in African Americans and Black Africans, even when administered at low doses.[569,584] Higher doses are usually required for drugs that block the renin–angiotensin–aldosterone system (RAAS).[585] Because multiple antihypertensive agents are usually required, use of a fixed-dose combination is recommended during initial antihypertensive drug therapy in African American patients with moderate or more severe hypertension.[296] Compared with Whites, Hispanics and Asians do not appear to have different BP-lowering responses to commonly used drugs.[581] However, Caribbean Hispanics, who have a higher proportion of African ancestry, may respond to antihypertensive agents in a manner similar to non-Hispanic Blacks.[583]

OBESE PATIENTS

Compression of the kidneys, activation of the renin-angiotensin-aldosterone system, and increased SNS activity[586] are common in obesity, as are other medical problems.[587] Obese patients with hypertension usually require several BP-lowering agents, prompting the use of fixed-dose combination agents. In addition to thiazides, combinations that include a CCB, ACE inhibitor, ARB, or β-blocker can be useful, especially in younger adults. β-Blockers may increase the likelihood of weight gain, and both β-blockers and diuretics can result in hyperglycemia. Mineralocorticoid receptor antagonists (MRA) may be helpful as a component of multidrug treatment.

TREATMENT OF HYPERTENSION IN ADULTS WITH CARDIOVASCULAR DISEASE

Choice of antihypertensive medication in adults with hypertension and CVD is often influenced by a patient's specific type of comorbidity.[523,588,589] Patients with stable ischemic heart disease, HFrEF, and heart failure with preserved ejection fraction (HFpEF) frequently require multiple agents including ACE inhibitors, ARB, and β-blockers. Other agents including thiazides, CCB, or MRA can be added to achieve an SBP/DBP <130/80 mm Hg. Nondihydropyridine CCBs should generally be avoided in patients with heart block or heart failure. Other agents prescribed to patients with CKD and hypertension for other indications that do not meet the regulatory requirements for labeling as antihypertensive agents, including SGLT2 inhibitors[590,591] and GLP-1 receptor agonists,[592,593] may contribute to BP lowering without additional AEs or pill burden.

TREATMENT OF HYPERTENSION IN ADULTS WITH KIDNEY DISEASE

In SPRINT, intensive antihypertensive treatment resulted in similarly beneficial effects for the primary CVD outcome and all-cause mortality in participants with and without

CKD at baseline.[526,527] There was no difference in the principal kidney outcome (reduction eGFR by 50% or more or ESKD in those with CKD at baseline) in participants randomized to more intensive or standard antihypertensive treatment[594] (17 vs. 16 participants), but the intensive treatment group had less incident albuminuria (64 vs. 85 participants). There were more reports of acute kidney injury (AKI) events, based on a modified KDIGO definition, in participants randomized to more intensive compared with standard treatment (179 vs. 109 participants), with the difference being more apparent in those without (82 vs. 38 participants) than with (97 vs. 71) CKD at baseline. However, most of the AKI events were mild, only identified on one occasion, and resolved during follow-up.[595] A network meta-analysis that pooled experience from 18 RCT (15,924 participants with eGFR <30 mL/min/m^2; 1293 deaths) documented a significant 14% reduction in mortality with more compared with less intensive antihypertensive therapy (average achieved SBP of 132 mm Hg and 140 mm Hg, respectively).[596]

Achieving BP control may be difficult in adults with hypertension and kidney disease because of the combined presence of vascular resistance and increased intravascular volume.[597] Glomerular circulation is optimal at one-half to two-thirds of systemic BP.[598] Normally, preglomerular vasoconstriction decreases the BP experienced by the glomerulus to a level that optimizes capillary filtration but is low enough to avoid mechanical injury to the filtering apparatus.[598,599] The efferent glomerular arteriole also serves an important purpose of vasoconstriction, when needed, to maintain adequate pressure for glomerular filtration. With the development of kidney disease, the afferent glomerular arteriole tends to provide less than optimal vasoconstriction, resulting in a higher than desired intraglomerular pressure. A clinical clue to this pathophysiology is microalbuminuria or larger amounts of protein in the urine. Under these circumstances, systemic BP should be reduced and specific antihypertensive agents should be used to minimize the risk of mechanical injury to the glomerulus.

Drugs that block the RAAS, such as ACE inhibitor and ARB,[600-603] provide an opportunity to reduce the progression of kidney disease as part of a BP-lowering strategy, especially in patients with heavy proteinuria and/or progressive reduction in eGFR as outlined earlier (see Tables 48.2 and 48.5). Renin inhibitors have a similar BP-lowering effect, reducing proteinuria and decreasing the progression from micro to macroproteinuria,[604] but are not as well established for renoprotection in RCT compared with ACE inhibitors or ARBs.[605] Additional medications, including diuretics, are almost always needed to achieve the desired level of BP control. For many years, loop diuretics were recommended instead of thiazide or thiazide-type diuretics when kidney function declines to an eGFR of 30 mL/min/1.73 m^2 or less. However, in an RCT conducted in patients with resistant hypertension and stage 4 CKD (mean eGFR 23 mL/minute/1.73 m^2), addition of chlorthalidone (12.5–50 mg/day) lowered 24-hour ambulatory BP by 10.5 mm Hg compared with placebo and reduced the urinary albumin/creatinine ratio by 50%.[606] The safety of CCB in patients with kidney disease has been questioned because of their preferential effects in dilating the afferent glomerular arteriole, and in some studies, CCBs have increased proteinuria, despite lowering systemic BP.[607,608] However, there is no clinical evidence that these drugs lead to progressive kidney disease when they are combined with ACE inhibitors or ARBs, which dilate the efferent glomerular arteriole.[609] Antiproteinuric agents such as ACE inhibitors and ARBs seem to slow the progression of kidney disease, at least in those with more pronounced CKD.[601,610] Dual RAAS blockade reduces proteinuria but has not provided incremental benefit for kidney disease progression in clinical trials of patients with diabetic kidney disease and is associated with a higher risk for an acute decline in kidney function and hyperkalemia.[183,611]

DIABETES

The first RCT to document benefit of more versus less intensive antihypertensive treatment was conducted in adults with type 2 diabetes mellitus, but the average achieved SBP/DBP in the more intensively treated group was only 144/82 mm Hg.[544] The more recent ACCORD BP trial primary analysis did not document an overall benefit of intensive (SBP goal <120 mm Hg) compared with standard (SBP goal <140 mm Hg) antihypertensive therapy in adults with type 2 diabetes mellitus.[612] The most likely explanation for this unexpected result was failure to meet the factorial design assumption of no interaction between the two treatments for the primary analysis that was employed in the main results manuscript. In contrast to the two-arm parallel design used in the SPRINT[613,614] and STEP[545] trials, the ACCORD BP[612] trial employed a two-by-two factorial design in which participants were randomized to two interventions: an intensive (glycosylated hemoglobin A1c target <6.0%) versus standard (glycosylated hemoglobin A1c target 7.0%–7.9%) glycemic control intervention and an intensive (SBP target <120 mm Hg) versus standard (SBP target <140 mm Hg) antihypertensive treatment intervention. The intensive glycemic control intervention was discontinued after 3.5 years due to an increased risk of death from any cause (HR, 1.35; 95% CI, 1.04–1.76). Like SPRINT, intensive BP-lowering therapy in the ACCORD BP standard glycemia arm reduced the hazard ratio of the composite primary CVD endpoint (HR, 0.77; 95% CI, 0.63–0.95).[613] In the ACCORD participants assigned to the intensive glycemic arm, there was no evidence that intensive antihypertensive treatment reduced the risk of the composite CVD endpoint during the period of intensive glycemic control (HR, 1.67; 95% CI, 1.13–2.45), but a benefit similar to the SPRINT experience was observed following discontinuation of the intensive glycemic intervention (HR, 0.79; 95% CI, 0.44–1.44).[613] SPRINT did not include patients with diabetes, but the benefits of intensive antihypertensive treatment in patients with prediabetes were similar to the overall findings.[614] On the basis of these and other findings, the ACC/AHA guideline recommends an SBP/DBP treatment target of <130/80 mm Hg. A network meta-analysis based on 30 RCTs, with a sample size of 59,934 participants and 7799 CVD events, demonstrated substantial benefit in patients treated more intensively, with the greatest benefit for CVD prevention in those with an achieved SBP 120 to 124 mm Hg (27% reduction in CVD risk compared with an SBP 130–134 mm Hg and 40% compared with 140–144 mm Hg).[615]

STRATEGIES FOR SELECTING THE OPTIMAL COMBINATION OF ANTIHYPERTENSIVE MEDICATION

A meta-analysis involving 40,000 treated patients evaluated the usefulness of low-dose combination treatments separately and in combination.[616] In 354 randomized, double-blind, placebo-controlled trials, the average BP was reduced by 20% with low-dose thiazide, β-blocker, ACE inhibitor, ARB, or CCB therapy. Thiazides and CCBs caused side effects infrequently when taken at a half-standard dose (2% and 1.6%, respectively) but more commonly with standard doses (9.9% and 8.3%, respectively). In those treated with β-blockers, side effects were relatively common at both low- and high-dosage levels (5.5% and. 7.5%, respectively). The only side effect observed with ACE inhibitor was cough (3.9%), which did not vary with dosage. ARBs were not associated with an excess of side effects when taken at a standard dose. The reduction in BP with combination therapy was additive, but the prevalence of AEs was not. Therefore combinations of two or three antihypertensive drugs at low doses seemed to be preferable to treatment with individual drugs at full doses. In the absence of cough, ACE inhibitors and ARBs can be used at higher doses because of a relative lack of other AEs. In a subsequent meta-analysis of 42 RCTs, the authors quantified the effect of drug combinations of agents from any two classes of thiazides, β-blockers, ACE inhibitors, and CCBs compared with the use of one drug alone and compared the effects of combining drugs with doubling dose. The extra BP reduction achieved by combining two drugs from different classes was approximately five times greater than that achieved when the dosage of one drug was doubled.[617]

The rationale for fixed-dose combination therapy is as follows.

- Combining drugs that act on different physiologic systems should:
  - Enhance BP lowering compared with monotherapy.
  - Mitigate counterregulatory responses that interfere with BP reduction. For example, diuretics can activate the RAAS system, which could attenuate BP lowering. This action can be mitigated by adding an ACE inhibitor or ARB. Likewise, the antihypertensive effect of a RAAS blocker can be enhanced by the natriuresis of diuretics and, to a lesser extent, CCB.
  - Decrease pill burden and improve medication adherence. RCTs have documented the need for multiple antihypertensive agents in most patients with hypertension. In the ALLHAT, for example, only one-third of the participants achieved the SBP/DBP <140/90 mm Hg target with monotherapy. The mean number of antihypertensive drugs used in RCTs with an SBP/DBP target of <140/90 mm Hg has been approximately two, and in RCTs with an SBP target of <120 mm Hg, it has been about three. A 2013–2016 NHANES report documented treatment with one, two, three, or four antihypertensive medications in 40.1%, 38.4%, 16.6%, and 4.3% of U.S. adults, respectively.[618] Overall, two or more medications were being used in almost 60% of those being treated with antihypertensive medications.
  - Decrease BP variability compared with monotherapy. Variability in visit-to-visit SBP is a predictor of CVD, independent of usual average BP.[619,620] CCBs and diuretics may reduce BP visit-to-visit variability, whereas β-blockers may increase it.[621]
  - Reduce AEs compared with monotherapy at a higher dose. For example, combining an ACE inhibitor or ARB with a CCB should reduce the incidence of CCB-related peripheral edema by up to 38%, and combining an ACE inhibitor or ARB with a diuretic should reduce the occurrence of diuretic-induced hypokalemia.

In the TRIUMPH trial, a higher percentage of those randomized to a once-daily, fixed-dose triple combination antihypertensive pill (20 mg of the ARB, telmisartan, 2.5 mg of the CCB, amlodipine, and 12.5 mg of the thiazide-type diuretic, chlorthalidone) achieved the target BP at 6 months compared with usual care (70% vs. 55%; $P < .001$).[560] There was no significant difference between the two treatment groups in participant withdrawal from their BP-lowering therapy because of AEs (6.6% vs. 6.8%).

PREFERRED COMBINATIONS

Renin–angiotensin–aldosterone system inhibitors and calcium channel blockers

Combining an ACEI or ARB with a CCB is logical and especially useful in patients with kidney disease and hypertension because CCBs decrease vascular responsiveness to Ang II and the synthesis and secretion of aldosterone.[314] Examples of such combinations in the United States include amlodipine with benazepril and trandolapril with verapamil. The ACCOMPLISH trial (N = 11,000) reported single-pill combination therapy with an ACE inhibitor and a CCB (benazepril–amlodipine) was superior (hazard ratio [95% CI] of 0.80 [0.62, 0.90]) to ACE inhibitor and thiazide (benazepril–hydrochlorothiazide) combination for the primary outcome (composite of CVD events or death).[609] However, the choice and dosage of the diuretic employed (hydrochlorothiazide) has been criticized.[622]

Renin–angiotensin–aldosterone system inhibitors and thiazide-type diuretics

By depleting intravascular volume, diuretics may activate the RAAS system, with the potential for vasoconstriction, as well as salt and water retention. RAAS blockade attenuates this response to diuretics and results in additional BP reduction. RASS inhibitor-diuretic combinations also tend to mitigate diuretic-induced hypokalemia and hyperglycemia. In the HYVET trial, the long-acting thiazide-type diuretic, indapamide, was combined with the ACE inhibitor, perindopril, in almost 75% of the participants. Active treatment reduced the incidence of CVD by 34% compared with placebo (stroke by 30% and heart failure by 64%).[564] Examples of ACE inhibitor-diuretic combinations in the United States include lisinopril, benazepril, enalapril, fosinopril, or quinapril with hydrochlorothiazide. Examples of ARB-diuretic combinations include losartan, valsartan, olmesartan, telmisartan, irbesartan, or candesartan with hydrochlorothiazide.

ACCEPTABLE COMBINATIONS

β-Blockers and diuretics

Many commercially available β-blocker–diuretic formulations exist (e.g., bisoprolol, metoprolol, propranolol, timolol with hydrochlorothiazide, and atenolol with chlorthalidone).

The addition of diuretics increases the effectiveness of β-blockers, especially in African Americans and others with low-renin hypertension. β-blockers and diuretics have overlapping adverse effects including the development of hyperglycemia, fatigue, and sexual dysfunction. A meta-analysis has reported a 32% increase in new-onset diabetes with diuretics and a 28% increase with β-blockers.[623] However, drug-related biochemical hyperglycemia may not be associated with the same risk as metabolically related diabetes mellitus. ALLHAT findings suggest that <20% of thiazide-associated biochemical diabetes is due to a diuretic drug effect per se.[624] Further, the risk associated with drug-associated diabetes or hyperglycemia was greater for lisinopril and amlodipine than chlorthalidone. In SHEP, participants randomized to chlorthalidone had a significantly lower risk of CVD mortality compared with placebo during long-term follow-up (mean = 14.3 years), especially those with thiazide-associated diabetes.[625]

Calcium channel blockers and thiazide-type diuretics

CCB and diuretic combinations provide partially additive antihypertensive effects.[626] This combination performed well in the VALUE trial, in which hydrochlorothiazide was administered as a second-step agent in about 25% of the participants randomized to the CCB, amlodipine, or ARB, valsartan.[627] Adverse effects included peripheral edema and hypokalemia, as well as a slight increase in the incidence of new-onset diabetes.

Calcium channel blockers and β-blockers

Additive BP reduction can be achieved with the combination of a β-blocker and dihydropyridine CCB, but this is usually reserved for unusual situations in which there are impediments to use of other combinations. Nondihydropyridine CCB such as verapamil and diltiazem should not be administered in combination with β-blockers because of the potential for heart block or severe bradycardia.

Dual calcium channel blockade

In a meta-analysis of 6 studies that included 153 patients, dual CCB therapy with dihydropyridine and nondihydropyridine agents provided additive antihypertensive effects without an increase in AEs.[628] However, long-term RCTs are needed to confirm treatment safety and CVD benefits.

UNACCEPTABLE OR INEFFECTIVE COMBINATIONS

Dual renin–angiotensin blockade

In the ALTITUDE trial, aliskiren or placebo was added to an ACE inhibitor or ARB in adults with CKD and type 2 diabetes mellitus.[184] The trial was stopped early after 32.9 months with no evidence that the addition of aliskiren to an ACE inhibitor or ARB improved CVD or kidney outcomes and resulted in an excess of AEs such as hypotension and hyperkalemia. In the ONTARGET trial, patients with CVD or high-risk type 2 diabetes mellitus were randomly assigned to treatment with an ACE inhibitor (ramipril), ARB (telmisartan), or a combination of both agents.[629] The ACE inhibitor and ARB combination resulted in an excess incidence of hypotension, hyperkalemia, and impaired kidney function. The 2013 Veterans Affairs Nephropathy in Diabetes (VA NEPHRON-D) RCT of losartan plus either lisinopril or placebo in type 2 diabetes and nephropathy was also stopped early due to more hyperkalemia and AKI in the group treated with both an ACE inhibitor and ARB, with no observed benefit on the primary composite endpoint (reduced estimated GFR, ESKD, or death) benefit.[611]

β-Blockers and central adrenergic agonists

Both β-blockers and central adrenergic agents, including clonidine and α-methyldopa, interfere with the SNS. The additive antihypertensive effect of this combination is unknown, but its use is not recommended because of the potential for bradycardia and heart block, especially in older adults, and rebound hypertension following abrupt discontinuation.

BEDTIME VERSUS MORNING ANTIHYPERTENSIVE DOSING

Typically, BP follows a circadian rhythm with a 10% to 20% decrease during the night. Lack of a reduction in BP at night, termed “nondipping,” is a CVD risk factor.[630,631] This is a rationale for recommending nighttime antihypertensive medications. Two trials by the same investigative group reported CVD benefits with bedtime antihypertensive drug therapy.[632,633] However, a review of these and six additional studies on the topic highlighted serious methodologic errors in each of the reports and culminated in a consensus statement by the International Society of Hypertension, endorsed by the World Hypertension League and the European Society of Hypertension, cautioning against the need for routine bedtime antihypertensive medication dosing until better quality data are available.[634] The consensus statement recommends use of long-acting antihypertensive medications administered as a once-a-day treatment.

RESISTANT HYPERTENSION

Resistant hypertension is a term used to characterize high BP that remains above goal despite the use of three optimally dosed drugs or the use of four or more medications to reach the BP goal.[635] Risk factors for resistant hypertension include older age, obesity, CKD, Black race, obesity, and diabetes mellitus.[523] An approach to the diagnosis and management of apparent resistant hypertension is presented in Fig. 48.6. It is important to exclude apparent resistant hypertension, which has a wide variety of causes (Table 48.26) but usually results from nonadherence to the prescribed medication regimen, white coat hypertension, or inaccurate BP measurement.

Medication nonadherence is common and can result from inadequate clinician-patient communication, lack of patient education, inconvenience or cost of prescriptions, misguided perceptions of drug side effects, and the challenge of adhering to complex and multidrug regimens. In a carefully performed drug metabolite screening study of patients with resistant hypertension at one U.S. academic center, complete and partial nonadherence was identified in 32% and 22%, respectively.[636] The corresponding percentages in a study conducted at an English academic center were 10% and 15%.[637] A second common explanation for apparent resistant hypertension is white coat hypertension, where office BP readings are high but out-of-office home BP or 24-hour ambulatory BP measurements

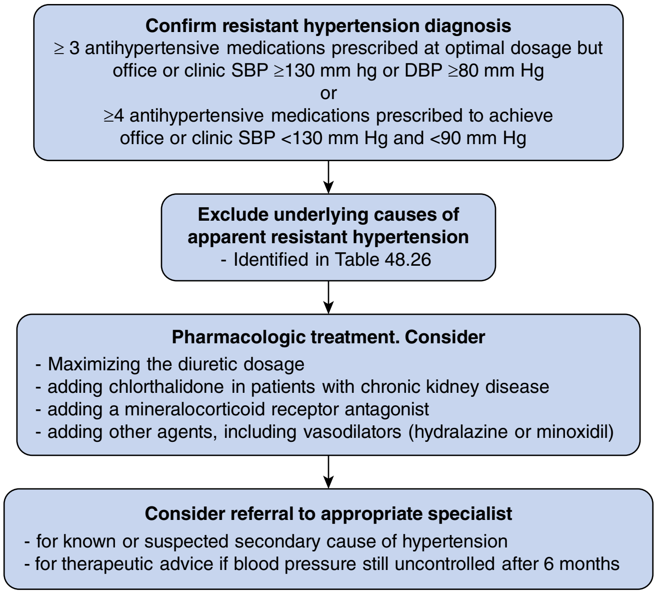

Fig. 48.6 Approach to management of apparent resistant hypertension.

Table 48.26 List of the Most Common Causes of Apparent Resistant Hypertension

Nonadherence to the treatment regimen
Pseudo resistance
- White coat hypertension
- Improper measurement of blood pressure, including:
  - Use of inappropriately small cuff

Pseudo hypertension (in older adults)
Lifestyle-related causes
- Excess sodium consumption, and/or insufficient potassium intake
- Poor quality diet, overweight and obesity
- Excess alcohol consumption
- Stress and other anxiety-producing stressors
- Inactivity

Drug-related causes
- Inadequate treatment
  - Inappropriate selection of antihypertensive drugs
  - Insufficient antihypertensive drug dosage
  - Inadequate attention to lifestyle counseling
- Treatment with agents that raise blood pressure
  - Sympathomimetics
  - Nasal decongestants
  - Appetite suppressants
  - Cocaine
  - Nonsteroidal antiinflammatory agents
  - Antidepressants
  - Oral contraceptives
  - Adrenal steroids
  - Cyclosporine and tacrolimus
  - Erythropoietin and other erythropoiesis-stimulating agents
  - Excess caffeine consumption
  - High intake of licorice or licorice products

Secondary causes of hypertension, including
- Endocrine disorders, especially hyperaldosteronism
- Kidney disease
- Renal artery stenosis
- Sleep apnea

are “normal.” The ACC/AHA BP guideline provides estimates for corresponding levels of office (clinic), home, and ambulatory BP measurements (Table 48.27).[523] Home BP measurements are more practical than ambulatory readings and have the added value of patient engagement. The need to wear ambulatory BP measurement devices for 24 hours, bring the device back to the clinic, and the limited reimbursement available through insurance limit the use of ambulatory BP measurements in the United States. Inaccurate BP measurement, especially use of a cuff that is too small, and pseudohypertension in older adults with arteriosclerosis should also be excluded.

Underlying lifestyle factors that contribute to resistant hypertension include poor diet, overweight/obesity, high-salt intake, physical inactivity, and excessive alcohol consumption. Volume overload is a relatively common cause of resistant hypertension. Resistant hypertension may be related to excessive salt intake and/or an inability to excrete sodium due to kidney disease or endocrine abnormalities. High dietary sodium intake offsets the BP-lowering effects of antihypertensive medications.[569] Some patients, including many African Americans and those with CKD and/or heart failure, are especially salt sensitive.[583] Counseling patients to avoid excessive sodium consumption and judicious use of thiazide-type or loop diuretics can be helpful in restoring antihypertensive efficacy.

Drug-related causes of resistant hypertension include the use of oral contraceptives, immunosuppressants (e.g., cyclosporine and tacrolimus), erythropoiesis-stimulating agents, certain antidepressants, and over-the-counter (OTC) preparations of sympathomimetics, such as nasal decongestants, appetite suppressants, and NSAIDs.[638] In addition, alcohol or cocaine use and large intake of licorice can complicate BP lowering. NSAIDs seem to counteract the antihypertensive activity of diuretics and ACE inhibitor 674 but not CCB.[639]

If the proceeding issues have been excluded, secondary causes of hypertension should be excluded. Endocrine causes of hypertension include primary hyperaldosteronism, Cushing disease, pheochromocytoma, hypothyroidism or hyperthyroidism, and hyperparathyroidism. Because aldosteronism is responsible for 15% to 20% of true resistant hypertension, all such patients should be screened by measuring the ratio of plasma aldosterone–to–plasma renin activity. In a large retrospective study of clinical centers in five continents, widespread use of plasma aldosterone concentration–to–plasma renin activity ratios increased the detection of primary aldosteronism from 5- to 15-fold.[640] Finding an underlying adrenal adenoma was rare but increased by up to sixfold across the centers. Most patients with nonadenomatous hyperaldosteronism respond well to treatment with an MRA.[641]

Kidney disease can usually be identified biochemically and/or by urinalysis. Kidney artery stenosis is usually due to fibromuscular disease or atherosclerosis. Fibromuscular dysplasia is classically found in young to middle-aged white women[642] and can often be managed successfully by percutaneous kidney angioplasty. In contrast, atheromatous renovascular disease usually presents in older adults with widespread atherosclerosis and several RCTs have failed to demonstrate the superiority of kidney revascularization compared with medical therapy for BP control, slowing the progression of kidney disease, or prevention of CVD.[643-645]

Table 48.27 Corresponding Average Values of Office/Clinic, Home, and Ambulatory Monitoring Systolic and Diastolic Blood Pressure Measurements (in mm Hg). Ambulatory Blood Pressure Measurement Values are Displayed for 24-Hour, Daytime, and Nighttime Readings

		Ambulatory		
Clinic/Office	**Home**	**24-hour**	**Daytime**	**Nighttime**
120/80	120/80	115/75	120/80	100/65
130/80	130/80	125/85	130/80	110/65
140/90	135/85	130/80	135/85	135/85
160/100	145/90	145/90	145/90	140/85

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The evidence for stenting of the atherosclerotic kidney artery in patients with stenosis >60% and difficult-to-control hypertension is weak.[523] The decision to undertake kidney revascularization in addition to medical management should be individualized and relatively infrequent.

The only other relatively common cause of secondary hypertension is sleep apnea.[523] However, the effects of continuous positive airway pressure (CPAP) on BP are modest: an average SBP decrement of 2.6 mm Hg compared with passive treatment in a meta-analysis of 31 RCTs.[646] CPAP treatment might be better indicated for quality of life rather than management of BP or CVD prevention.[647]

"Refractory hypertension" refers to patients whose BP is uncontrolled with maximal or near-maximal medical therapy (five or more drugs given at maximally tolerated doses).[648] There may be a role for novel drug therapies and device therapies to lower BP in such patients. Selected novel antihypertensive drug and device treatments have been discussed in several review articles.[649] Device therapies proposed for control of high BP include kidney denervation (RDN) and carotid baroreceptor stimulation.[556] RDN is the best-proven device therapy, yielding an average SBP reduction of slightly more than 6 mm Hg in sham-controlled RCTs. Research studies suggest it may be useful as a treatment option in selected patients an eGFR >40 mL/min/1.73 m^2 who have uncontrolled BP despite adequate combination antihypertensive drug therapy or controlled hypertension with debilitating treatment side effects and poor quality of life. It may also prove useful as additional treatment in certain patients with resistant hypertension. In November 2023, the U.S. FDA approved two RDN devices: Medtronic's Symplicity Spyral Renal Denervation System and the Paradise Ultrasound RDN system from Recor Medical, Inc. and its parent company Otsuka Medical Devices.

DRUG TREATMENT OF HYPERTENSIVE URGENCIES AND EMERGENCIES

Hypertensive urgency and hypertensive emergency are distinct syndromes. A hypertensive emergency is a clinical syndrome in which marked elevation in BP results in ongoing target-organ damage in the body (Table 48.28). The syndrome can be manifested by neurologic (encephalopathy, retinal hemorrhage, papilledema, stroke); cardiac (acute MI or heart failure); vascular (aortic dissection, thrombotic microangiopathy); or kidney (AKI) damage. Any delay in control of BP may lead to irreversible sequelae including death.

Table 48.28 Hypertensive Emergencies

- Hypertensive encephalopathy
- Acute aortic dissection
  - Central nervous system bleeding
  - Intracranial hemorrhage
  - Thrombotic cerebrovascular accident
  - Subarachnoid hemorrhage
- Acute left ventricular failure refractory to conventional medical therapy
- Myocardial ischemia or infarction associated with persistent chest pain
- Accelerated or malignant hypertension[a]
- Toxemia of pregnancy – eclampsia
- Renal failure or acute kidney injury
- Hypertension associated with hyperadrenergic states
  - Pheochromocytoma
  - Interaction between monoamine oxidase inhibitors and tyramine-containing foods
  - Interaction between an α-adrenergic agonist and nonselective β-adrenergic antagonist
  - After abrupt withdrawal of clonidine
  - After severe body burns
  - Neurogenic hypertension
- Hypertension in the surgical patient[a]
  - Associated with postoperative bleeding
  - After open heart or vascular surgery
  - Preceding emergency surgery
  - After kidney transplantation
  - Hypertension in a diabetic patient with retinal hemorrhage

[a]Considered by some authors to be a hypertensive urgency.

In contrast, hypertensive urgency is a clinical situation in which a patient may have a marked elevation in BP (e.g., >200/130 mm Hg) but no evidence of ongoing target-organ damage. Such a patient can be treated with rapid-onset oral drugs, such as captopril or clonidine, and be

Table 48.29 Parenteral Drugs Used in the Treatment of Hypertensive Emergencies

Drug	Dosage	Onset of Action	Peak Effect	Duration of Action
Clevidipine	1–2 mg/h (16 mg/h maximum)	2–4 min	5–6 min	15 min
Enalaprilat	0.625–5.0-mg bolus over 5 min q6h	5–15 min	1–4 h	6 h
Esmolol	250–500 μg/kg/min × 1 (loading dose), then 50-100 μg/kg/min × 4 (maintenance); maintenance dosage may be increased to maximum of 300 μg/kg/min	1–2 min	5 min	0–30 min
Fenoldopam	0.01–1.6-μg/kg/min constant infusion	5–15 min	30 min	5–10 min
Hydralazine	10–20 mg as slow intravenous injection or 0.05–0.15 mg/min continuous infusion	1–5 min	10–80 min	3–6 h
Labetalol	20 mg bolus then 0.5 to 2 mg/minute infusion	5 min	10 min	3–6 h
α-Methyldopa	250–500 mg bolus every 6 h (2 g maximum)	2–3 h	3–5 h	6–12 h
Nicardipine	5–15 mg/h	5–10 min	45 min	50 h
Nitroglycerine	5–100 μg/min infusion	1–2 min	2–5 min	3–5 min
Nitroprusside	0.25–10 μg/kg/min infusion	Immediate	1–2 min	2–5 min
Phentolamine	0.5–1.0 mg/min infusion or 2.5–5.0 mg bolus q5–15 minutes	Immediate	3–5 min	10–15 min
Trimethaphan	0.5–10 mg/min infusion bolus over 2 min (300 mg maximum)	Immediate	1–2 min	5–10 min

observed cautiously, with the administration of long-acting medications on an outpatient basis as progressive restoration of a more appropriate BP level is attained.

Apart from careful history and physical examination, laboratory workup should include serum chemistries including creatinine, serum troponins, complete blood count and peripheral blood smear if hemolysis is suspected, chest radiograph, electrocardiogram, brain imaging if neurologic symptoms are present, chest computed tomography, or transesophageal echocardiography if aortic dissection is suspected. The decision about whether to hospitalize the patient in an intensive care unit and use intravenous medication or to observe the patient carefully and use oral medications to facilitate better BP control depends in large part on the presence or absence of ongoing target-organ injury.

TREATMENT OF HYPERTENSIVE EMERGENCIES

Acute lowering of BP should be avoided in hypertensive emergencies to decrease the risk of ischemic neurologic damage. The target BP is usually <180/<120 mm Hg in the first hour and <160/110 mm Hg in the following 23 hours. The exception is aortic dissection, where the SBP should be reduced to <120 mm Hg in 20 minutes to reduce shear pressure on the aorta. In those with intracranial hemorrhage presenting with SBP >220 mm Hg, it should be lowered to <220 mm Hg and slowly reduced to a goal of 140 to 160 mm Hg. In those presenting in the acute phase of ischemic stroke and candidates for reperfusion therapy, BP should be lowered to <180/105 mm Hg. If they are not candidates for reperfusion therapy, BP should be maintained <220/120 mm Hg.

Parenteral drugs used in the treatment of hypertensive emergencies are summarized in Table 48.29. The choice of antihypertensive drug depends on the presentation.

In acute ischemic or hemorrhagic stroke, intravenous labetalol, nicardipine, and clevidipine are generally recommended and nifedipine should be avoided. Intravenous nitroprusside and nitroglycerin are also not preferred agents because of considerations of increased intracranial pressure. In hypertensive encephalopathy without stroke, nicardipine, clevidipine, fenoldopam, and nitroprusside are commonly used.

Apart from loop diuretics, agents that reduce both preload and afterload such as intravenous nitroglycerine or nitroprusside are used in those with hypertensive emergencies with acute heart failure. β-Blockers (including labetalol) that diminish cardiac contractility and peripheral vasodilators (such as hydralazine) that increase cardiac workload should be avoided.

In the setting of hypertensive emergency with acute MI, β-blockers (intravenous metoprolol or esmolol) that reduce cardiac contractility should be used along with intravenous nitroglycerine, nicardipine, or clevidipine to reduce blood pressure.

In aortic dissection, to reduce the shear stress on the aorta, both the heart rate (to <60 beats/min) and BP (<120/80 mm Hg) should be quickly reduced. This is usually achieved with a combination of intravenous β-blocker and nitroprusside or clevidipine.

In hypertensive emergencies with sympathetic overactivity such as cocaine/methamphetamine intoxication or pheochromocytoma, β-blockers should be avoided because of concerns about unopposed α activity resulting in coronary vasoconstriction and systemic hypertension. In these cases, intravenous phentolamine, an α-blocker, or intravenous nitroprusside is used.

Fenoldopam, a dopamine agonist, preserves kidney blood flow. Hence it is a preferred agent in those with hypertensive emergencies presenting with AKI. In hypertensive emergencies in pregnancy, hydralazine or labetalol are commonly used.

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